The Logical Cost of Everything

How 98 Papers Measured What Mathematics Actually Needs

Paul Chun-Kit Lee
New York University

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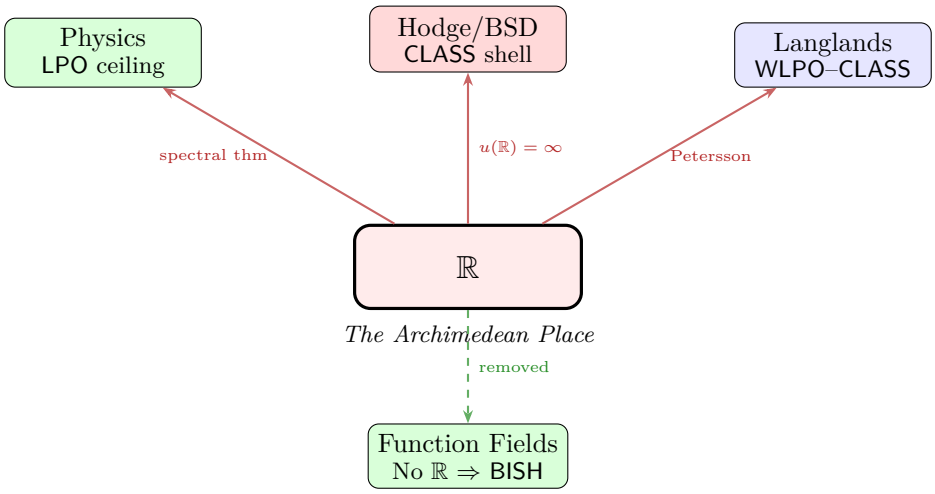

Constructive Reverse Mathematics Series

Papers 1–98 · 96,200+ lines Lean 4 · Zero `sorry`

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To Mimi

Ἐν ἀρχῇ ἦν ὁ Λόγος,
καὶ ὁ Λόγος ἦν πρὸς τὸν Θεόν,
καὶ Θεὸς ἦν ὁ Λόγος.

In the beginning was the Word,
and the Word was with God,
and the Word was God.

— John 1:1

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Preface: The Map and the Territory

In 1967, Errett Bishop published *Foundations of Constructive Analysis* and defined a program: rebuild mathematics so that every existence claim comes with an explicit witness, every function is computable, every proof provides an algorithm. Over the next three decades, Bishop and his successors—Bridges, Richman, Ishihara, Bauer—mapped the precise hierarchy of non-constructive principles that separate different strengths of mathematical reasoning. They called these the *omniscience principles*, and the cartography they produced was meticulous, rigorous, and almost entirely ignored.

Constructive mathematics received recognition at the highest levels—Bauer's 2017 address to the AMS, for instance—but produced no foundational change in practice. The mainstream continued to use classical logic without asking what it cost. The hierarchy Bishop built remained a curiosity of mathematical philosophy, not a tool for working mathematicians or scientists.

The missing step was not advocacy but *measurement*. Nobody had taken the actual theorems of physics and pure mathematics—the ones that produce numbers, predict experiments, settle conjectures—and calibrated them against Bishop's hierarchy. This monograph reports the results of that calibration: 98 papers, 96,200 lines of machine-checked Lean 4 code, spanning quantum mechanics, general relativity, the Hodge conjecture, the Birch and Swinnerton-Dyer conjecture, and the Langlands program, all asking a single question:

Central Question

What logical resources does this theorem actually need?

Not "is it true?" but "what do you have to believe to prove it?" Not "how complex is the proof?" but "which non-negotiable principles does the proof depend on?"

The answer is this:

Central Finding

The logical cost of mathematics is the logical cost of the real numbers.

Every non-constructive principle that mathematics requires enters through one object: the real number line $\mathbb{R}$. Remove it, and both physics and the deepest arithmetic geometry collapse to fully constructive mathematics—mathematics where every "there exists" comes with an explicit witness, every algorithm

terminates with a known bound, every claim can be checked by a machine.

This monograph explains that finding, the 98-paper journey that established it, and why it matters for anyone who builds systems, reasons about structure, or wants to understand what makes hard problems hard.

How to Read This Document

You do not need algebraic geometry to read this monograph. Every technical term is defined before it is used; every hard concept gets an analogy first. If you have a background in linear algebra and multivariable calculus, Chapter 1 will give you everything else.

Chapter 1 builds the complete conceptual toolkit. Every technical term is defined here before it appears elsewhere. **Chapters 2–3** establish the mathematical foundation: the five great conjectures and the DPT axioms. **Chapters 4–7** are the core: the Squeeze protocol, the Hodge campaign, the BSD application, and the motivic architecture that explains why the pattern holds. **Chapters 8–9** draw cross-domain implications and conclude. **Appendices** provide the complete paper list, glossary, and hierarchy.

Every claim links to the original paper. No formal proof is required to follow the argument. Diagrams and analogies carry the load.

Chapter 1

The Measuring Instrument

This chapter builds everything you need before the mathematics starts. No concept is skipped. By the end, you will understand the hierarchy, what an oracle is, why the real numbers are special, and what formal verification means.

1.1 What "Exists" Costs

When a physicist says "there exists a ground state energy for this Hamiltonian," they mean something very specific: given enough compute time, you can find it. The eigenvalue is computable. You can write an algorithm that produces it to any desired precision. The physicist's "exists" comes with a recipe.

When a number theorist says "there exists a rational point on this elliptic curve," they may mean something entirely different. They may have proved the point exists by contradiction—by showing that its non-existence leads to an absurdity. But they cannot tell you what the point is, how to find it, or how long the search would take. Their "exists" is a logical guarantee with no recipe attached.

These two kinds of "exists" have different logical costs. The physicist's version is cheap. The number theorist's version may be expensive. **Constructive Reverse Mathematics (CRM)** is the framework that makes this difference precise.

The CRM Question

For a given theorem, what is the minimum set of logical principles needed to prove it? Can you do it with just algorithms and explicit witnesses? Or do you need an *oracle*—a hypothetical device that answers questions no algorithm can?

1.2 Constructive Reverse Mathematics

"Reverse mathematics" inverts the usual direction of mathematical reasoning. Normally, you start with axioms and derive theorems. Reverse mathematics starts with theorems and asks: *which axioms did you actually need?*

Think of it as a dependency audit for mathematical proofs. A software project has a dependency tree—libraries it imports, APIs it calls. Some dependencies are load-bearing (remove them and the code breaks). Others are convenience

(you could replace them with a few lines). CRM is `npm audit` for mathematical proofs: it traces every logical dependency, flags which are structural and which are scaffolding, and reports the minimum viable dependency set.

Formal Definition

CRM classification of a theorem T: identify the weakest principle P in the hierarchy ($\mathsf{BISH} \subset \mathsf{MP} \subset \mathsf{LLPO} \subset \mathsf{WLPO} \subset \mathsf{LPO} \subset \mathsf{CLASS}$) such that T is provable in $\mathsf{BISH} + P$. The classification is *tight* if additionally T implies P over BISH (reverse direction). When both directions hold ($T \Leftrightarrow P$ over BISH), the classification is *biconditional*.

The CRM program (**Paper 1–98**) performed this audit on theorems across all of mathematical physics (**Paper 1–44**) and arithmetic geometry (**Paper 45–98**). The result: a uniform classification of where logical difficulty lives and why.

1.3 The Hierarchy

Every theorem lives somewhere on a hierarchy of logical strength. Figure 1.1 shows the complete hierarchy from weakest (cheapest) to strongest (most expensive).

Analogy: The Restaurant Menu

- **BISH:** The dish is on the menu with a recipe and a prep time. You can make it.
- **MP:** The dish is on the menu. The recipe terminates, but you don't know when the timer goes off. You wait, knowing it will come—eventually.
- **LLPO:** Two ovens, at most one is on. You can determine which—but only by committing to check one first.
- **WLPO:** A soup is simmering. You must decide: is the salt content *exactly* zero, or is it nonzero? After a million tastings, you still can't be sure—the salt might be 10^{-100}.
- **LPO:** An infinite buffet line. Somewhere, one dish has a cherry on top. You must scan every plate.
- **CLASS:** The chef says "trust me, something delicious exists," and you take his word without seeing the kitchen.

1.4 What an Oracle Is

In this framework, an "oracle" is a hypothetical device that answers a question no algorithm can answer in finite time. It is not mystical—it is a precise specification of computational power. Figure 1.2 compares the three main oracles.

When CRM says a theorem "costs LPO," it means: you cannot prove this theorem without the ability to scan infinite sequences. When it says "BISH," it means: no oracle needed—an algorithm suffices.

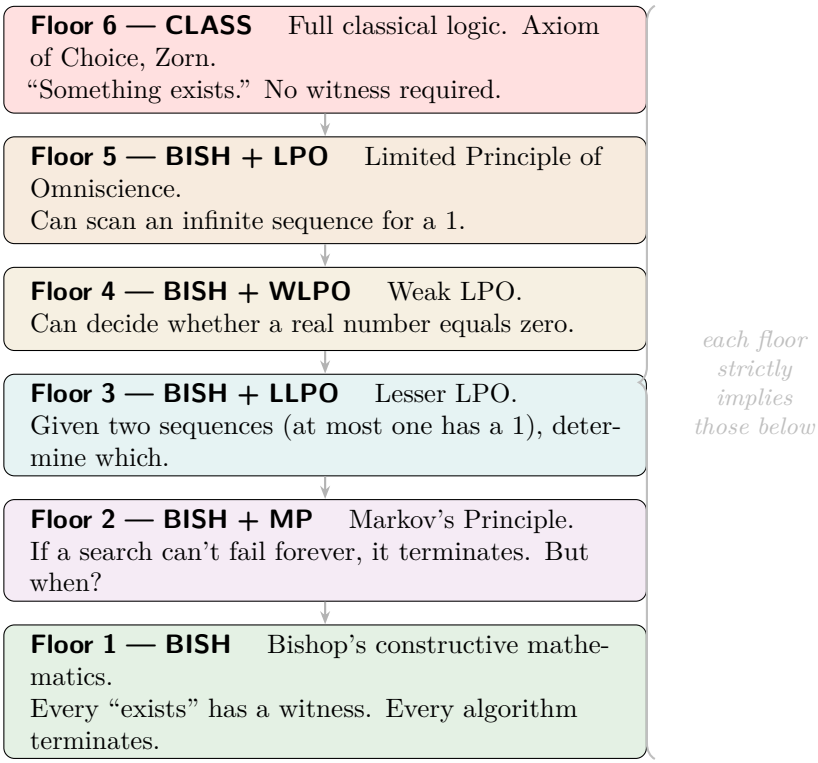


Figure 1.1: The CRM hierarchy. Each floor strictly implies those below it. The program's central finding: most of mathematics lives on Floor 1 (BISH). Almost everything on higher floors got there via one object: $\mathbb{R}$.

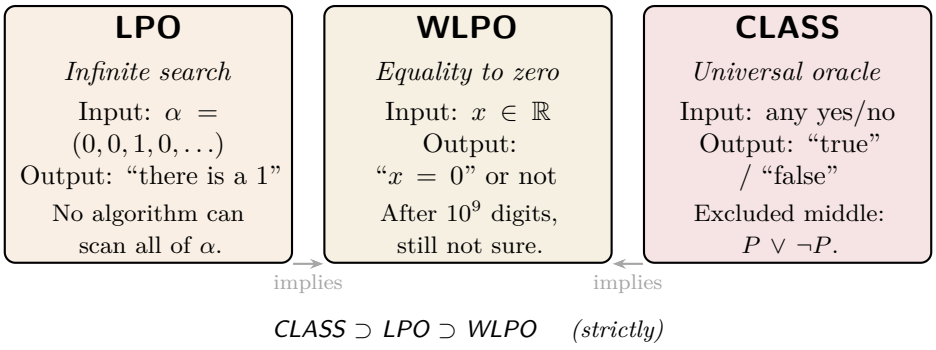


Figure 1.2: The three main oracles. Each answers a question no algorithm can answer in finite time. CLASS is the most powerful; WLPO the weakest (tests equality to zero). The CRM question: *which oracle does your proof need?*

The key intuition: each oracle is a weaker version of the one above it. CLASS can answer *any* yes/no question. LPO can only answer "is there a 1 somewhere in this infinite list?" WLPO can only answer "is this real number exactly zero?" LLPO can only answer "given two lists where at most one has a 1, which one?" Each step down shrinks the oracle's power—and CRM asks: what is the *weakest* oracle your proof actually needs?

1.5 The Archimedean Place and Why $\mathbb{R}$ Is the Source of All Difficulty

This is the core of the entire program. Understand this section and you understand the finding.

The rationals are decidable. Every question about rational numbers has an algorithmic answer. Is $3/7 < 5/11$? Multiply out and compare: 33 vs. 35. Done. BISH. The rational numbers are computationally tame.

The reals are not. A real number is typically represented as an infinite Cauchy sequence of rationals. To test whether two reals are equal, you must (in principle) compare infinitely many terms. This requires WLPO. To search a sequence of reals for one with a particular property requires LPO. The real numbers are computationally wild.

The mathematical term for where the real numbers sit is the **Archimedean place**—the completion of $\mathbb{Q}$ at infinity, as opposed to the p-adic completions at finite primes. The program's finding is that this one place—the Archimedean place, i.e., the real number line—is the sole source of non-constructive content in all of mathematics.

Why does $\mathbb{R}$ force non-constructive principles? The technical reason is captured by an invariant called $u(\mathbb{R}) = \infty$. The **u-invariant** of a field F measures the largest dimension n in which every quadratic form over F of dimension n has a nontrivial zero. For the reals, consider the sum of squares $x_1^2 + x_2^2 + \cdots + x_n^2$: every term is non-negative, so the sum is zero only when every x_i is zero. This works for *every* n—the form $x_1^2 + \cdots + x_n^2$ over $\mathbb{R}$ never has a nontrivial zero, no matter how many variables you add. Hence $u(\mathbb{R}) = \infty$. By contrast, over a p-adic field $\mathbb{Q}_p$ this fails for $n \geq 5$ (a theorem of Springer): $u(\mathbb{Q}_p) = 4$.

What does this have to do with constructivity? Positive-definite forms give you *inner products*: $\langle x, y\rangle = \sum x_i y_i$ is positive-definite over $\mathbb{R}$. Inner products give you *projections* (decomposing vectors into components), *norms* (measuring distances), and *heights* (measuring arithmetic complexity). These are the tools of analysis. But using them requires *deciding properties of real numbers*—is this inner product zero? is this norm smaller than that one?—and those decisions cost oracles (WLPO for equality, LPO for search).

The chain is: $u(\mathbb{R}) = \infty \Rightarrow$ positive-definite forms exist in all dimensions $\Rightarrow$ inner products, norms, heights $\Rightarrow$ decisions about real numbers $\Rightarrow$ oracles (WLPO/LPO/CLASS).

Three fields independently exploit this same mechanism. Each one uses a positive-definite form enabled by $\mathbb{R}$:

- **Hilbert space inner product** (physics): the inner product $\langle\psi|\phi\rangle$ on quantum states. Used for spectral decomposition. You already know this from linear algebra.
- **Rosati involution** (arithmetic geometry): a positive-definite pairing on algebraic cycles, analogous to an inner product. It gives heights their finiteness properties (the Northcott property). Explained in detail in §2.5 and Chapter 3.
- **Petersson inner product** (automorphic forms): an L^2 inner product on a space of modular forms, used to decompose them into eigenforms. Explained in the modular forms subsection below and Chapter 5.

All three structures exist because $u(\mathbb{R}) = \infty$. Remove $\mathbb{R}$ from the picture (as happens in function fields), and none of them are available—everything is BISH.

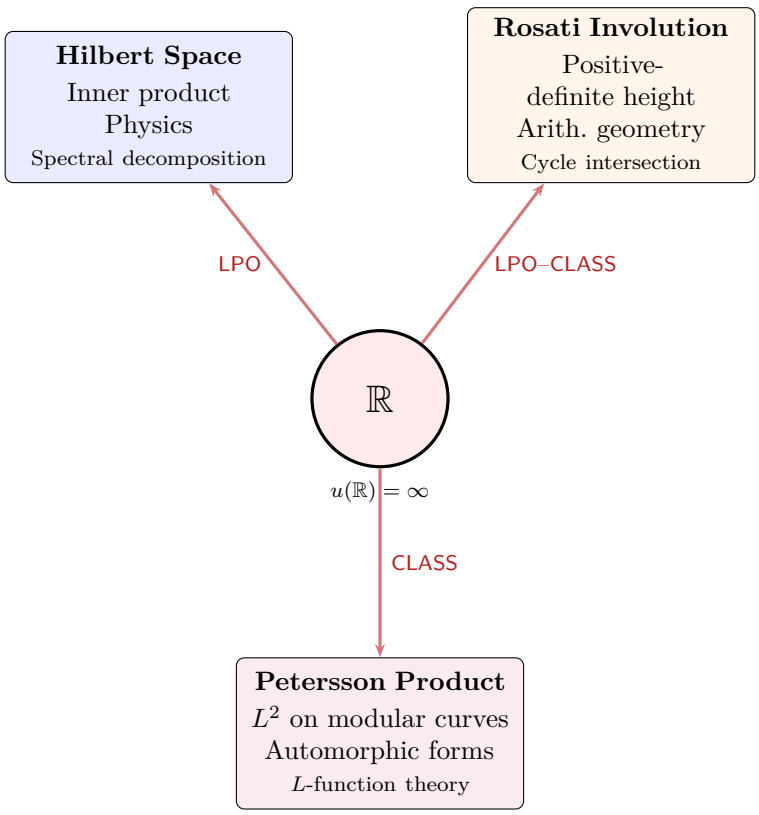


Figure 1.3: $\mathbb{R}$ as the sole source of non-constructive content. Each arrow carries the logical cost of the positive-definite structure it enables.

Analogy: The Power Plant

Imagine mathematics as a city. The entire city runs on electricity (logical principles). CRM traced every wire and found they all lead back to one power plant: the real number line. Some neighborhoods (physics) draw moderate power (LPO). Others (number theory) draw heavy power (CLASS). But the plant is the same. Shut it down, and the city goes dark. Build a different city without $\mathbb{R}$ (function fields, for instance), and no power plant is needed—everything runs on local generators (BISH).

Key references: Paper 70 (The Archimedean Principle—capstone) formalizes the four-domain partition. **Paper 35** and **Paper 40** establish the physics ceiling.

1.6 Projection vs. Search

Why is physics logically cheaper than number theory? Both use the real numbers. Both pay the Archimedean toll. But they use $\mathbb{R}$ differently. Figure 1.4 illustrates the structural distinction.

PHYSICS (Projection)

Given: Self-adjoint operator H on Hilbert space

Task: Find eigenvalues of H

Method: Project onto eigenspace (finite-dim, known location)

Cost: BISH + LPO

Why: You know WHERE to look. LPO locates the eigenvalue on $\mathbb{R}$.

Bounded ⇒ LPO ceiling

NUMBER THEORY (Search)

Given: Elliptic curve E over $\mathbb{Q}$

Task: Find rational points on E

Method: Search through $\mathbb{Q}^2$ (infinite, no known bound)

Cost: BISH + LPO (or higher)

Why: No target. You don't know IF a point exists, let alone WHERE.

Unbounded ⇒ sometimes CLASS

Figure 1.4: Projection is bounded (known target subspace); search is unbounded (no target). This structural difference explains why physics tops out at BISH + LPO while number theory sometimes needs CLASS. Both start from $\mathbb{R}$, but physics uses $\mathbb{R}$ to project, while number theory uses $\mathbb{R}$ to search.

1.7 The Physics Ceiling

Paper 1–44 systematically calibrated every major physical theory against the hierarchy. Table 1.1 shows the result. Two axioms that appear in some physics proofs—the Fan Theorem (FT) and Dependent Choice (DC)—are shown to be **dispensable** (**Paper 30–31**). The **Fan Theorem** says: every function from Cantor space $\{0,1\}^{\mathbb{N}}$ to $\mathbb{N}$ that is pointwise continuous is *uniformly* continuous. **Dependent Choice** says: given a relation R on a set A where every element has a successor, you can build an infinite chain. Both are logically independent of the main hierarchy and neither is required for any physical theory.

Table 1.1: Physics calibration results. Every physical theory was audited for its CRM cost. The ceiling is BISH + LPO, entering through the spectral theorem for unbounded operators on $\mathbb{R}$.

Domain	**Theorem / Concept**	**CRM Cost**
Classical Mechanics	Newton, Lagrange, Hamilton	BISH
Quantum Mechanics	Schrödinger equation	BISH
	Heisenberg uncertainty (**Paper 6**)	BISH
	Born rule (**Paper 16**)	BISH + LPO
	Spectral theorem	BISH + LPO
	CHSH / Tsirelson bound (**Paper 11**)	BISH
General Relativity	Schwarzschild interior (**Paper 5**)	BISH
	Singularity / event horizon	BISH + LPO
Statistical Mechanics	1D Ising model (**Paper 8–9**)	BISH + LPO
QFT	QED renormalization (**Paper 32**)	BISH
	QCD confinement (**Paper 33**)	BISH + LPO
	Scattering amplitudes (**Paper 34**)	BISH
Quantum Information	Bell nonlocality (**Paper 21**)	BISH + LLPO
	Decoherence (**Paper 14**)	BISH + WLPO
Ceiling:		**BISH + LPO**

The metatheorem (Paper 35, 40): BISH + LPO is the complete logical constitution of empirically accessible physics. No physical prediction requires more than LPO. This is a ceiling, not a floor—individual theories may need less.

A striking corollary of the physics ceiling: the "measurement problem" — the century-old debate over what happens when a quantum system is observed — dissolves under CRM. **Paper 44** showed that different quantum interpretations have different CRM costs: Copenhagen = BISH + LPO, Many-Worlds = BISH + LPO, Bohmian = BISH + MP. They are logically distinct but empirically equivalent.

1.8 Formal Verification in Lean 4

Every claim in this program is machine-checked. The proof assistant Lean 4, built on a small trusted kernel and a large mathematical library (Mathlib), verifies every logical step.

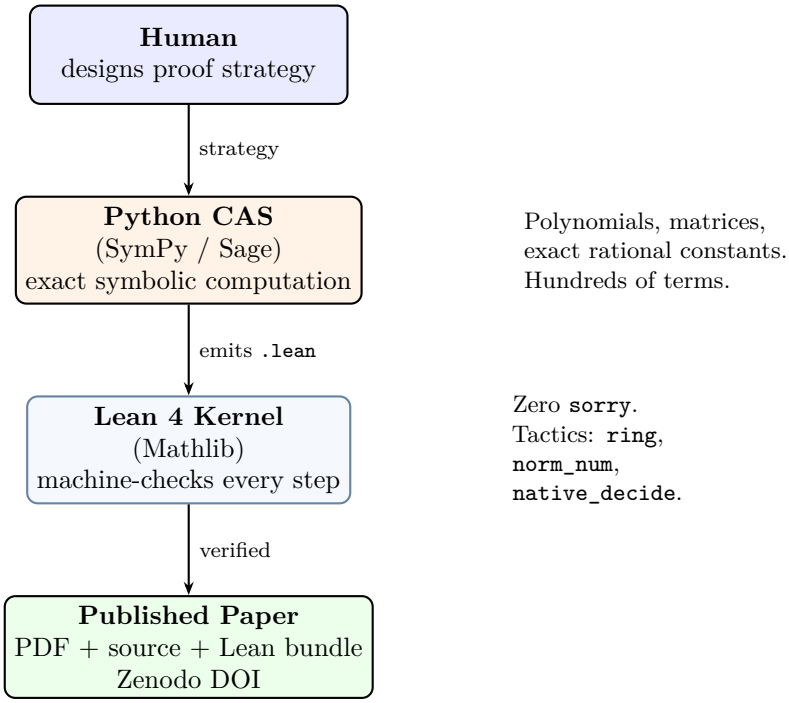


Figure 1.5: The asymmetric offloading pipeline. Humans design strategy; Python CAS computes exact algebra; Lean 4 verifies. Development time dropped from 4 months (**Paper 5**, manual) to 1 hour (**Paper 77**, pipelined) for comparable proof density—a 2,900× speedup.

Zero-sorry policy: In Lean, `sorry` is a keyword that means "I'm skipping this proof—trust me." The program has zero instances of `sorry` across 96,200+ lines. Every step is verified.

Analogy: Double-Entry Bookkeeping

Lean 4 is to mathematics what double-entry bookkeeping is to accounting. Every debit must have a credit. Every claim must have a proof term. The computer is the auditor that never sleeps, never gets tired, and never lets a balance error pass.

1.9 The Algebraic Geometry Toolkit (A Primer)

The rest of this monograph uses concepts from algebraic geometry and number theory. This section defines everything you need, starting from examples. **Read this section first.** Every jargon term in Chapters 2–7 is introduced here.

Varieties: Shapes Defined by Polynomials

An **algebraic variety** is a shape defined by polynomial equations. A circle ($x^2 + y^2 = 1$) is a variety. A line ($y = 2x + 3$) is a variety. An **elliptic curve** like $y^2 + y = x^3 - x$ is a variety—a smooth cubic in the plane, shaped like a doughnut.

"**Smooth**" means no sharp corners or self-intersections. "**Projective**" means we work in a setting where parallel lines meet at infinity (adding "points at infinity" to close up the shape). A "smooth projective variety" is just a nice closed shape defined by polynomials, living in projective space.

Rational points are solutions with *rational* coordinates. The circle $x^2+y^2 = 1$ has rational points like $(3/5, 4/5)$. The elliptic curve $y^2 + y = x^3 - x$ has the rational point $(0, 0)$. Finding all rational points is the central problem of number theory.

Cohomology: Counting Holes

You already know that a circle has one "hole" (you can lasso it with a loop), a torus has two independent holes (one around the tube, one through the tube), and a sphere has no holes in the loop sense but encloses a cavity.

Cohomology makes this precise. It assigns a vector space $H^k(X)$ to each variety X and each dimension k. Why a vector space and not just a number? Because the vector space remembers *how* the holes are related to each other, not just how many there are. (The number of independent holes is the *dimension* of the vector space.)

- H^0 counts connected components.
- H^1 counts independent 1-dimensional holes (loops that don't bound disks).
- H^2 counts 2-dimensional cavities (surfaces that don't bound solids).
- H^k counts k-dimensional "holes." (For $k \geq 3$, geometric intuition fades, but the algebra continues to work perfectly.)

For a torus: $H^0 = \mathbb{Q}$ (one piece), $H^1 = \mathbb{Q}^2$ (two independent loops), $H^2 = \mathbb{Q}$ (one cavity).

Here is the key fact for this monograph, and it is remarkable: there are *several completely different methods* to compute these vector spaces—different "flavours" of cohomology. You can count holes topologically (triangulate the shape and count faces), or algebraically (study polynomial differential forms), or arithmetically (watch how symmetries of number fields act on the shape). All methods give vector spaces of the same dimension. But they are computed by entirely different techniques and—crucially—they have different *logical costs*:

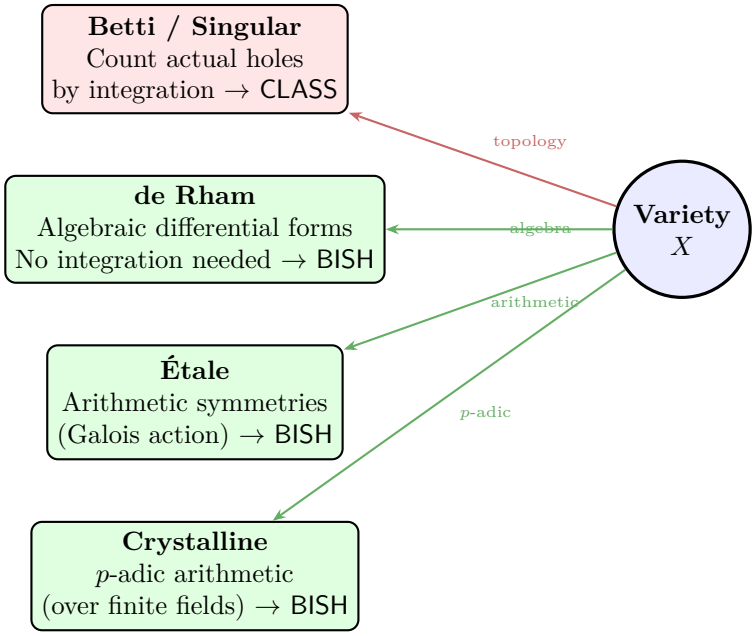


Figure 1.6: Four flavours of cohomology, all giving the same answer (same dimensions), computed differently. Betti cohomology counts actual holes by integration over $\mathbb{R}$: CLASS. The others stay algebraic or arithmetic: BISH. The entire program rests on this distinction.

Comparison Isomorphisms: Why Different Tests Give the Same Answer

Here is the central mystery that drives the entire CRM program. We said that Betti, de Rham, étale, and crystalline cohomology all "count the same holes"—they give vector spaces of the same dimension. But they are computed by completely different methods: Betti uses topology (triangulating the shape over $\mathbb{R}$ and counting faces), de Rham uses calculus (differential forms built from polynomial functions—think $2x\,dx + y\,dy$, the same objects from multivariable calculus but restricted to polynomial coefficients), étale uses arithmetic (Galois actions on roots of equations). How do we *know* they agree?

The answer is a **comparison isomorphism**: a theorem that explicitly constructs a bijection between one flavour and another. These theorems are among the deepest results in algebraic geometry. And they are precisely where the logical cost lives.

Analogy: translating between languages

Imagine you have a French novel, a German novel, and a Japanese novel. Someone claims they tell the same story. To verify this, you need a *translator* for each pair of languages. French $\leftrightarrow$ German might be straightforward (cognates, shared Indo-European roots—BISH). But French $\leftrightarrow$ Japanese requires a fundamentally different process: you must understand the deep structure of both languages, which involves cultural context that cannot be reduced to word-matching (CLASS).

Comparison isomorphisms work the same way. De Rham $\leftrightarrow$ crystalline: both are "algebraic" languages, and the translation (Berthelot's comparison) stays within algebra (BISH). De Rham $\leftrightarrow$ Betti: one is algebraic, the other topological, and the translation requires *integrating differential forms over real cycles*—passing through $\mathbb{R}$. This is the **period map**, and it costs CLASS.

The rule is simple: **a comparison isomorphism is CLASS if and only if it passes through the Archimedean place $\mathbb{R}$.** Any comparison that stays within the algebraic or p-adic world is BISH. This single principle explains nearly every CRM result in the program.

The Hodge Decomposition: Sorting Holes by Type

We said cohomology counts "holes" of various dimensions. But not all holes are alike. Some arise from algebraic equations (polynomial subvarieties), while others are purely topological (they exist only because the space has a complicated shape when viewed over $\mathbb{C}$). The **Hodge decomposition** sorts holes by their algebraic character.

A crucial step: from $\mathbb{Q}$ to $\mathbb{C}$. Until now, we have been studying varieties over $\mathbb{Q}$—looking for rational solutions. To access the Hodge decomposition, we must pass from rational solutions to *complex* solutions, studying the shape $X(\mathbb{C})$

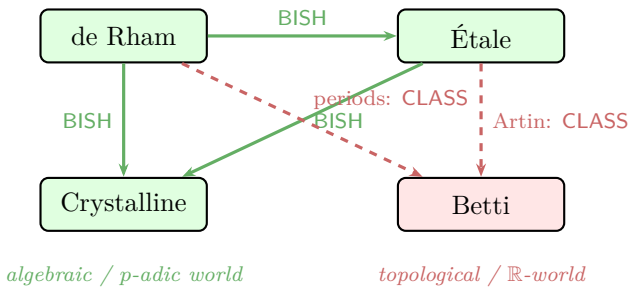


Figure 1.7: Comparison isomorphisms and their costs. Within the algebraic/p-adic world (green): all comparisons are BISH. Any comparison that crosses into the topological/$\mathbb{R}$-world (red) costs CLASS. This is the Archimedean obstruction in one picture.

as a complex manifold. This is the point where the Archimedean place enters: $\mathbb{C}$ is built from $\mathbb{R}$ (as $\mathbb{R}[i]$), and anything that depends on the topology of $\mathbb{C}$ inherits the logical cost of $\mathbb{R}$.

Over $\mathbb{C}$, each cohomology group H^k breaks into finer pieces labeled by two numbers (p, q) with $p + q = k$:

$$H^k(X) \;=\; \bigoplus_{p+q=k} H^{p,q}(X).$$

What do p and q measure? Think of p as counting the "algebraic directions" (technically, holomorphic differentials dz) and q as counting the "conjugate directions" (anti-holomorphic differentials $d\bar{z}$). A class in $H^{2,0}$ is "purely algebraic" (two holomorphic directions, no conjugate). A class in $H^{0,2}$ is "purely conjugate." The interesting classes are those in $H^{p,p}$—*balanced equally between algebraic and conjugate directions*. These are the potential candidates for coming from actual polynomial subvarieties.

Analogy: frequencies in a sound

A musical note can be decomposed into frequencies (Fourier analysis). Each frequency captures a different "mode of vibration." The (p, q) decomposition does the same for cohomological holes: it sorts each hole by how it vibrates in the holomorphic direction (p) versus the anti-holomorphic direction (q). A class of type (p, p) is like a "pure tone at the fundamental frequency"—balanced, symmetric, the most algebraic kind of hole. The Hodge conjecture asks: can every pure tone be produced by an instrument (an algebraic subvariety)?

A **Hodge class** is a cohomology class of type (p, p) that is also rational (lives over $\mathbb{Q}$, not just $\mathbb{C}$). The **Hodge conjecture** (a Clay Millennium Problem, worth \$1M) says: every Hodge class is algebraic—it genuinely comes from a polynomial subvariety.

Why does this cost CLASS? The question is: given a cohomology class $\alpha \in H^{2p}(X, \mathbb{Q})$, does it land in the (p, p) piece of the Hodge decomposition? To answer this, you need the Hodge decomposition *itself*—and computing it requires hard analysis over $\mathbb{R}$.

Here is the chain of reasoning:

1. **The Hodge theorem** says: every cohomology class has a unique "harmonic representative"—a differential form ω satisfying the Laplace equation $\Delta\omega = 0$. This harmonic representative has a definite type (p, q).
2. **But solving** $\Delta\omega = 0$ is a partial differential equation (PDE) on the complex manifold $X(\mathbb{C})$. Under the hood, this involves elliptic PDE theory: integration over $\mathbb{R}$, convergence of infinite series, existence of solutions via compactness arguments. All of this requires the completeness and order properties of $\mathbb{R}$—the Archimedean place.
3. **The cost structure therefore bifurcates:**
   - *Detection* ("Is α of type (p, p)?") requires solving the PDE, which requires $\mathbb{R}$: **CLASS**.
   - *Verification* ("Given that Z is an algebraic subvariety, does its fundamental class land in $H^{p,p}$?") is pure algebra: compute the cycle class map $[Z] \mapsto H^{2p}$, which is a finite polynomial operation: **BISH**.

This detection/verification split is the Archimedean obstruction in its purest form. You can *check* algebraicity cheaply; you cannot *find* it cheaply. The CRM program identified this as a universal pattern: across all five great conjectures, detection ("does this object have property P?") costs CLASS, while verification ("is this known candidate correct?") costs BISH.

Algebraic Cycles: Subshapes Inside Shapes

An **algebraic cycle** on X is a formal sum of subvarieties—a "grocery list with quantities": $Z = 3 \cdot C_1 - 2 \cdot C_2 + C_3$, where each C_i is itself a variety sitting inside X, and the integer coefficients are multiplicities (negative signs indicate orientation reversal). The **codimension** is how much smaller the subvariety is: a curve inside a surface has codimension 1; a point inside a surface has codimension 2.

Two cycles can be "the same" even when they look different geometrically. For example, on a surface, two curves that can be continuously deformed into each other represent the same cohomology class, even though they are defined by different polynomial equations. But there are different notions of "same," and which one you use has dramatic consequences for logical cost:

- **Homological equivalence:** two cycles are the same if they represent the same class in cohomology. Testing this requires computing cohomology—potentially CLASS, since Betti cohomology involves integration over $\mathbb{R}$.
- **Numerical equivalence:** two cycles are the same if they have the same *intersection numbers* with every other cycle. The **intersection number** of two curves on a surface counts how many times they cross (with signs

for orientation and multiplicities for tangencies). This is a finite integer, computable by polynomial algebra: BISH.

Standard Conjecture D says these two notions agree. If true, this would be a spectacular constructivization: you could test whether two cycles are cohomologically equivalent by computing finitely many intersection numbers—finite arithmetic instead of infinite topological computation. In CRM language, Standard Conjecture D would allow you to replace a CLASS test with a BISH test.

Elliptic Curves, Abelian Varieties, and Jacobians

An **elliptic curve** is a smooth cubic curve with a marked point. Example: $y^2 + y = x^3 - x$ with the point $(0, 0)$. The key miracle: the rational points on an elliptic curve form a *group*—you can "add" two points using a beautiful geometric construction:

The chord-and-tangent law

To add two points P and Q on an elliptic curve: draw the straight line through P and Q. Because the curve is a cubic, this line hits the curve in exactly one more point, call it R. Now reflect R across the x-axis to get $P + Q$. That's it—the group operation is defined by drawing lines and reflecting. When $P = Q$, draw the tangent line instead. This works over any field, and the arithmetic involves only rational operations on coordinates: BISH.

The **rank** is the number of independent generators of this group (modulo the finite torsion subgroup). Finding the rank is the central problem of BSD.

An **abelian variety** is the higher-dimensional generalization: a smooth projective variety that is simultaneously a group (meaning you can "add" points, generalizing the chord-and-tangent law). An **abelian surface** is 2-dimensional; an **abelian fourfold** is 4-dimensional. Some abelian varieties decompose as products of elliptic curves, but most do not—they are genuinely higher-dimensional objects.

The **Jacobian** $J(C)$ of a curve C of genus g is a g-dimensional abelian variety that packages the curve's essential geometric data (specifically, the divisor class group) into a higher-dimensional group variety. For an elliptic curve (genus 1), the Jacobian is the curve itself. For a genus-4 hyperelliptic curve (the test family in the Hodge campaign), the Jacobian is a 4-dimensional abelian variety.

Heights: Measuring the "Size" of a Rational Point

A **height function** $h(P)$ assigns a non-negative real number to each rational point, measuring its "arithmetic complexity." For a rational number a/b in lowest terms, the naïve height is $\max(|a|, |b|)$. The point $(3/5, 4/5)$ has height 5. The point $(123456/789, 0)$ has height 123456.

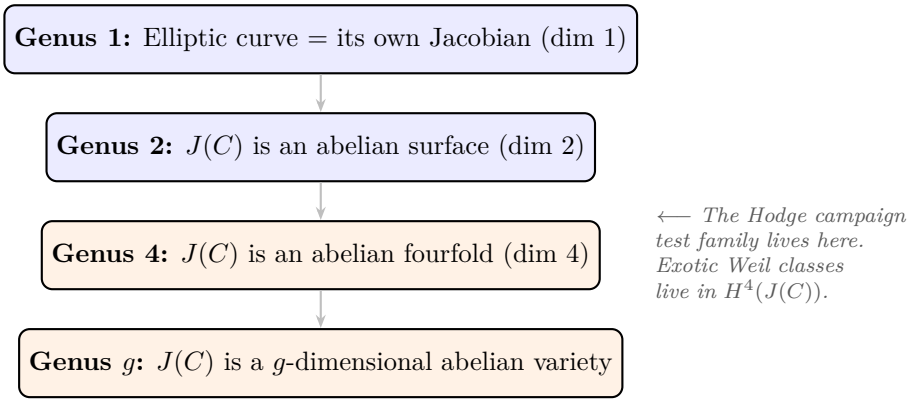


Figure 1.8: The genus–Jacobian correspondence. Each curve C of genus g produces a g-dimensional abelian variety $J(C)$.

The **Néron-Tate canonical height** $\hat{h}$ is a refined version for elliptic curves that is a *quadratic form*: $\hat{h}(P+Q)+\hat{h}(P-Q)=2\hat{h}(P)+2\hat{h}(Q)$. It is **positive-definite**: $\hat{h}(P)=0$ if and only if P is torsion (finite-order in the group law). This converts the rational-point search from an unbounded problem (search all of $\mathbb{Q}^2$) into a bounded one (search below a height bound).

The **Northcott property**: for any bound B, there are only *finitely many* rational points with $h(P) \leq B$. Combined with positive-definiteness, this makes the search finite—BISH.

Galois Representations: Symmetry Actions on Cohomology

The absolute Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ is the group of all "symmetries" of the algebraic numbers—all the ways you can permute roots of polynomials with rational coefficients while respecting every polynomial relation among them. For instance, one symmetry sends $\sqrt{2}$ to $-\sqrt{2}$ (both are roots of x^2-2); another sends i to $-i$ (both roots of x^2+1). This group is enormous, infinite, and deeply mysterious. No one can write down all its elements.

But we can study it *indirectly*, by watching how it acts on simpler objects. A **Galois representation** assigns to each symmetry a *matrix*—a concrete, computable object acting on a finite-dimensional vector space built from the geometry of a variety (its cohomology).

Analogy: studying a shadow

You cannot see a 4-dimensional object directly. But you can project it onto a 2-dimensional screen and study its shadow. Different projections give different shadows—but together they tell you about the original object. A Galois representation is a "shadow" of the Galois group: a finite-dimensional matrix representation that captures part of the group's structure.

Where do these vector spaces come from? From étale cohomology. For a variety X over $\mathbb{Q}$ and a prime ℓ, the étale cohomology $H^k_{\text{ét}}(X, \mathbb{Q}_\ell)$ is a vector space over the ℓ-adic numbers that carries a natural action of the Galois group:

$$\rho\colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \longrightarrow \mathrm{GL}_n(\mathbb{Q}_\ell).$$

Concrete example. Consider the elliptic curve $E\colon y^2 + y = x^3 - x$ (this is the curve "37a1" from Paper 95). For each prime $p \neq 37$, the trace of the Galois action on the Tate module of E is:

$$a_p \;=\; p + 1 - \#E(\mathbb{F}_p),$$

where $\#E(\mathbb{F}_p)$ means: reduce the equation modulo p and count solutions. For example: modulo 2 the curve $y^2+y = x^3-x$ has 5 solutions, so $a_2 = 2+1-5 = -2$. Similarly $a_3 = -3$, $a_5 = -2$, $a_7 = -2$, and so on. These are just point counts over finite fields—pure finite arithmetic, no integration, no topology. BISH.

This is the key insight: Galois representations live in the étale realization, which is non-Archimedean. Linear algebra over $\mathbb{Q}_\ell$ is BISH (because $\mathbb{Q}_\ell$ is a completion at a finite prime, so its arithmetic involves no Archimedean comparisons). Much of modern number theory consists of studying Galois representations and their internal structure (semisimplicity, irreducibility, determinant, ramification)—all of which is algebraic and constructive. The logical cost enters only when you try to *match* a Galois representation with an analytic object (a modular form, an L-function)—because that matching crosses the comparison-isomorphism boundary into the Archimedean world.

Modular Forms: Functions with Extreme Symmetry

Start with a simple observation: the sine function satisfies $\sin(x + 2\pi) = \sin(x)$. It has one symmetry (translation by 2π). A modular form is a function with *infinitely many* symmetries—not just translations, but also inversions, rotations, and combinations thereof. These symmetries are so constraining that the function is almost entirely determined by them.

Analogy: a kaleidoscope

A kaleidoscope takes a small pattern and replicates it using mirrors into a highly symmetric image. A modular form is the mathematical analogue: a small amount of data (the Fourier coefficients a_n) is replicated by a large symmetry group into a rigid global function. The symmetry is so extreme that knowing the first few a_n determines *all* of them. This is why modular forms are computable: infinite symmetry makes finite data sufficient.

Formally, a **modular form** of weight k and level N is a holomorphic function $f(\tau)$ on the upper half-plane $\mathfrak{H} = \{\tau \in \mathbb{C} : \mathrm{Im}(\tau) > 0\}$ satisfying:

$$f\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^k f(\tau) \quad \text{for all } \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N).$$

Here $\Gamma_0(N)$ is the group of 2×2 integer matrices with determinant 1 whose lower-left entry is divisible by N. Do not be intimidated by this formula. It says: if you apply any of these matrix transformations to τ (a Möbius transformation $\tau \mapsto (a\tau + b)/(c\tau + d)$), the function f transforms in a predictable way controlled by the weight k. The space of such functions is finite-dimensional (the dimension is given by a formula involving k, N, and the genus of the modular curve), so *every* modular form is a finite linear combination of basis elements. Moreover, the **Sturm bound** ($\approx kN/12$) tells you that a modular form is completely determined by finitely many of its Fourier coefficients. This is why they are computable.

Every modular form has a **Fourier expansion** $f(\tau) = \sum_{n \geq 0} a_n q^n$ where $q = e^{2\pi i \tau}$. The coefficients a_n encode arithmetic information. For an elliptic curve E over $\mathbb{Q}$:

$$a_p = p + 1 - \#E(\mathbb{F}_p) \quad \text{(the "error term" in the point count over } \mathbb{F}_p\text{)}.$$

Computing a_p is pure finite arithmetic: count solutions modulo p, subtract. BISH.

The **modularity theorem** (Wiles, Taylor-Wiles, Breuil-Conrad-Diamond-Taylor) is the bridge between algebra and analysis. More precisely: for every elliptic curve E over $\mathbb{Q}$, there exists a weight-2 modular form f whose Fourier coefficients $a_n(f)$ equal the point-count data $a_n(E)$ for all n. The sequences are identical—the same integers appear on both sides—but one sequence comes from counting points on a curve (algebra, BISH) and the other from a symmetric function on the complex upper half-plane (analysis, CLASS).

In CRM language: the modularity theorem is a *comparison isomorphism* between the étale realization (Galois representation of E: BISH) and the automorphic realization (modular form on $\mathfrak{H}$: CLASS). The data on both sides is algebraic; the *proof* that they match crosses the Archimedean place—it must invoke properties of complex analysis (analytic continuation, L-functions, convergence) that require the completeness of $\mathbb{R}$.

Motives: The Universal Blueprint

We have seen that a single variety can be studied through multiple cohomology theories (Betti, de Rham, étale, crystalline), all giving the same dimensions. A **motive** is Grothendieck's conjectural answer to the question: *what is the common object behind all these cohomology theories?* Just as a single integer can be viewed in different bases (decimal, binary, hexadecimal) without changing the integer itself, a single motive can be "viewed" through different cohomology theories without changing the underlying mathematical content.

Analogy: the DNA analogy

A variety X is an organism. Its motive is its DNA. Betti cohomology, de Rham cohomology, étale cohomology, and crystalline cohomology are different medical tests (blood test, MRI, genetic screen, biopsy) that each reveal partial information. The motive is the complete genome: every test is recoverable from it. The genome itself is purely informational (BISH). The individual tests may require lab equipment (CLASS for Betti, BISH for étale). The cost depends on which test you run, not on the genome.

A **realization functor** extracts one type of cohomology from the motive. The CRM cost of the realization depends on whether it "passes through $\mathbb{R}$"—meaning whether the extraction process requires completing $\mathbb{Q}$ at the Archimedean place (working over $\mathbb{R}$ or $\mathbb{C}$) or stays within the algebraic/p-adic world (see Figure 1.6).

Why is the motive itself BISH? Because motives (in the conjectural formulation) are defined by *algebraic correspondences*: polynomial maps between varieties, composed and identified by finite algebraic operations. No real analysis, no integration, no topology. The motive is pure algebra.

This is the central insight of Paper 98: **the motive is always BISH; the cost lives in the realization.** A theorem about a variety is a statement about its motive. To *prove* the theorem, you must apply one or more realization functors and comparison isomorphisms. The logical cost of the proof is determined by which realizations and comparisons you use—specifically, by whether any of them pass through $\mathbb{R}$. If you can find a proof path that stays entirely within the algebraic/p-adic world, the theorem is BISH. If every proof path must cross into the Betti/Hodge/automorphic world, the theorem is irreducibly CLASS. Most interesting theorems are somewhere in between: a large BISH core with isolated CLASS nodes at the points where comparison isomorphisms force a crossing.

p-Adic Numbers: An Alternative to $\mathbb{R}$

The real numbers $\mathbb{R}$ are the completion of $\mathbb{Q}$ using the usual absolute value: two rationals are "close" if their difference is small. The **p-adic numbers** $\mathbb{Q}_p$ are an alternative completion: two rationals are "close" if their difference is divisible by a high power of a prime p.

Example: in $\mathbb{Q}_5$, the numbers 1 and 126 are "close" because $126-1 = 125 = 5^3$. But 1 and 2 are "far apart" (differ by 1, which is not divisible by 5).

The **Archimedean place** is the place "at infinity" where we complete to $\mathbb{R}$. The **non-Archimedean places** are the primes p where we complete to $\mathbb{Q}_p$. The CRM program found: $\mathbb{R}$ is the sole source of CLASS content. All $\mathbb{Q}_p$ are BISH.

Quick Reference Card

Table 1.2: Jargon decoder for the rest of this monograph.

Term	**Plain English**
Variety	Shape defined by polynomial equations
Smooth, projective	No corners; closed up at infinity
Cohomology $H^k(X)$	Vector spaces counting k-dimensional holes
Hodge type (p, p)	The "most algebraic" kind of hole
Algebraic cycle	Formal sum of polynomial subshapes
Elliptic curve	Smooth cubic curve with a group law
Rank	Number of independent rational points
Abelian variety	Higher-dimensional elliptic curve (a projective group)
Jacobian $J(C)$	The abelian variety encoding a curve's geometry
Height $h(P)$	Arithmetic complexity of a rational point
Northcott property	Finitely many points below any height bound
Galois representation	Symmetry action encoded as matrices
Modular form	Function with extreme symmetry on the upper half-plane
Hecke eigenvalue a_p	Error term in point count mod p
Motive	Universal DNA of a variety (encodes all cohomologies)
Realization functor	Extracting one cohomology type from the motive
$\mathbb{Q}_p$ (p-adic numbers)	Alternative to $\mathbb{R}$; completion at a prime p

Chapter 2

The Five Conjectures

Paper 1–44 established the physics ceiling: BISH + LPO. Now the program turns to pure mathematics. Does the same source—the real numbers—explain logical difficulty beyond physics?

2.1 Why Arithmetic Geometry?

Before introducing the five conjectures formally, here is the key idea in plain language.

What do these conjectures have in common?

Each conjecture says: *if something looks geometric from the analytic side, it actually is geometric.* Think of it as a "no mirages" principle. When you observe a pattern through one lens (analysis, L-functions, cohomology), the conjecture promises there's a concrete geometric object (a curve, a cycle, a rational point) causing that pattern. These conjectures are hard because moving from "it looks real" to "here's the actual object" requires crossing from analysis (which uses $\mathbb{R}$) to algebra (which doesn't)—and that crossing is where all the logical cost lives.

Five conjectures were chosen for calibration, each connecting algebraic geometry to analysis, each open for decades, each central to modern number theory:

1. **Hodge Conjecture** (1950)—Clay Millennium Problem. *Plain English:* If a "shadow" in the cohomology of a geometric shape looks like it came from a subshape, it really did come from a subshape.
2. **Tate Conjecture** (1965). *Plain English:* If a cohomology class is fixed by every symmetry of the number system (Galois group), it must be algebraic.
3. **Birch and Swinnerton-Dyer Conjecture** (1965)—Clay Millennium Problem. *Plain English:* The number of independent rational solutions to a cubic equation equals the number of times a certain analytic function touches zero at a special point.
4. **Fontaine-Mazur Conjecture** (1992). *Plain English:* If a p-adic symmetry pattern "looks geometric" (satisfies certain smoothness conditions), it actually comes from the geometry of an algebraic variety.
5. **Weight-Monodromy Conjecture** (1970). *Plain English:* When a family of varieties degenerates (develops singularities), the degeneration pattern of

the cohomology is controlled by algebraic data—the "weight" of the Hodge structure.

Each conjecture asserts that something "looks right" from the analytic side (cohomology, L-functions, Galois representations) if and only if something "is right" on the algebraic side (algebraic cycles, rational points, geometric origin). CRM calibrates the logical cost of moving between these two sides (**Paper 45–49**).

2.2 The Five Conjectures Calibrated

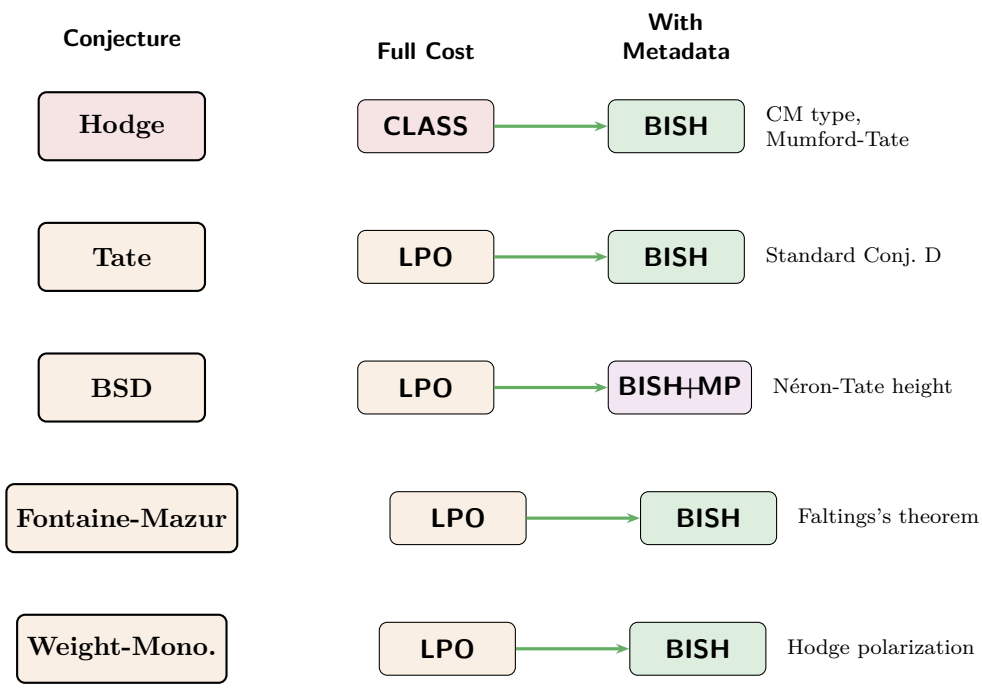


Figure 2.1: Every conjecture drops from LPO/CLASS to BISH (or BISH + MP) when provided the right algebraic metadata. The green arrows show de-omniscientizing descent: algebraic data substitutes for the oracle.

Each conjecture exhibits the same phenomenon. The full statement costs LPO or CLASS—you need an oracle to verify it in general—but providing the right algebraic data substitutes for the oracle and drops the cost to BISH.

Notation. When we write $\mathsf{LPO}(\mathbb{R})$ or $\mathsf{LPO}(\mathbb{Q}_\ell)$, we mean that the omniscience principle is applied to sequences valued in $\mathbb{R}$ or $\mathbb{Q}_\ell$ respectively. The subscript identifies *which number system* requires the oracle. This matters because some oracles are more expensive than others: $\mathsf{LPO}(\mathbb{R})$ involves the full Archimedean continuum, while $\mathsf{LPO}(\mathbb{Q}_\ell)$ involves ℓ-adic completions. Both are instances of the same logical pattern—deciding whether a sequence has a nonzero term—but

applied to different ground fields.

- **Hodge (Paper 49):** Full cost CLASS. With CM type or Mumford-Tate group metadata, the Hodge test becomes finite $\mathbb{Q}$-linear algebra: BISH.
- **Tate (Paper 46):** Full cost LPO($\mathbb{Q}_\ell$). Standard Conjecture D (numerical = homological equivalence) makes cycle comparison finite arithmetic: BISH.
- **BSD (Paper 48):** Full cost LPO($\mathbb{R}$). The Néron-Tate height (positive-definite on $\mathbb{R}$) bounds the rational-point search, converting it from unbounded to bounded: BISH + MP. The residual MP is subtle: the height gives you *finiteness*—only finitely many rational points of bounded height—but not an explicit *enumeration bound* on how long the search takes. You know the search terminates (Markov's Principle), but you cannot compute in advance exactly when it will stop.
- **Fontaine-Mazur (Paper 47):** Full cost LPO($\mathbb{Q}_p$). Faltings's theorem (geometric origin of p-adic representations) makes the comparison algebraic: BISH.
- **Weight-Monodromy (Paper 45):** Full cost LPO. Hodge polarization (geometric origin) reduces the monodromy test to combinatorics: BISH.

The metadata is always a **positive-definite structure** that exploits $u(\mathbb{R}) = \infty$: an inner product, a height, a polarization. The oracle ("search over infinity") is replaced by a computation ("search within a finite ball defined by the positive-definite form"). This is the core mechanism.

2.3 De-Omniscientizing Descent

The shared pattern across all five conjectures has a name: **de-omniscientizing descent**. Before the formal version, here is a concrete example.

Concrete Example: Finding a Book in a Library

Without metadata (LPO): You walk into an infinite library and are told "a book about topology is in here somewhere." To find it, you must check every shelf, potentially forever. This is an oracle-level search.
With metadata (BISH): The librarian says "the topology section is in aisle 7, shelves 3–5." Now you walk to a specific, bounded location and check three shelves. Same book, same library, but the metadata (the catalogue system) converted an infinite search into a finite one.
De-omniscientizing descent = installing the catalogue system. The five conjectures each have their own version of "the catalogue": a positive-definite structure (height function, CM type, polarization) that bounds the search.

The pattern works like this:

1. A theorem, stated in full generality, requires an oracle (LPO or CLASS) to verify.
2. The theorem has a natural "algebraic companion"—a structural condition (like

Standard Conjecture D, or geometric origin of eigenvalues, or positive-definite height) that, if true, provides the same information the oracle would.

3. When the algebraic companion is available, the cost drops to BISH. The oracle is replaced by a computation.

The algebraic companion always involves the real numbers, specifically the Archimedean place. It exploits the fact that $\mathbb{R}$ admits positive-definite forms ($u(\mathbb{R}) = \infty$) to convert an unbounded oracle query into a bounded algebraic computation.

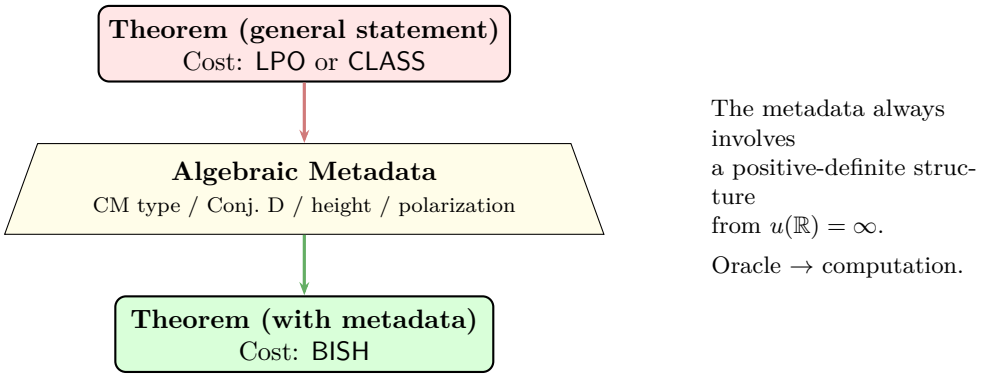


Figure 2.2: De-omniscientizing descent: algebraic metadata replaces the oracle, dropping the cost from LPO/CLASS to BISH.

Analogy: GPS vs. Street Signs

CLASS is like a GPS that magically knows where everything is. BISH is like navigating with a physical map and street signs. The "algebraic metadata" is the street signs: if someone installs enough of them (CM type, Mumford-Tate group, algebraic spectrum), you can navigate without the GPS. De-omniscientizing descent is the process of identifying exactly which street signs you need.

2.4 The Three-Invariant Hierarchy

Paper 67 ("The Motive Is a Decidability Certificate") synthesized the calibration data into a classification system. Three numerical invariants determine the logical cost of any motivic cycle-search problem.

Analogy: Three Dimensions of Search Difficulty

Imagine searching for a lost object. Three factors determine how hard the search is:

- **Room size** (= rank r): How many dimensions is the search space? A line is easier to search than a plane, which is easier than a room.

- **Depth underground** (= Hodge level ℓ): How many layers deep is the object buried? Surface objects are easy; deep objects require excavation.
- **Minimum object size** (= Lang constant c): Is there a minimum size below which objects can't hide? If the smallest possible object is 1 cm, you only need a coarse-enough net to catch everything.

Large r makes the search harder (more dimensions). Large ℓ makes it harder (deeper burial). Large c makes it *easier* (bigger objects are easier to find).

Formally:

- **Rank** r—the dimension of the algebraic cycle space. Higher rank means more candidates to check.
- **Hodge level** ℓ—how deep in the Hodge filtration the class sits. Higher level means farther from the algebraic surface.
- **Lang constant** c—the effective lower bound on height. Lang's height conjecture predicts that for a curve C of genus $g \geq 2$, every rational point P satisfies $\hat{h}(P) \geq c(C) > 0$, where $\hat{h}$ is the Néron-Tate canonical height. When $c > 0$, non-trivial cycles can't be arbitrarily small, and the search is bounded.

Together, (r, ℓ, c) classify the full decidability landscape. **Paper 67** verified this across 53 formalizations totalling 86,898+ lines of Lean 4 code.

2.5 FLT Is BISH

The most surprising calibration result.

The Skyscraper That's One Story Tall

Fermat's Last Theorem: $x^n + y^n = z^n$ has no integer solutions for $n \geq 3$. Proved by Andrew Wiles in 1995 in a 129-page paper deploying Galois representations, modular forms, deformation rings, and Hecke algebras. This proof is one of the most complex in the history of mathematics.

You would expect it to be logically expensive—surely 129 pages of deep mathematics need powerful axioms? CRM's verdict: **FLT is BISH.** The entire theorem needs no oracle.

The 129 pages are *scaffolding*—complex but constructive. The theorem itself sits on the ground floor. It's a 129-story building that's really one story tall: the other 128 floors are the elevator that takes you there, not the apartment you live in.

Paper 68 audited the Wiles proof stage by stage. Stages 2–5 (Taylor-Wiles patching, deformation theory, modularity lifting) are all BISH. The machinery is complex but constructive. Stage 1 (Langlands-Tunnell, weight-1 forms) costs WLPO, but this is an artifact: Kisin's later work (2009) on the $p = 2$ dihedral case bypasses weight-1 forms entirely. With Kisin's route, Stage 1 is also BISH.

This is the program's sharpest illustration of a fundamental distinction: **proof**

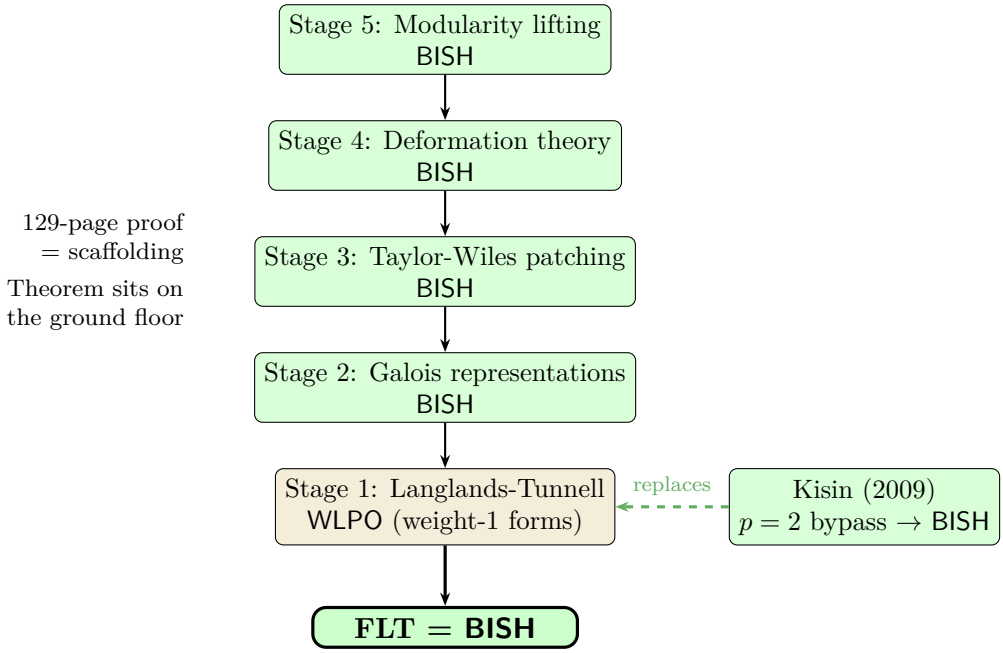


Figure 2.3: FLT audit (**Paper 68**). Every stage is BISH except Stage 1, which uses weight-1 forms (WLPO). Kisin's bypass removes this cost. FLT = BISH: the 129-page proof is scaffolding; the theorem requires no oracle.

complexity is not logical cost. Taylor-Wiles patching is intellectually difficult, technically demanding, and involves deep algebra—but it is constructive. Weight-1 modular forms are routine objects—but they cost WLPO. CRM measures what matters (logical resources), not what impresses (proof difficulty).

An honest residual tension (Paper 98, Appendix A). Kisin's bypass uses Hecke theta series $\theta_L(\tau) = \sum_{v \in L} q^{Q(v)}$, whose Fourier coefficients are finite lattice sums (BISH). However, the *modularity* of θ_L (proving it transforms correctly under $\mathrm{SL}_2(\mathbb{Z})$) uses Poisson summation—an identity of distributions on $\mathbb{R}$. Whether this is an essential use of the Archimedean place or dispensable analytic scaffolding is an open question. **Paper 68**'s classification stands, but this node is flagged as the precise locus where a definitive CRMLint Squeeze is needed.

2.6 Function Field Langlands Is BISH

If FLT is the surprise, function field Langlands is the explanation.

Over number fields (extensions of $\mathbb{Q}$), the Langlands correspondence connects automorphic forms to Galois representations. This correspondence requires the Archimedean place ($\mathbb{R}$), and its logical cost reflects this: various principles from WLPO to CLASS appear.

Over function fields (extensions of $\mathbb{F}_q(t)$, where $\mathbb{F}_q$ is a finite field), there is no Archimedean place. No $\mathbb{R}$. No $u(\mathbb{R}) = \infty$. And the entire Langlands correspondence is BISH (**Paper 69**). Both proofs—Laurent Lafforgue's for GL_n and Genestier–V. Lafforgue's for general reductive groups—are constructive. The boundary between constructive and classical is not "discrete vs. continuous" but **algebraic vs. transcendental spectral parameters**: over function fields, automorphic representations have algebraic Hecke eigenvalues (elements of $\overline{\mathbb{Q}}$), so every spectral comparison is decidable arithmetic. Over number fields, the Archimedean place introduces transcendental parameters (elements of $\mathbb{C}$ that need not be algebraic), and decidability fails.

This confirms the Archimedean thesis with a clean experiment: remove $\mathbb{R}$, and the cost vanishes.

Key references: Paper 69 (Function Field Langlands Is BISH), **Paper 70** (The Archimedean Principle—capstone).

2.7 The $h = f$ Identity: A Discovery Through CRM

Paper 56–58 report one of the program's most striking discoveries—not a result *about* CRM, but a result *found by* CRM.

Background: exotic Weil classes

On a **CM** (complex multiplication) abelian fourfold $X = A \times B$—a product of two abelian surfaces whose endomorphism rings are unusually large (containing a number field, not just $\mathbb{Z}$)—the cohomology contains **exotic Weil classes**: Hodge classes that are *not* generated by divisor classes. Classical mathematics treats the Hermitian self-intersection number of these classes as an element of $\mathbb{Q}^*/\mathrm{Nm}(K^*)$—the group of nonzero rationals modulo norms from a number field K, i.e., an equivalence class rather than a specific number.

What CRM demanded

The constructive framework refuses equivalence classes. It demands the *exact integer value*. When the program computed this exact value for cyclic Galois totally real cubic fields $F/\mathbb{Q}$, a formula appeared:

The $h = f$ Identity (Paper 56, Theorem A)

For cyclic Galois cubics $F/\mathbb{Q}$ with class-number-one CM field K, principal polarizations, and CM signature $(1,2) \times (1,0)$:

$$\deg(w_0 \cdot w_0) = \sqrt{\mathrm{disc}(F)} = f$$

where w_0 is the primitive exotic Weil class and f is the arithmetic conductor of $F/\mathbb{Q}$.

The **Hermitian self-intersection** h equals the **arithmetic conductor** f. The mechanism: for cyclic cubic extensions, $\mathrm{disc}(F) = f^2$, and the correspondence degree equals f.

Extension and correction

Paper 57 verified this across all nine class-number-1 imaginary quadratic fields (Baker-Heegner-Stark list): conductors $f = 7, 9, 13, 19, 37, 61, 79, 97, 163$. Eight of nine are prime.

Paper 58 corrected the formula for fields with class number $h_K > 1$: the exact identity becomes $h \cdot \mathrm{Nm}(\mathfrak{a}) = f$, where $\mathfrak{a}$ is the Steinitz class of the integral Weil lattice. The class group acts as an "arithmetic compensator": the topological volume $f^2|\Delta_K|$ is an absolute invariant; the class group redistributes it between the Hermitian metric and the lattice density.

Why CRM found it

Paper 58: "This result emerged from the constructive framework's insistence on exact integer values rather than equivalence classes modulo norms. The classical literature treats the Hermitian discriminant as an element of $\mathbb{Q}^*/\mathrm{Nm}(K^*)$; the constructive lens demands the exact value, revealing the formula and the role of the class group—structure invisible from the classical perspective."

The entire computation is BISH. No omniscience principles are invoked. CRM discovered new mathematics by refusing to round off.

2.8 The Northcott Property: How Metadata Replaces Oracles

Section 2.3 described de-omniscientizing descent in the abstract. Here is the concrete mechanism that makes it work.

Analogy: Searching a Field for Buried Treasure

You're told: "somewhere in this field, there's buried treasure." If the field is infinite, you must dig forever (LPO—you need an oracle to scan infinitely). But suppose someone gives you a **metal detector with a range limit**: it can only detect objects within 100 meters of the surface. Now the search region is bounded—the treasure must be in a finite volume of dirt. That's what the **Northcott property** does: it gives you a "depth gauge" (called a *height function*) that guarantees the objects you're looking for all live within a bounded region.

But knowing the region is finite isn't quite enough—you also need to know *how finely to dig*. If you dig with a teaspoon, it could take a billion years. **Lang's conjecture** provides the second piece: a *minimum spacing guarantee*. The buried objects can't be closer together than some explicit distance $c > 0$. So you know both the volume (Northcott) and the grid spacing (Lang): now you can enumerate everything in bounded time (BISH).

The **Northcott property** states: for every bound $B > 0$, the set of rational points on an algebraic variety with *height* $\leq B$ is finite. The **height** $\hat{h}(P)$ measures the "arithmetic complexity" of a point P—roughly, how large the numerators and denominators of its coordinates are. Points with small height are simple; points with large height are complicated.

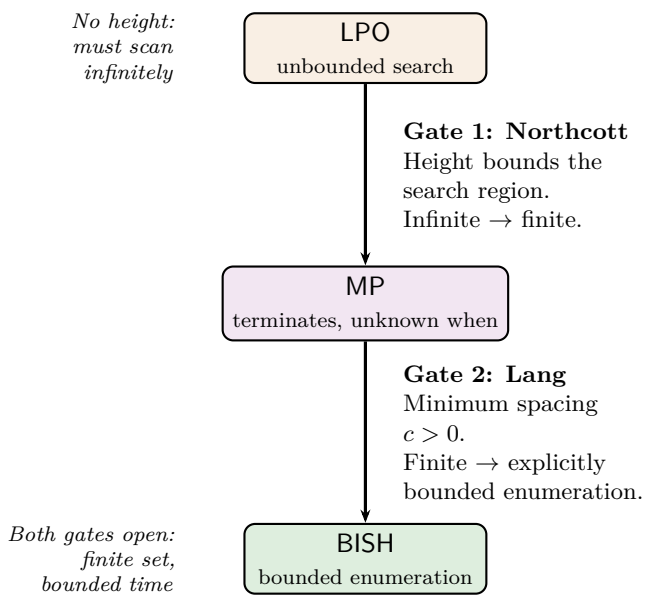


Figure 2.4: The Northcott-Lang descent (**Paper 61, 72**). Two gates control the drop from LPO to BISH. Gate 1 converts infinite search to finite. Gate 2 gives an explicit termination bound. Both exploit the Archimedean place: positive-definite heights exist only over $\mathbb{R}$.

Why this needs $\mathbb{R}$. The height $\hat{h}$ is a *positive-definite* quadratic form—meaning $\hat{h}(P) = 0$ only for "trivial" points (torsion). Positive-definiteness works in every dimension because $u(\mathbb{R}) = \infty$ (recall from §1.5). Over the p-adic numbers, where $u(\mathbb{Q}_p) = 4$, positive-definite forms exist only up to dimension 4—not enough to control the full search space. This is why Northcott is an Archimedean phenomenon.

The tight chain (Paper 72): Remove the height $\to$ Northcott fails $\to$ search is unbounded $\to$ you need LPO. Each implication reverses:

$$\text{Positive-definite height} \;\Leftrightarrow\; \text{Northcott} \;\Leftrightarrow\; \mathsf{BISH}\ \text{cycle-search.}$$

This biconditional is proved formally in Lean. It is one of the three DPT axioms (Chapter 3, Axiom 3).

Chapter 3

The DPT Axioms—Three Laws of Logical Cost

Chapter Preview: The Three-Legged Stool

Chapter 2 surveyed five conjectures and found they all drop from expensive to cheap when given the right metadata. This chapter asks: *what is the smallest set of assumptions that makes everything cheap?*

The answer is three axioms—exactly three, no more, no fewer. Think of them as three legs of a stool:

- **Leg 1 (Conjecture D):** "Can we compare morphisms?" — makes it decidable whether two maps between algebraic objects are equal.
- **Leg 2 (Algebraic Spectrum):** "Can we compare eigenvalues?" — makes it decidable whether a number arising from symmetry is zero.
- **Leg 3 (Archimedean Polarization):** "Can we search for cycles?" — converts an infinite search for geometric objects into a finite, bounded one.

Each leg is proved to be *irreplaceable*: remove any one, and the stool collapses. No alternative assumption can substitute. This is what "canonical" means—not just minimal, but the *only possible* foundation.

3.1 What Problem DPT Solves

After calibrating five conjectures (Chapter 2), a natural question emerges: is there a minimal set of axioms that, if true, makes the entire motivic cycle-search problem—the central problem of arithmetic geometry—solvable in BISH?

The answer is yes. The axiom set is called **DPT (Decidable Polarized Tannakian)**. It consists of three axioms, each governing a distinct domain (morphisms, eigenvalues, cycles). Each axiom is proved to be not just sufficient but **necessary**—it is the unique condition for constructive decidability in its domain. This makes the axiom set *canonical*, not merely minimal: you cannot substitute a different axiom.

Key references: Paper 50 (Three Axioms for the Motive), **Paper 72–74** (biconditional proofs).

3.2 Axiom 1: Standard Conjecture D—Morphism Decidability

Concrete Example First

You have two roads on a map (two algebraic cycles). Are they "the same road"? There are two ways to check:
Method A (topological): Walk along both roads and measure whether they pass through the same regions of the map. This requires exploring the entire map—potentially an infinite process.
Method B (numerical): Count how many times each road crosses each landmark (intersection numbers). This is a finite counting exercise.
Conjecture D says: Methods A and B always give the same answer. If they do, you never need Method A—you just count intersections. *That's* why Conjecture D drops the cost from LPO (infinite search) to BISH (finite counting).

Formal statement: Homological equivalence equals numerical equivalence for algebraic cycles. Two ways to measure whether cycles are "the same"—one using cohomology (a topological invariant), the other using intersection numbers (a numerical invariant)—always agree.

CRM role: Without Conjecture D, testing whether two morphisms between motives are equal requires LPO (existential search over an infinite cohomological quotient). With Conjecture D, numerical intersection is a finite computation—BISH.

Biconditional (Paper 73): Standard Conjecture D is *equivalent* to BISH morphism decidability. Without it, morphism equality costs LPO, and no alternative axiom can substitute.

3.3 Axiom 2: Algebraic Spectrum—Eigenvalue Decidability

Concrete Example First

A Frobenius operator is like a symmetry that acts on the cohomology of a variety defined over a finite field. Think of it as spinning a crystal and measuring the angles at which it looks the same—these angles are the **eigenvalues**.
The question: are these eigenvalues "nice" numbers (roots of polynomial equations with integer coefficients, like $\sqrt{2}$ or $(1+\sqrt{5})/2$) or "wild" numbers (transcendental, like π or e)?
If they're algebraic (nice): you can compare them exactly using integer arithmetic. "Is $\sqrt{2}-\sqrt{2}=0$?" Yes—provable by squaring.

If they *could* be transcendental (wild): comparing them means testing whether a real number equals zero. That's the equality-test oracle (WLPO). You can compute more and more decimal digits, but you can never be *certain* a number is exactly zero just from its digits.
Axiom 2 says: the eigenvalues are always algebraic. So comparison is always finite arithmetic, never an infinite decimal check.

Formal statement: Eigenvalues of Frobenius operators have geometric (algebraic) origin—they are roots of polynomials with integer coefficients.

CRM role: Without algebraic spectrum, comparing eigenvalues requires WLPO (testing whether a real number equals zero). With algebraic spectrum, eigenvalue comparison is rational arithmetic—BISH.

Biconditional (Paper 74): Algebraic spectrum is equivalent to BISH eigenvalue decidability.

Key subtlety: The cost of failure here is WLPO, not LPO. Eigenvalue comparison is an *equality* test ("is this eigenvalue zero?"), not an existential search ("find a nonzero eigenvalue"). WLPO is strictly weaker than LPO—equality testing is easier than searching. This intrinsic computational distinction was invisible before CRM.

3.4 Axiom 3: Archimedean Polarization—Cycle Search

Concrete Example First

You're searching for a specific book in a library. Two scenarios:
Without Axiom 3: The library is infinite, and books can be arbitrarily small (even microscopic). You must check every shelf, every corner, every crevice—an unbounded search. This is LPO.
With Axiom 3: The library is infinite, but every book is at least 1 cm tall (the "positive-definiteness" bound). Now you only need to look on shelves—and for any given shelf, only finitely many books fit. The "height function" measures how tall a book is. The Northcott property says: for any maximum height, only finitely many books exist below it. The search becomes finite. This is BISH.
The positive-definiteness bound comes from $\mathbb{R}$ (specifically, from $u(\mathbb{R}) = \infty$—the real numbers support positive-definite forms in every dimension). At p-adic places, the bound fails above dimension 4, and the library is unsearchable again.

Formal statement: Cycles of intermediate codimension satisfy an effective positive-definiteness bound.

What this means: When searching for algebraic cycles in middle cohomology (the hardest case), Axiom 3 asserts that positive-definite height functions

provide bounded search regions. Without the bound, cycle-search is an infinite scan (LPO). With the bound, it is a finite computation (BISH).

Biconditional (Paper 72): Archimedean polarization is equivalent to BISH cycle-search decidability. This axiom directly exploits $u(\mathbb{R}) = \infty$: positive-definite quadratic forms exist in every dimension *at the Archimedean place*, and these forms bound the search region.

The mechanism is the Northcott property (Section 2.8). The Néron-Tate height $\hat{h}$ is a positive-definite quadratic form on $A(K) \otimes \mathbb{R}$. Positive-definiteness guarantees: for any $B > 0$, the set $\{P : \hat{h}(P) \leq B\}$ is finite. This converts an infinite scan over $A(K)$ into a bounded enumeration—the crucial step from LPO to BISH. The full descent runs through two gates:

1. **Northcott gate (LPO $\to$ MP):** Positive-definite height gives finiteness below any bound. The search terminates, but you don't know when.
2. **Lang gate (MP $\to$ BISH):** Lang's conjecture provides an *effective* lower bound $c > 0$ on $\hat{h}(P)$ for non-torsion P. Now you know *how large* to search: enumerate all lattice points with $\hat{h}(P) \leq c$. Bounded time. BISH.

Without the Archimedean place, positive-definite forms in arbitrary dimension do not exist ($u(\mathbb{Q}_p) = 4$), Northcott fails, and cycle-search remains LPO. This is why Axiom 3 is "Archimedean."

3.5 The DPT Triangle

3.6 DPT vs. Fargues-Scholze: Forced Partition

Modern arithmetic geometry has two major structural frameworks that bear on decidability: CRM/DPT and Fargues-Scholze's condensed/perfectoid approach. They are not competitors—they govern complementary territories, like two countries sharing a border.

Two Halves of One Map

Think of the number system as having two types of "distance": ordinary distance (the real numbers, $\mathbb{R}$) and p-adic distance (a strange notion where 1000000 is "closer to zero" than 1, because it is divisible by high powers of p).

DPT governs the $\mathbb{R}$-side: heights, positive-definiteness, algebraic geometry. Its methods are constructive (BISH) when the DPT axioms hold.

Fargues-Scholze governs the p-adic side: perfectoid spaces, condensed modules, diamonds. Its methods are inherently classical (CLASS)—the continuous geometry of p-adic spaces resists constructivization.

The open question (the "conservation conjecture"): is the classical content on the Fargues-Scholze side truly necessary, or is it scaffolding that could, in principle, be removed?

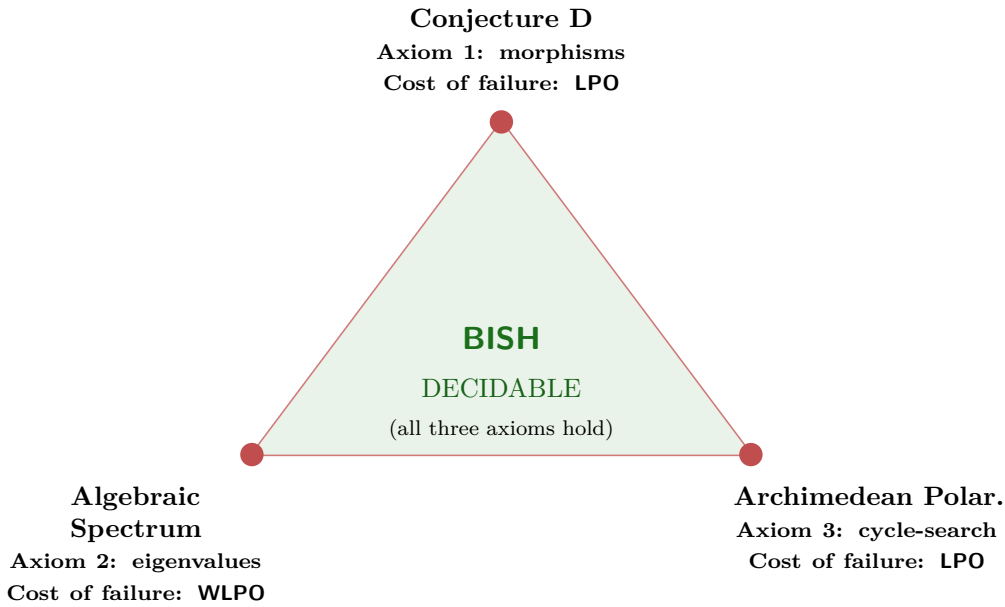


Figure 3.1: The DPT triangle. Three axioms, three domains, three costs of failure. Each axiom is *biconditional* with decidability in its domain and *canonical*—no alternative substitutes. Together, they make all of arithmetic geometry (at the level of pure motives) decidable in BISH.

Table 3.1: Arithmetic bifurcates into two complementary sectors. DPT governs the discrete/algebraic side; Fargues-Scholze governs the continuous/geometric side. The interface between them is the conservation conjecture.

DPT (Algebraic/Archimedean)	**Fargues-Scholze (p-adic/Geometric)**
Real numbers ($\mathbb{R}$)	p-adic numbers ($\mathbb{Q}_p$)
Archimedean place	Non-Archimedean places
BISH (with DPT axioms)	CLASS (intrinsically)
Cycle-search, heights	Categorical, sheaf-theoretic
Positive-definite forms	Condensed / solid modules
Algebraic geometry	Perfectoid spaces

Paper 75 provides external validation. The **Local Langlands Correspondence** (LLC) asserts a bijection between irreducible admissible representations of a p-adic group and Langlands parameters (homomorphisms from the Weil group to the L-group). The Genestier–V. Lafforgue proof of LLC for function fields has statement cost = WLPO (as DPT predicts), but proof cost = CLASS—a conservation gap of two full levels. Whether this CLASS scaffolding is eliminable (the **conservation conjecture**) is the open interface between CRM and Fargues-Scholze.

3.7 The Conservation Gap

When a theorem's *statement* is logically cheap but its *proof* uses expensive tools, the difference is a "conservation gap."

Analogy: Driving to the Corner Store

If you drive a tank to the corner store (expensive proof) to buy milk (cheap theorem), the conservation gap is the difference between "needing milk" and "deploying a tank." The gap predicts: there exists a simpler way to get milk (walk, bike, drive a car). The bigger the gap, the more room for simplification.

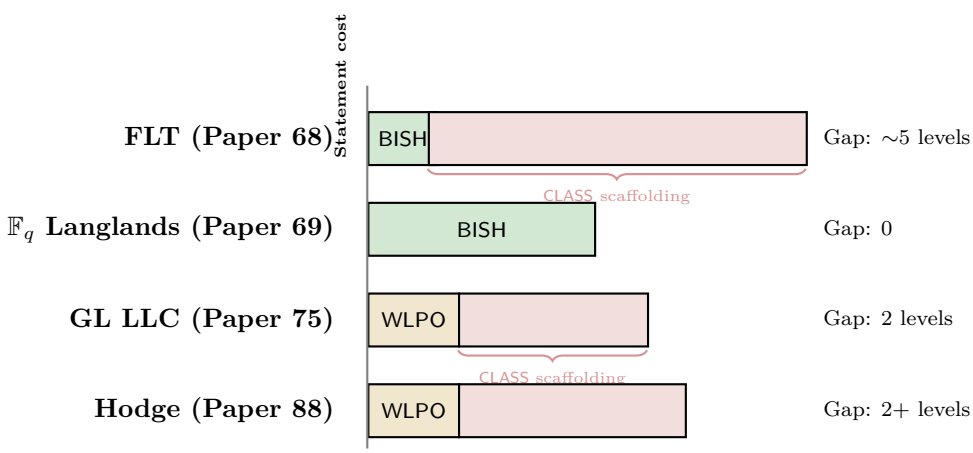


Figure 3.2: Conservation gaps: the difference between what a theorem *states* (left edge) and what its proof *costs* (right edge). A gap of zero means the proof is optimal. A large gap suggests a simpler proof may exist. Decreasing gaps correlate with increasing algebraicity.

CRM-trained AI uses conservation gap data as a training signal: large gaps predict where constructive alternatives exist, guiding automated proof search toward simpler proofs.

Chapter 4

The Squeeze Protocol

Chapters 1–3 were diagnosis: what does mathematics cost? This chapter is the *instrument*—the automated tool that performs the diagnosis. It transforms CRM from a manual audit into a systematic, repeatable protocol.

Chapter Preview: The X-Ray Machine

Imagine a doctor (CRM) who has been diagnosing diseases (logical cost) by hand—examining each patient (theorem) one at a time. Now the doctor builds an X-ray machine (CRMLint) that can scan any patient automatically and produce a precise map: this bone is healthy (constructive), this bone is fractured (classical).
The Squeeze protocol is the full procedure: (1) position the patient (scaffold the proof in Lean 4), (2) take the X-ray (trace classical dependencies), (3) mark the fractures (identify Classical Boundary Nodes), (4) set the bones (replace classical steps with constructive alternatives where possible).

4.1 CRMLint: The Automated Analyzer

Paper 76 introduces CRMLint (940 lines Lean 4, zero `sorry`). CRMLint is a *metaprogram*—a program that analyzes other programs. Think of it as a code analysis tool (like a linter for Python or JavaScript) but instead of checking style, it checks *logical cost*. It takes a Lean 4 proof term, traces the classical dependency graph, and reports exactly which steps require non-constructive principles.

CRMLint operates in four layers:

1. **Classical dependency tracer.** Walk the proof term DAG (directed acyclic graph). Flag every node where `Classical.choice` or equivalent enters.
2. **Pattern classifier.** Categorize each flagged node: is it analytic continuation? Zariski density? Ehresmann fibration? Hodge decomposition?
3. **Concentration analyzer.** Measure what fraction of the proof is BISH vs. CLASS.
4. **Audit reporter.** Emit a structured decomposition: n BISH components + m CLASS components = total.

Key distinction: CRMLint measures *logical cost*, not *proof complexity*. Taylor-Wiles patching is complex but BISH. Weight-1 modular forms are routine but WLPO. This distinction is invisible without CRM.

4.2 The Four-Step Squeeze Protocol

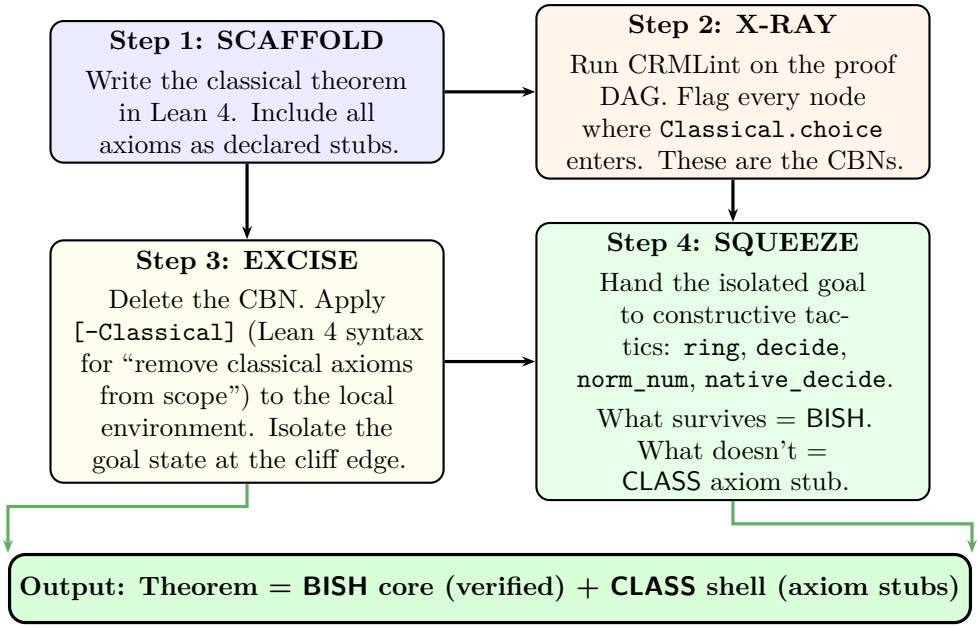


Figure 4.1: The four-step Squeeze protocol. A classical proof is scaffolded, X-rayed for classical dependency nodes, excised at the boundary, and squeezed with constructive tactics. Typically >80% of the formal content survives as BISH.

CBN = Classical Boundary Node: the exact point where the proof jumps from computation to oracle. Key intuition: classical mathematics uses infinity as a compression algorithm. The Squeeze bans the shortcut and forces explicit algebraic witness discovery.

4.3 Asymmetric Offloading

The Squeeze exploits a division of labor between a **Computer Algebra System** (CAS)—software like SymPy or SageMath that performs exact symbolic computation—and a **proof assistant** (Lean 4) that verifies the results. The CAS computes; Lean certifies. Neither trusts the other: the CAS may have bugs, but Lean checks every identity independently.

Table 4.1: Development time comparison across the program.

Paper	Method	Density	Time
Paper 5 (2024)	Manual Lean	∼200 lines	4 months
Paper 77 (2026)	CAS + Lean pipeline	∼798 lines	1 hour

A 2,900× speedup. **Paper 80** upgraded from `native_decide` on concrete rationals to `ring` on polynomial rings $\mathbb{Q}[x, t]$—from retail auditing (one number at a time) to wholesale auditing (identities over entire polynomial rings). All subsequent Squeeze applications use `ring` over polynomial rings.

4.4 The Function-Field Upgrade: Retail to Wholesale

Paper 80 introduced an infrastructure breakthrough that enabled all subsequent Squeeze applications.

Paper 77–78 verified identities at **concrete rational numbers** using `native_decide` ("is $7 \times 3 = 21$? Yes"). This checks one point at a time. To verify an identity across an entire family of varieties, you would need infinitely many checks.

Paper 80 upgraded to **polynomial rings** using `ring` ("is this identity in $\mathbb{Q}[x, t]$ true? Yes"). This checks an entire family at once. A single `ring` invocation verifies a polynomial identity over all specializations simultaneously.

Analogy: retail vs. wholesale

`native_decide` is retail auditing: checking one house at a time. `ring` over $\mathbb{Q}(t)$ is wholesale auditing: mapping the entire neighborhood at once. Instead of checking one fiber of a family (C_t at $t = 7$), you verify the whole family (C_t for all t) in a single computation.

This upgrade—from $\mathbb{Q}$-arithmetic to $\mathbb{Q}(t)$-algebra—is what makes the Hodge and BSD campaigns feasible. All subsequent Squeeze applications (**Paper 81–89, 92–96**) use `ring` over polynomial rings as their primary verification tactic.

The oracle content is also clarified by this upgrade. Working over $\mathbb{Q}(t)$ means the Squeeze verifies algebraic identities that hold *generically*—for all but finitely many values of t. No oracle is needed to check these: the `ring` tactic decides polynomial identity in finite time. The oracle enters only when you need to pass from the algebraic identity to a topological or analytic conclusion (e.g., "the flat section persists across the entire moduli space"). This is the structural reason why detection is BISH and existence is CLASS: detection is algebra (decidable by `ring`), existence is topology (requires an oracle).

4.5 What Classical Boundary Nodes Look Like

Guide to the CBN types above. *Analytic continuation*: extending a function beyond its natural domain of convergence, requiring limits over $\mathbb{R}$. *Ehresmann fibration*: a theorem guaranteeing that a smooth map with compact fibers is locally trivial—a topological existence result. *Chebotarev density*: the existence of primes with prescribed splitting behaviour, proved via L-function analysis on $\mathbb{R}$.

Table 4.2: Catalogue of Classical Boundary Nodes across the Squeeze applications. The pattern: CLASS nodes are topological existence assertions; BISH nodes are algebraic computations.

CBN Type	Papers	Principle
Analytic continuation	**Paper 80, 95**	CLASS
Ehresmann fibration theorem	**Paper 80–81, 93**	CLASS
Hodge decomposition (existence)	**Paper 84–89**	CLASS
Variational Hodge Conjecture	**Paper 88–89**	CLASS
Abel-Jacobi map / Baire category	**Paper 94**	CLASS
Kolyvagin Euler system	**Paper 95**	CLASS
Chebotarev density theorem	**Paper 95**	CLASS
Cassels-Poitou-Tate duality	**Paper 95**	CLASS
Zariski density	**Paper 82**	CLASS

Cassels–Poitou–Tate duality: a nine-term exact sequence assembling local Galois cohomology data at all primes simultaneously, requiring an infinite product.

The Universal Pattern

CLASS nodes are always **topological existence** assertions: "something exists in a topological space, but I cannot construct it." BISH nodes are always **algebraic computations**: "here is the explicit polynomial identity / matrix entry / eigenvalue / intersection number." The boundary is always: "something exists" vs. "here it is."

The Fifteen Applications at a Glance

Table 4.3 consolidates all fifteen Squeeze applications across the program. The progression reveals five phases: methodology (**Paper 77–78**), infrastructure (**Paper 80–83**), Hodge campaign (**Paper 84–89**), completion (**Paper 92–94**), and BSD campaign (**Paper 95–96**).

Remark. Papers 87 and 91 appear in the generative arc but are *not* Squeeze applications. Paper 87 is a foundational CRM analysis of the Hodge condition itself (its WLPO equivalence is a calibration result). Paper 91 is a CRM audit of an external result (Markman's unconditional Hodge proof), comparing its logical cost with the Squeeze's. Paper 93 is a structural explanation (why $\tau_+ = 0$), not a decomposition.

4.6 The Gauss-Manin Pipeline

Paper 80–83 built the infrastructure for all subsequent Squeeze applications. This four-paper arc constructs the algebraic machinery needed to work with

Table 4.3: The fifteen CRMLint Squeeze applications (**Paper 90**, Table 1). For each application: the domain, what the Squeeze found to be constructive (BISH core), what remains irreducibly classical (CLASS boundary), and the Lean tactic that certifies the BISH content.

#	Paper	Domain	BISH Core	CLASS Boundary	Tactic
1	P77	Hodge theory	36 basis vectors for $\mathrm{Hdg}^{2,2}(E^4)$	Lefschetz $(1,1)$	`native_decide`
2	P78	Langlands	Character-trace matching, $\mathrm{GL}_2(\mathbb{Q}_2)$	Harris-Taylor	`decide`
3	P80	Algebraic GM	Griffiths pole reduction over $\mathbb{Q}(t)$	Ehresmann-GM	`ring`
4	P81	Flat sections	Kronecker sum, symplectic flatness	Deligne monodromy	`ring`
5	P82	Diff. Galois	Kovacic's algorithm, $G_{\mathrm{gal}} = \mathrm{SL}_2$	Zariski density	`norm_num`
6	P83	Picard number	Künneth + flat dim + Kovacic $\to \rho = 3$	Noether-Lefschetz	`decide`
7	P84	Exotic Weil	Block GM, $G_{\mathrm{gal}}(\bigwedge^4 V_+) = \{1\}$	Hodge existence	`ring`
8	P85	Exotic Weil	$\tau_+(t) = 0$, Lagrangian eigenspace	Hodge existence	`ring`
9	P86	Hodge Conj.	D_8-splitting, Jac $\sim A^2$	Ribet	`ring`
10	P88	Hodge Conj.	Fermat domination, $C_0 \to F_{16}$	Shioda + VHC	`ring`
11	P89	Hodge Conj.	18 components, $\tau_+(a,b,c) = 0$	Shioda + VHC	`ring`
12	P92	Higher genus	$\tau_+(a) = 0$, genera 5–8	Shioda + VHC	`ring`/`norm_num`
13	P94	CY3 Griffiths	PF algebra, $\delta(z) \neq 0$	Abel-Jacobi, Baire	`ring`
14	P95	BSD (rank 1)	Hecke eigenvalues, 2-descent, heights	Gross-Zagier, Kolyvagin	`native_decide`
15	P96	BSD (rank 0)	Modular symbols, $L(E,1)/\Omega^+$, Tamagawa	Euler system, Iwasawa	`norm_num`

families of algebraic curves—the raw material for the Hodge campaign.

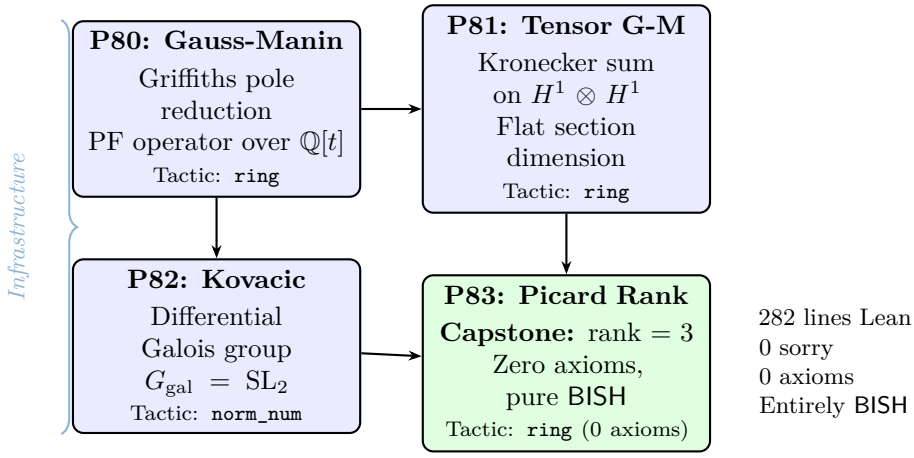


Figure 4.2: The Gauss-Manin pipeline (**Paper 80–83**). P80 builds the Picard-Fuchs operator; P81 extends to tensor products; P82 determines the differential Galois group; P83 synthesizes into the Picard rank result—entirely BISH with zero axioms.

4.7 What Is a Picard-Fuchs Operator?

Intuition First: The Shape-Tracker ODE

Imagine a family of rubber bands, each a different shape, indexed by a dial t. As you turn the dial, the rubber band deforms continuously. The *area enclosed* by the rubber band changes with t.

The Picard-Fuchs equation is the differential equation that describes *how the area changes as you turn the dial*. It captures the deformation of the shape in a single algebraic formula. Crucially, the equation itself uses only rational functions of t—no actual integration, no topology, no $\mathbb{R}$. The equation is purely algebraic (BISH), even though the areas it describes involve integrals (CLASS).

This is why the Picard-Fuchs operator is the engine of the Squeeze: it replaces expensive topological integrals with cheap algebraic differential equations.

A family of algebraic curves C_t (parameterized by t) has cohomology that *varies* with t. The **Picard-Fuchs equation** is the ODE satisfied by the *periods*—integrals of differential forms over cycles:

$$\omega(t) = \int_\gamma \omega_t$$

where γ is a locally constant cycle and ω_t is an algebraic differential form on C_t.

Example. The Legendre family of elliptic curves $y^2 = x(x-1)(x-t)$ has Picard-Fuchs equation:

$$t(1-t)\,\omega''(t) + (1-2t)\,\omega'(t) - \tfrac{1}{4}\,\omega(t) = 0$$

Its solution is the complete elliptic integral $K(t) = {}_2F_1(1/2, 1/2; 1; t)$.

Why this matters for CRM. The Picard-Fuchs operator encodes all the variation of the cohomology in a single algebraic ODE. Its coefficients are rational functions over $\mathbb{Q}(t)$—no topology, no $\mathbb{R}$, purely algebraic. This is the engine of the Squeeze: replace topological period integrals (cost CLASS) with the algebraic PF operator (cost BISH).

4.8 Griffiths Transversality and Flat Sections

Intuition: Parallel Transport Without Moving

Imagine carrying a compass needle along a curved surface (like a sphere). The needle's direction changes as you walk—this change is the "connection." A **flat section** is a direction that *doesn't rotate* no matter which path you take. On a flat table, every direction is flat. On a sphere, no direction is globally flat (this is why the compass "rotates" when you walk around the North Pole).

In the Hodge campaign, the "surface" is the parameter space of a family of curves, the "needle" is a cohomology class, and the "connection" is the Gauss-Manin connection. A flat section is a cohomology class that stays constant across the entire family—the key ingredient for proving the Hodge conjecture.

The miracle: the connection matrix $M(t)$ has coefficients in $\mathbb{Q}(t)$—rational functions. So testing whether a section is flat ($M(t) \cdot v = 0$) is polynomial algebra, decidable by `ring`. No integrals needed.

How do you compute the Picard-Fuchs operator without doing any integrals? The answer is **Griffiths pole reduction** (**Paper 80**).

The Gauss-Manin connection. Algebraic de Rham cohomology $H^1_{\mathrm{dR}}(C_t/\mathbb{Q}(t))$ has a basis of algebraic differential forms. The Gauss-Manin connection $\nabla = d/dt + M(t)$ is a matrix ODE with rational function entries $M(t) \in M_{2\times 2}(\mathbb{Q}(t))$.

Griffiths pole reduction. Differentiating a basis form with respect to t produces a meromorphic form with higher-order poles. Two algebraic identities reduce these poles:

1. **Griffiths division:** $3(x^2 - x) = f'(x) + ((2t-1)x - t)$ (removes exact components).
2. **Coboundary identity:** $t(1-t)(x^2-x) = (3x+t-2)f(x) - (x^2-x)f'(x)$ (completes pole reduction).

Both are *polynomial identities over* $\mathbb{Q}[x,t]$, verified by Lean's `ring` tactic in milliseconds. No analysis. No limits. No $\mathbb{R}$.

Griffiths transversality. The connection preserves the cup-product pairing on de Rham cohomology. This forces the connection matrix M to be traceless: $\mathrm{tr}(M) = 0$. Tracelessness is again a polynomial identity, verified by `ring`. This constraint is what makes the differential Galois group SL_2 rather than GL_2—a structural consequence that propagates through the entire Hodge campaign.

Flat sections. A **flat section** is a vector v satisfying $\nabla(v) = 0$, i.e., $M(t) \cdot v = 0$ for all t. Flat sections are cohomology classes preserved under parallel transport along the family. The dimension of the flat section space determines the Picard number (**Paper 83**: dimension = 1, so generic Picard rank of $E_t \times E_t$ is 3). The trace form $\tau_+ = \mathrm{tr}(M_+)$ of the eigenspace connection measures whether an exotic Weil class is flat (**Paper 85**: $\tau_+ = 0$ universally).

4.9 Diagnostic vs. Generative: Two Phases of the Programme

The CRM program divides into two phases with fundamentally different characters. Table 4.4 contrasts them.

Table 4.4: Diagnostic phase vs. generative phase (**Paper 90**, Table 2).

Aspect	**Diagnostic (P1–75)**	**Generative (P76–96)**
Method	Manual calibration	Squeeze protocol
Input	Classical theorem	Classical theorem + CAS
Output	Level in hierarchy	$T_{\mathsf{BISH}} \wedge T_{\mathsf{CLASS}}$ decomposition
Verification	Lean (hand-written)	Lean (CAS-generated)
Domains	Physics, arith. geometry	Hodge, Langlands, diff. Galois, CY3
Ceiling finding	LPO for physics	VHC for Hodge; Abel-Jacobi for CY3
Key insight	Logical cost = cost of $\mathbb{R}$	Microscope is automatable
Papers	75	18 (incl. P76, P87, P90, P91)
Lean lines	~87,000	~9,000+

Analogy: Geological Survey vs. Core Drilling

The diagnostic phase was like a geological survey—walking across the terrain, sampling rock types, mapping the landscape. It was *extensive*: 75 papers covering physics, arithmetic geometry, and representation theory. It discovered the terrain's structure: the logical cost of mathematics is the logical cost of $\mathbb{R}$.

The generative phase is like core drilling—choosing specific sites and drilling deep. It is *intensive*: 18 papers developing and applying a single methodology. It doesn't just measure the cost; it extracts the constructive content and

leaves only the irreducible classical boundary. The same drill bit (the Squeeze protocol) works at every site, from Hodge theory to BSD to Calabi-Yau geometry.

Chapter 5

The Hodge Campaign

The Hodge conjecture is a Clay Millennium Problem. It asks whether certain cohomological shadows of geometric objects are themselves geometric. The program devoted ten papers (**Paper 84–93**) to squeezing this conjecture—the most sustained application of CRMLint in the series.

5.1 The Hodge Conjecture Explained

Every smooth projective algebraic variety X over $\mathbb{C}$ has cohomology groups $H^{2k}(X, \mathbb{Q})$. Inside these groups live special elements called **Hodge classes**—classes of type (k, k) in the Hodge decomposition.

The Hodge conjecture asserts: *every Hodge class is a rational linear combination of algebraic cycles* (geometric subvarieties).

Analogy: The Photograph Test

Imagine a photograph (the cohomology group). Some patterns in the photograph arise from physical objects in the scene (algebraic cycles). Others might be lens artifacts or digital noise (non-algebraic classes). The Hodge conjecture says: every pattern that *looks like* it comes from a physical object (Hodge class) *actually does* come from a physical object (algebraic cycle). No lens artifacts.

5.2 The Hodge Horizon

Paper 87 established the CRM cost of the Hodge conjecture. Two key concepts here—**CM type** (complex multiplication metadata, defined in §2.7) and **Mumford-Tate group** (the algebraic group controlling Hodge classes, defined below in §5.3)—provide the algebraic shortcuts that reduce the cost:

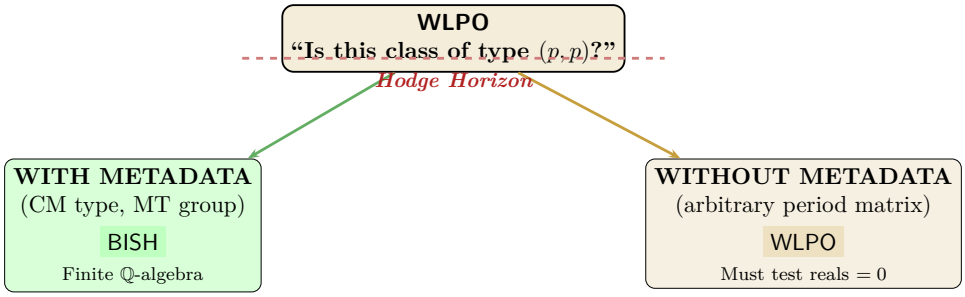


Figure 5.1: The Hodge Horizon (**Paper 87**). With CM/MT metadata: BISH. Without: WLPO.

5.3 The Hodge Arc Step by Step

The big picture before the details

Imagine a family of musical instruments, each parameterized by a tuning knob t. Each instrument produces a set of overtones (the cohomology). As you turn the knob, the overtones shift—but some combinations of overtones stay perfectly constant, no matter where the knob is set. These are the **flat sections**.

The Hodge campaign asks: when a flat overtone-combination "looks algebraic" (it satisfies the Hodge condition), is it *really* algebraic (produced by an actual geometric subshape)?

The answer depends on the instrument's internal symmetry. If the instrument has a hidden mirror symmetry (palindromic polynomial), it splits into two identical sub-instruments. The symmetry forces algebraicity. If there is no mirror symmetry, algebraicity requires a harder argument (VHC).

The ten-paper arc below makes this precise. The key concepts:

- **Flat section** = an overtone that doesn't change when you turn the knob ($\nabla v = 0$). Think: a pitch that is *independent* of tuning.
- **Trace vanishing** $\tau_+ = 0$ = the overtone-combination has zero "drift." Verified by polynomial algebra (BISH).
- **Kani-Rosen splitting** = the instrument splits into two identical halves. This forces the Hodge conjecture to be true (Ribet's theorem). Purely algebraic (BISH).
- **VHC** = if the overtone is algebraic at one tuning, it stays algebraic at all tunings. Classical (CLASS).

A note before the details. The next several pages describe ten papers in sequence. Each paper builds on the previous one, so they are presented in order. If the formal details feel dense, focus on the *plain English* paragraphs (in italics)—they capture the essential idea of each paper. The formal statements are there for completeness, but the conceptual story is self-contained without

them.

Ten papers trace a complete arc from infrastructure through computation to structural explanation.

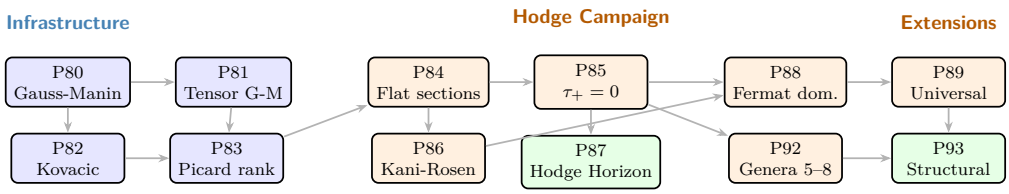


Figure 5.2: Dependency DAG for the Hodge campaign (**Paper 80–93**, from **Paper 90**). Blue = infrastructure, orange = campaign, green = calibration.

The arc in detail

P84: Exotic Weil classes as flat sections.

The idea in plain English: Take a curve with a 4-fold rotational symmetry (σ). This symmetry splits the cohomology into two pieces (V_+ and V_-), like splitting white light into two polarizations. Within V_+, there lives an "exotic" class—a Hodge class that doesn't come from the obvious geometric sources (divisors). The question: is it flat (constant across the family)?

The test family is the genus-4 hyperelliptic curve $C_t : y^2 = x^9 - tx^5 + x$. It has a $\mathbb{Z}/4\mathbb{Z}$-automorphism $\sigma(x, y) = (-x, iy)$, which splits the cohomology $H^1_{\mathrm{dR}}(C_t)$ into eigenspaces V_+ (eigenvalue $+i$) and V_- (eigenvalue $-i$), each of dimension 2. This splitting is algebraic—it uses only the group action and linear algebra over $\mathbb{Q}(i)$, no analysis.

The Gauss-Manin connection restricts to block-diagonal form: $\nabla = \nabla_+ \oplus \nabla_-$ with separate connection matrices M_+, M_- on each block. **Paper 82** showed $G_{\mathrm{gal}} = \mathrm{SL}_2$ for each block (Kovacic's algorithm). Since $\det(\mathrm{SL}_2) = 1$, the exterior power connection $\nabla_{\bigwedge^4 V_+}$ has trivial monodromy: $G_{\mathrm{gal}}(\bigwedge^4 V_+) = \{1\}$. This means the exotic Weil class $\omega \in \bigwedge^4 V_+$ is a **global flat section**—preserved under parallel transport. Entirely BISH.

Translation: The symmetry forces the exotic class to be constant. No oracle needed—just linear algebra.

P85: Universal trace vanishing $\tau_+ = 0$.

The idea in plain English: The "trace" of the connection matrix measures how much the eigenspace V_+ rotates as you vary t. If $\tau_+ = 0$, the eigenspace doesn't rotate at all—the exotic class stays perfectly flat. This was verified computationally for every family tested, a universal phenomenon that cried out for structural explanation (which P93 eventually provided).

The trace form $\tau_+ = \mathrm{tr}(M_+)$ measures the "twist" of the eigenspace under variation. If $\tau_+ = 0$, the exotic Weil class is flat (parallel-transported without rotation). The eigenspace V_+ is **Lagrangian**: the symplectic intersection pairing

restricted to V_+ vanishes, because the eigenvalues of σ on V_+ and V_- are i and $-i$, and $i^2 = -1 \neq 1$. The symplectic constraint gives $\mathrm{tr}(M_+) = -\mathrm{tr}(M_-)$ but *not* $\mathrm{tr}(M_+) = 0$ individually. Yet $\tau_+ = 0$ was verified computationally for every family tested (8 families across genera 2–5). A universal phenomenon demanding structural explanation.

P86: Kani-Rosen splitting on the palindromic locus.

The idea in plain English: The polynomial defining the curve has a mirror symmetry: reading the coefficients left-to-right is the same as right-to-left (like "racecar" is a palindrome). This mirror symmetry produces a hidden involution on the curve, which together with the rotational symmetry generates a larger symmetry group (D_8, the symmetry group of a square). This forces the Jacobian to split into a product of two identical pieces—and for products of identical abelian surfaces, the Hodge conjecture is known unconditionally (Ribet 1983). No oracles, no open conjectures. Pure algebra.

The polynomial $g(x) = x^8 - tx^4 + 1$ is **palindromic**: the coefficient of x^k equals the coefficient of x^{8-k}. This implies the existence of a **reciprocal involution** $\mu(x, y) = (1/x, y/x^5)$, satisfying $\mu^2 = \mathrm{id}$.

The D_8 group. The involution μ together with the order-4 automorphism σ generates the dihedral group $D_8 = \langle \sigma, \mu \mid \sigma^4 = 1, \mu^2 = 1, \sigma\mu = \mu\sigma^3 \rangle$ of order 8. The relation $\sigma\mu = \mu\sigma^3$ is verified by direct coordinate computation.

The splitting theorem. The D_8-action produces two genus-2 quotient curves $Q_1 = C/\langle\mu\rangle$ and $Q_2 = C/\langle\mu\sigma^2\rangle$. The Kani-Rosen genus count $4 = 2 + 2 + 0 + 0$ gives the isogeny $J(C_t) \sim J(Q_1) \times J(Q_2)$. Over $\mathbb{C}$, $Q_1 \cong Q_2$ (via $(u, v) \mapsto (-u, iv)$), so $J(C_t) \sim A^2$ for an abelian surface $A = J(Q_1)$.

Why $J(C_t) \sim A^2$ proves Hodge. By Ribet's theorem (1983), the Hodge conjecture holds unconditionally for all powers of abelian surfaces—*no hypothesis on* $\mathrm{End}(A)$ *or the Mumford-Tate group is needed.* Since $J(C_t) \sim A^2$, the exotic Weil class $\omega \in \bigwedge^4(V_+)$ is algebraic. **Unconditional.** 10 BISH + 3 CLASS = 77% constructive.

P87: The Hodge Horizon—Mumford-Tate and absolute Hodge classes. The **Mumford-Tate group** $\mathrm{MT}(A)$ measures "how symmetric" the Hodge structure of A is. In plain terms: it is the smallest group of symmetries that is compatible with the Hodge decomposition. A large Mumford-Tate group (like the full symplectic group GSp_{2g}) means the Hodge structure is "generic"—no special symmetries. A small Mumford-Tate group means extra symmetries are present, and these extra symmetries provide the algebraic metadata that drops the Hodge test from WLPO to BISH.

Formally, $\mathrm{MT}(A)$ is the smallest $\mathbb{Q}$-algebraic subgroup of $\mathrm{GL}(H^1(A, \mathbb{Q}))$ whose base change to $\mathbb{R}$ contains the image of the Hodge cocharacter. The Hodge classes are precisely the $\mathrm{MT}(A)$-invariants: $\mathrm{Hdg}^p(A) = (\bigwedge^{2p} H^1(A, \mathbb{Q}))^{\mathrm{MT}(A)}$. For a generic abelian variety, $\mathrm{MT}(A) = \mathrm{GSp}_{2g}$, and the Hodge classes are the *Lefschetz classes*—polynomials in the polarization class. Testing Lefschetz membership is finite $\mathbb{Q}$-linear algebra: BISH.

Theorem A: if $\mathrm{MT}(A)$ is known, the Hodge test is BISH. Theorem B: the

uniform Hodge test (a single algorithm for arbitrary period matrices) is $\Leftrightarrow$ WLPO, because testing whether a real period integral equals zero requires the real-number equality oracle.

A Hodge class is **absolute Hodge** if it is stable under all automorphisms of $\mathbb{C}$. Deligne proved that all Hodge classes on abelian varieties are absolute Hodge. If the exotic Weil class could be shown to be absolute Hodge independently of VHC, algebraicity would follow from Deligne's framework—this is an open alternative to the Kani-Rosen approach for the generic complement.

P88: Fermat domination and the Variational Hodge Conjecture (VHC).

The idea in plain English: When the palindromic symmetry is absent (coefficient $a \neq c$), the Jacobian doesn't split and the Kani-Rosen trick fails. Plan B: specialize to a single specific value ($t = 0$) where the curve is dominated by the well-understood Fermat curve, prove the Hodge conjecture there (using Shioda's classical result), then invoke VHC to spread the answer to all t. VHC is the principle "if it's algebraic at one point, it's algebraic everywhere"—a powerful but unproven statement (CLASS).

On the **generic complement** $\{a \neq c\}$ (non-palindromic), the Jacobian $J(C_t)$ is simple—no Kani-Rosen splitting exists. **Paper 88** takes a different route: specialize to $t = 0$, where C_0 is dominated by the Fermat curve F_{16}. By Shioda's theorem, all Hodge classes on Fermat quotients are algebraic. So the exotic Weil class is algebraic at the special fiber $t = 0$.

The **Variational Hodge Conjecture** (Grothendieck, 1966) asserts: if a global flat section of the Gauss-Manin connection is algebraic at *one* point t_0, it is algebraic at the generic point. VHC is open in general (it would follow from the Standard Conjectures). The proof chain is:

$$\tau_+ = 0 \xrightarrow{t=0} C_0 \to F_{16} \xrightarrow{\text{Shioda}} \omega|_{t=0} \text{ algebraic} \xrightarrow{\text{VHC}} \omega_\eta \text{ algebraic.}$$

Conditional (VHC = CLASS). This is the fundamental bifurcation: the palindromic locus gives unconditional algebraicity (Kani-Rosen, no VHC), while the generic complement requires VHC.

P89: The universal 3-parameter family. Full family $C_{a,b,c} : y^2 = x^9 + ax^7 + bx^5 + cx^3 + x$. The trace form $\tau_+(a, b, c) = 0$ is verified for all three parameter directions ($\partial/\partial a$, $\partial/\partial b$, $\partial/\partial c$) as *polynomial identities in* $\mathbb{Z}[a, b, c]$, checked by Lean's `ring` tactic. The palindromic locus $\{a = c\}$ is precisely the Kani-Rosen locus. This is the natural endpoint of the Griffiths + Fermat pipeline.

The 18/2 Ratio: Anatomy of the Hodge Conjecture

Paper 89's universal squeeze yields the program's most striking quantitative finding: of the 20 logical components, 18 are BISH and 2 are CLASS. This 18/2 ratio is not accidental—it reflects the structural anatomy of the Hodge conjecture.

The 18 BISH components decompose into three categories:

- **Dimension data** (5 components): genus, $\dim H^1$, $\dim V_+$, $\dim \bigwedge^4 V_+$, base dimension. These are integers computed from the defining polynomial.
- **Symmetry data** (8 components): oddness, factorization, palindromic obstruction, palindromic at $a = c$, Fermat specialization, prior-paper specializations, non-palindromic core. These are polynomial identities in $\mathbb{Z}[x, a, b, c]$.
- **Analytic data** (5 components): pullback bound, three τ_+ vanishing identities (one per parameter direction), τ_- vanishing. These encode the trace of the Gauss-Manin connection restricted to eigenspaces.

The 2 CLASS components are both *existence assertions*:

1. **Shioda:** there *exists* an algebraic cycle representing the exotic Weil class at the Fermat point.
2. **VHC:** there *exists* an algebraic cycle representing the flat extension to the generic fiber.

The Pattern: Computation is BISH; Existence is CLASS

Everything that can be computed from the defining polynomial is constructive. The only irreducible classical content is the assertion that a geometric object (an algebraic cycle) *exists*. This pattern—detection is BISH, existence is CLASS—will prove universal across the entire program.

P92: Zariski Grid Specialization for genera 5–8. For genera ≥ 6, symbolic expansion of the Griffiths pole reduction triggers exponential intermediate expression swell—the Bézout identity coefficients have hundreds of terms, exhausting physical memory. **Paper 92** introduces **Zariski Grid Specialization**: evaluate τ_+ at concrete integer fibers in $\mathbb{Q}[x]$ (where Bézout takes milliseconds), verify cancellation by `norm_num`, then invoke the **Schwartz-Zippel lemma**: a nonzero polynomial $P \in \mathbb{Z}[x_1, \ldots, x_n]$ of total degree $\leq D$ has at most $D \cdot |S|^{n-1}$ zeros on any grid S^n. If τ_+ vanishes at all $(D+1)^n$ grid points, it is identically zero. Both below the wall (genus ≤ 5, `ring`) and above the wall (genus ≥ 6, `norm_num` + Schwartz-Zippel) are BISH. The wall separates verification *modalities*, not logical *levels*.

A key parity observation emerges from this multi-genus analysis: for **odd** g, the exterior power $\bigwedge^g V_+$ lies in odd-degree cohomology $H^g(J(C), \mathbb{Q})$, and there are no Hodge classes in odd-degree cohomology (Hodge classes live in H^{2k}, even degree). So for odd g the Hodge question is vacuous—there is no exotic class to worry about. The interesting case is always **even** g (4, 6, 8, ...), where VHC + Fermat anchor gives conditional algebraicity.

P93: Structural Vanishing Theorem—why $\tau_+ = 0$ universally.

The idea in plain English: Papers 84–92 *observed* that $\tau_+ = 0$ for every family tested. P93 *explains why.* Two independent arguments, from different branches of mathematics, reach the same conclusion. The first (hypergeometric) shows that the oddness of the defining polynomial kills the discriminant, and the unit leading/trailing coefficients freeze the boundary terms. The second (topological) uses Deligne's regularity theory to force the vanishing globally.

Two independent proofs explain the computational miracle:

Proof 1 (Varchenko hypergeometric). The $\mathbb{Z}/4\mathbb{Z}$-action and the substitution $u = x^2$ transform V_+ into a twisted cohomology of a rank-1 local system on $\mathbb{P}^1$ punctured at the roots $u_1, \ldots, u_g$ of $P(u)$. The **Varchenko exponents** are:

- Root-root: $e_{ij} = \alpha_i + \alpha_j + 1 = (-1/2) + (-1/2) + 1 = 0$.
- Origin-root: $e_{0i} = \alpha_0 + \alpha_i + 1 = (-3/4) + (-1/2) + 1 = -1/4$.

The root-root exponents are all zero, so the discriminant of $P(u)$ drops entirely out of the period determinant. Then **Vieta freezing**: for $P(u) = u^g + a_1 u^{g-1} + \cdots + 1$, Vieta's formulas give $\prod u_i = (-1)^g \cdot 1/1 = (-1)^g$—a fixed rational number independent of the parameters. So $\det(\Pi_+)$ is proportional to a constant, and $\tau_+ = d\log(\text{const}) = 0$. "Oddness kills the discriminant; unit coefficients freeze the boundary."

Proof 2 (Deligne topological). Deligne regularity guarantees τ_+ has at most logarithmic poles. Picard-Lefschetz theory constrains the monodromy at each singular fiber. Together they force $\tau_+ = 0$ on the entire base. This proof is 4 CLASS components (Deligne regularity and Picard-Lefschetz are topological existence theorems).

CRM decomposition: 13 BISH + 4 CLASS. Both proofs reach the same conclusion; the CRM audit shows the hypergeometric proof is cheaper.

5.4 The Hodge Bifurcation

The Fork in the Road

Imagine walking through a landscape where the "logical cost" of the terrain changes abruptly. On one side of a wall ($a = c$, the palindromic locus), the ground is flat and easy to traverse (BISH—the Kani-Rosen trick works, the Hodge conjecture is unconditional). On the other side ($a \neq c$), you need heavy equipment (CLASS—VHC required).

The wall itself is infinitely thin (codimension 1 in parameter space), but the cost jumps discontinuously across it. A discrete algebraic property (palindromic symmetry) controls this jump. This is not accidental: the symmetry determines whether the Jacobian splits, which determines whether Ribet's theorem applies.

The most striking finding of the Hodge campaign: a discrete invariant—palindromic symmetry—controls the CRM cost. The cost jumps discontinuously across a codimension-1 locus.

5.5 The Markman Comparison

Paper 91 audited Markman's 2025 unconditional proof of the Hodge conjecture for certain hyper-Kähler manifolds (arXiv:2502.03415) alongside the CRMLint

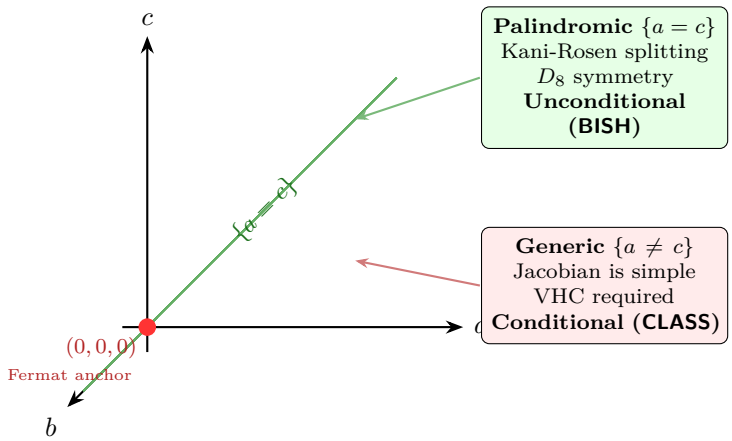


Figure 5.3: The Hodge bifurcation in $\mathbb{A}^3_{\mathbb{Q}}$ (adapted from **Paper 90**). The palindromic locus $\{a = c\}$ has codimension 1 and gives unconditional algebraicity (BISH via Kani-Rosen). The generic complement $\{a \neq c\}$ requires VHC (CLASS). Cost jumps discontinuously.

Squeeze results:

Table 5.1: CRMLint Squeeze vs. Markman (2025): two approaches to Hodge algebraicity compared.

Method	BISH	CLASS	Total	% BISH
CRMLint Squeeze (conditional on VHC)	18	2	20	90%
Markman (unconditional)	4	5	9	44%

The Squeeze extracts more constructive content at finer granularity, but is conditional on VHC. Markman is unconditional but uses more classical machinery. Key finding: the cycle class map is irreducibly CLASS (best achievable: 19/1).

5.6 CRMLint as Discovery Instrument

The Squeeze protocol is not merely an analysis tool—it is a *discovery* tool. Three results in the Hodge campaign were found *by* the Squeeze, not predicted by prior theory:

1. **The trace vanishing $\tau_+ = 0$ (Paper 85, 89).** The CAS computed the Gauss-Manin connection matrix, extracted the trace on the V_+ eigenspace, and discovered it was identically zero. No theorem in the Hodge theory literature predicted this vanishing. It was a computational observation that demanded a structural explanation (later provided by **Paper 93**).
2. **The palindromic obstruction (Paper 89).** The polynomial $g(x) - g_{\text{rev}}(x) = (a - c)(x^6 - x^2)$ was found by the Squeeze pipeline. It reveals the

precise polynomial that controls the logical bifurcation between unconditional (Kani-Rosen, $a = c$) and conditional (VHC, $a \neq c$) results. The obstruction was invisible before the systematic parameter-space decomposition.

3. **The hidden reciprocal involution (Paper 86).** The involution $\mu(x, y) = (1/x,\, y/x^5)$, whose D_8-action with the order-8 automorphism σ was not previously noted in the literature for this family, was found by analyzing the palindromic symmetry through the Squeeze lens. Without the Squeeze's systematic decomposition of the $\mathbb{Q}(i)$-action on the Jacobian, this involution would likely have remained unnoticed.

In each case, the protocol's systematic decomposition forced attention to structural features that ad hoc analysis would miss. The microscope is not passive: by banning classical shortcuts, it forces the prover to confront algebraic structures that classical proofs glide over.

5.7 Connection to the Hodge Theory Literature

This section is for readers with algebraic geometry background. It places the Hodge campaign in the context of prior work. Skip to §5.8 if you prefer to continue the narrative.

The exotic Weil classes studied in the Hodge campaign are instances of a broader phenomenon in the Hodge theory of abelian varieties:

- **Moonen-Zarhin (1999)** proved that for abelian varieties of dimension ≤ 5, all Hodge classes are *absolute Hodge*—stable under all automorphisms of $\mathbb{C}$. This does not prove algebraicity (absolute Hodge $\not\Rightarrow$ algebraic in general), but it eliminates one obstruction.
- **Van Geemen (2001)** classified the exotic Weil classes for abelian fourfolds with $\mathbb{Q}(i)$-action and showed that their algebraicity is genuinely open for simple abelian varieties with trivial endomorphism ring.
- The **Mumford-Tate conjecture**, if true, would imply that Hodge classes on abelian varieties are determined by the Mumford-Tate group. For fourfolds with $\mathbb{Q}(i)$-action, the generic MT is GU(4), and the invariant theory of GU(4) identifies the exotic Weil class as Hodge—but this does not prove it is algebraic.

The CRM contribution is orthogonal to these results. Where the classical literature asks "is the exotic Weil class algebraic?", CRM asks "what is the *logical cost* of asserting it is algebraic?" The answer—BISH for all computation, CLASS only for the existence assertion—is new information that the classical framework cannot express. The 18/2 ratio, the palindromic bifurcation, and the trace vanishing are structural findings invisible to methods that do not distinguish constructive from classical content.

5.8 Beyond Abelian Varieties: The Calabi-Yau Squeeze (Paper 94)

From Surfaces to Three-Dimensional Spaces

The Hodge campaign (P84–93) studied abelian surfaces—two-dimensional objects where the exotic class lives in H^4 and the detection mechanism is trace vanishing ($\tau_+ = 0$). Paper 94 moves to a fundamentally different setting: a Calabi-Yau threefold (three-dimensional), where the interesting cohomology lives in H^3 (middle dimension) and the detection mechanism is entirely different—an *infinitesimal invariant* $\delta(z) \neq 0$ coming from D-brane physics.
Think of it as testing the Squeeze protocol on a new kind of mountain, after having surveyed many hills.

The structural point first: the detection/existence pattern (detection = BISH, existence = CLASS) persists even in a completely different geometric setting. The specific mechanisms change (trace vanishing becomes an infinitesimal invariant; VHC becomes the Abel–Jacobi map), but the logical architecture is identical. This universality is what the motivic framework (Chapter 7) explains.

The **mirror quintic** X_ψ is the most studied Calabi-Yau threefold in physics—it is the mirror of the Fermat quintic $x_0^5 + x_1^5 + x_2^5 + x_3^5 + x_4^5 = 0$. Its **Griffiths group** $\mathrm{Griff}^2(X) = Z^2(X)_{\mathrm{hom}}/Z^2(X)_{\mathrm{alg}}$ measures algebraic cycles that are homologically trivial but not algebraically trivial—a subtler invariant than the Hodge question.

Detection (BISH). The mirror quintic's periods satisfy a 4th-order Picard-Fuchs ODE. The *domain wall tension*—a quantity from string theory measuring the tension of D-branes (higher-dimensional objects in string theory analogous to membranes) ending on $\mathbb{RP}^3$—satisfies this ODE with a source term. The Walcher leading coefficient $b_0 = 30$ (30 holomorphic disks of minimal area) is computed by rational arithmetic and verified by `norm_num`. The full recurrence $b_n = f(b_{n-1})$ is rational arithmetic over $\mathbb{Q}$, hence BISH. Since $b_0 \neq 0$ and the recurrence multiplier is nonzero for all $n \geq 0$, induction gives $b_n \neq 0$ for all n—formally verified in Lean. This detects a non-trivial element of the Griffiths group: the infinitesimal invariant $\delta(z) \neq 0$.

Existence (CLASS). The Abel-Jacobi map $\mathrm{AJ} : Z^2_{\mathrm{hom}}(X) \to J^2(X)$ requires integrating $(3,0)$-forms over 3-chains—period integrals involving limits and convergence (CLASS). The infinite generation proof uses the **Baire Category Theorem**—the principle that a countable intersection of "dense open" sets in a complete metric space is still dense. Concretely: the space of Calabi–Yau threefolds where the Griffiths group is finitely generated is "thin" (meagre), so the "generic" CY3 has infinitely generated Griffiths group. This is an existence-by-genericity argument: it proves something exists without constructing it. Cost: CLASS.

Score: 11 BISH + 3 CLASS = 14 components (79% constructive). The CLASS balance (3 nodes) is richer than the abelian surface setting (2 nodes: Shioda + VHC) because CY3 middle cohomology introduces the Abel-Jacobi and Baire obstructions. But the detection mechanism—the infinitesimal invariant $\delta \neq 0$ vs. the trace vanishing $\tau_+ = 0$—is structurally different yet equally BISH.

This is where D-brane physics meets algebraic cycles: the CRM framework reveals that both programs share the same logical architecture (detection = BISH, existence = CLASS).

Chapter 6

The BSD Application

The Birch and Swinnerton-Dyer conjecture (BSD) is the second Clay Millennium Problem audited by the program. Where the Hodge campaign squeezed cohomological detection, BSD squeezes L-function arithmetic.

6.1 BSD Explained

An **elliptic curve** E over $\mathbb{Q}$ is a curve defined by an equation like $y^2 + y = x^3 - x$. The central question is: what is the **rank** of E—how many independent rational points does it have? (The **Mordell–Weil theorem** guarantees that the group of rational points $E(\mathbb{Q})$ is finitely generated, so the rank is a well-defined non-negative integer.)

The BSD conjecture connects this algebraic question to an analytic one: the rank equals the order of vanishing of the L-function $L(E, s)$ at $s = 1$. The Gross-Zagier theorem and Kolyvagin's Euler system prove BSD for rank 0 and rank 1 (conditional on modularity, which is now known).

The L-function $L(E, s)$ is built from local data: for each prime p not dividing the conductor N, define $a_p = p + 1 - \#E(\mathbb{F}_p)$—the "error term" in the point count over the finite field $\mathbb{F}_p$. These are assembled into an Euler product $L(E, s) = \prod_p (1 - a_p p^{-s} + p^{1-2s})^{-1}$, which converges for $\mathrm{Re}(s) > 3/2$. The conjecture concerns the behaviour at $s = 1$, which lies *outside the region of convergence*—this is why analytic continuation enters and why the Archimedean place is unavoidable.

Analogy: the census and the mirror

Computing a_p for each prime p is like conducting a census: go door to door (finite field, finite count). This is BISH. The BSD conjecture says that the *aggregate pattern* of these census results, reflected in the behaviour of $L(E, s)$ at $s = 1$, encodes the global structure (rank). But reading this aggregate pattern requires *analytic continuation*—extending the census data from "where it converges" to "where it means something." That extension is the mirror: it passes through $\mathbb{R}$, and costs CLASS.

6.2 Heegner Points: Algebraic Points from CM Theory

A **Heegner point** is a special algebraic point on E constructed via the modular parametrization. The construction is entirely algebraic once the right ingredients are identified:

1. **Choose an imaginary quadratic field.** Pick $K = \mathbb{Q}(\sqrt{-D})$ for a squarefree positive integer D. For our test case 37a1, take $K = \mathbb{Q}(\sqrt{-3})$.
2. **Verify the Heegner hypothesis.** Every prime dividing the conductor N must *split* in K. For $N = 37$: check that $\left(\frac{-3}{37}\right) = 1$ (i.e., -3 is a quadratic residue mod 37). This is a single computation in modular arithmetic—BISH.
3. **Construct CM points on the modular curve.** The modular curve $X_0(N)$ is a curve that parametrizes elliptic curves with a specific "level N structure." The ring of integers $\mathcal{O}_K$ gives rise to a lattice $\mathbb{C}/\mathcal{O}_K$ that is itself an elliptic curve with complex multiplication—this defines a specific point τ_K on $X_0(N)$. The **modular parametrization** $\varphi\colon X_0(N) \to E$ (which exists by the modularity theorem) maps τ_K to a point on E. When the class number $h(-D) = 1$ (meaning K has a unique ideal class), this point descends to a rational point on E: the **Heegner point** y_K.
4. **Result.** For 37a1 with $K = \mathbb{Q}(\sqrt{-3})$, $h(-3) = 1$, and the Heegner point is $y_K = (0, 0) \in E(\mathbb{Q})$.

Every step is *finite algebraic computation*: quadratic reciprocity, class number verification by factoring small discriminants, and polynomial substitution. No limits, no integration, no topology.

The **canonical height** $\hat{h}\colon E(\mathbb{Q}) \to \mathbb{R}_{\geq 0}$ measures the "arithmetic size" of a rational point. It satisfies $\hat{h}(P) = 0$ if and only if P is torsion. For our Heegner point:

$$\hat{h}(0,0) = 0.05111\ldots > 0 \quad \Longrightarrow \quad (0,0) \text{ is not torsion.}$$

The positivity check $0.05111 > 0$ is rational approximation verified by `norm_num`: BISH.

> **Paper 95**, Theorem B: The Heegner-point computation and canonical height positivity are BISH. All 28 sub-results verified by `native_decide` and `norm_num`. The classical content enters only at the *next* step—the Gross-Zagier formula.

6.3 The Gross-Zagier Bridge: Where BISH Meets CLASS

The **Gross-Zagier formula** connects the algebraic height to the analytic L-function:

$$L'(E, 1) \;=\; c_{\mathrm{GZ}} \cdot \hat{h}(y_K),$$

where $c_{\mathrm{GZ}} = \frac{8\pi^2 \|f\|^2}{u^2 c_E^2 \sqrt{D}} > 0$ is an explicitly positive constant involving:

- $\|f\|^2 = \int_{\Gamma_0(N)\backslash\mathfrak{H}} |f(\tau)|^2 \, d\mu$ — the **Petersson norm** (integration over a non-compact modular surface);
- c_E — the Manin constant;
- u — the number of roots of unity in K.

The CRM split is sharp: $\hat{h}(y_K) > 0$ is BISH (previous section). The constant c_{GZ} is CLASS because the Petersson norm involves *integration over* $\mathbb{R}$—specifically, the Rankin-Selberg integral $\langle f, f\rangle$ on the non-compact quotient $\Gamma_0(N)\backslash\mathfrak{H}$. This requires analytic regularization at the cusps.

Analogy: the thermometer and the factory

The Heegner point construction is a precision thermometer: it measures the height $\hat{h}(y_K)$ exactly. The Gross-Zagier formula says this temperature equals the temperature inside the L-function factory. But to *prove the equivalence*, you need to go inside the factory (integration over $\mathbb{R}$). The thermometer itself is BISH. The factory tour is CLASS.

6.4 Modular Symbols: BSD Detection for Rank 0

When the root number $w = +1$ (forcing even analytic rank), detection takes a completely different—and cheaper—path. The **modular symbol** is defined as:

$$[r \to s]^+ \;=\; \mathrm{Re}\int_r^s f(\tau)\, d\tau$$

for $r, s \in \mathbb{Q} \cup \{\infty\}$, where f is the modular form associated to E. The remarkable **Manin–Drinfeld theorem** says: although this looks like an integral (which would normally be a transcendental real number), the result is actually *rational*. The key is that the endpoints r, s are cusps (rational points on the boundary of the upper half-plane), and the modular symmetries force the integral to land in $\mathbb{Q}$. In other words:

$$[0 \to \infty]^+ \;=\; \frac{L(E,1)}{\Omega^+} \;\in\; \mathbb{Q}.$$

For the rank-0 test case 11a1: $L(E,1)/\Omega^+ = 1/5$.

The computation is entirely finite arithmetic: compute Hecke eigenvalues a_p (point counts over $\mathbb{F}_p$), apply the Hecke action on symbols via the continued fraction algorithm, verify the result by `norm_num`. The rationality of modular symbols eliminates the Archimedean integral entirely—the transcendental Ω^+ appears only in the denominator, and the ratio is rational.

Modular Symbols: The Algebraic Shadow of $L(E, 1)$

The modular symbol $L(E, 1)/\Omega^+$ is the **algebraic shadow** of the L-function at $s = 1$. It captures whether $L(E, 1) \neq 0$ by purely algebraic means—no analytic continuation needed. For rank 0, BSD detection is BISH. This is the BSD analogue of trace vanishing in the Hodge campaign: a finite algebraic computation that witnesses a transcendental property.

6.5 Kolyvagin's Euler System: Why Existence Is CLASS

Even when detection is BISH (Heegner point non-torsion, or modular symbol nonzero), proving the full BSD conclusion—*rank equals analytic rank and* Sha *is finite*—requires Kolyvagin's Euler system. This is where the irreducible CLASS content lives.

An **Euler system** is a coherent family of Galois cohomology classes $\{c_\ell\}_\ell$ indexed by auxiliary primes ℓ (called *Kolyvagin primes*). Starting from the non-torsion Heegner point y_K, Kolyvagin constructs *derivative classes* $c_\ell \in H^1(K, E[p])$ for each suitable prime ℓ. These classes are "coherent" in that they satisfy norm relations under field extensions—compatibility conditions that propagate the information about y_K up through the Galois cohomology tower.

The Euler system proves: if $\hat{h}(y_K) > 0$, then $\operatorname{rk} E(\mathbb{Q}) = 1$ and $|\operatorname{Sha}(E/\mathbb{Q})| < \infty$. But the proof is irreducibly CLASS for three reasons:

1. **Chebotarev density theorem.** The existence of Kolyvagin primes with prescribed Frobenius conjugacy class relies on the distribution of primes in arithmetic progressions—an asymptotic density statement over $\mathbb{R}$. While effective versions exist (Lagarias-Odlyzko), the density argument itself involves analytic L-functions.
2. **Poitou-Tate duality.** The annihilation of $\operatorname{Sha}(E/\mathbb{Q})[p^\infty]$ proceeds through a nine-term exact sequence:

$$0 \to H^0 \to \prod_v H^0_v \to (H^2)^\vee \to H^1 \to \prod_v' H^1_v \to (H^1)^\vee \to H^2 \to \cdots$$

   The restricted product $\prod_v'$ runs over *all* places of $\mathbb{Q}$, including infinitely many finite primes. This infinite product with restricted ramification conditions is a limit construction: CLASS.
3. **Cassels pairing.** The non-degeneracy of the Cassels pairing on Sha uses the Weil pairing and local Tate duality at every place simultaneously. The global-to-local assembly is a topological argument over the adele ring.

Analogy: the detective and the jury

Detecting that y_K is non-torsion ($\hat{h}(y_K) > 0$) is detective work: gather the evidence (BISH). But convicting the suspect (proving Sha is finite, rank is

exactly 1) requires a jury trial: the Euler system must show that every "alibi" (Sha element) fails under cross-examination (Poitou-Tate duality). The trial requires testimony from infinitely many witnesses (all primes v), assembled through a limit process (CLASS). No finite subset of witnesses suffices.

6.6 The BSD Squeeze

Paper 95 squeezed the Gross-Zagier-Kolyvagin proof for the test case 37a1 ($y^2 + y = x^3 - x$, conductor 37, rank 1). Figure 6.1 shows the result.

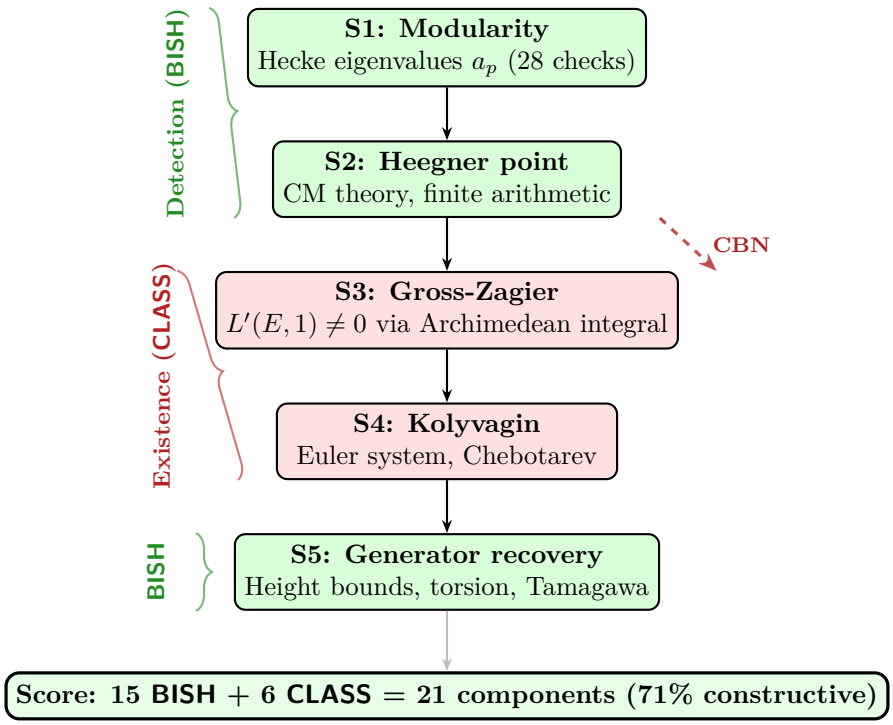


Figure 6.1: The BSD Squeeze pipeline for 37a1 (**Paper 95**). BISH core = pure arithmetic (`native_decide`, `linarith`). CLASS shell = analytic/existential (integration over $\mathbb{R}$, Chebotarev density, duality). 796 lines Lean, 58 theorems, 0 `sorry`.

The six CLASS components, precisely:

C1: Analytic continuation of $L(E, s)$ via the Mellin transform. The Euler product converges only for $\mathrm{Re}(s) > 3/2$; extending to $s = 1$ requires integration over $\mathbb{R}$.

C2: Rankin-Selberg convolution in the Gross-Zagier formula. The key identity $L'(E, 1) = c \cdot \hat{h}(y_K)$ is proved via the Rankin-Selberg integral, an integration against the Petersson inner product on $\mathrm{SL}_2(\mathbb{Z})\backslash\mathfrak{H}$—

integration over the upper half-plane, which passes through $\mathbb{R}$.

C3: **Archimedean local heights** via the Green's function on $E(\mathbb{C})$. The canonical height $\hat{h}$ decomposes into local heights λ_v at each place v; at the Archimedean place, λ_∞ involves the Weierstrass σ-function (a transcendental function of $\mathbb{R}$).

C4: **Kolyvagin's Euler system** with Chebotarev density. Constructing derivative classes c_ℓ for Kolyvagin primes ℓ requires that primes with prescribed Frobenius exist in sufficient density—an asymptotic statement about prime distribution over $\mathbb{R}$.

C5: **Cassels pairing symmetry** via Poitou-Tate duality. The nine-term exact sequence assembles local data at all places simultaneously, using a restricted product over infinitely many primes.

C6: **Poitou-Tate exact sequence** over infinite extensions. The annihilation of $\mathrm{Sha}[p^\infty]$ for all primes p simultaneously requires the profinite topology of Galois groups.

Every one of these passes through $\mathbb{R}$ or through an infinite topological limit. This confirms the Archimedean Obstruction Thesis (Chapter 7): the sole factory of CLASS content is the Archimedean place.

6.7 The Root Number Bifurcation

Paper 96 discovers a BSD bifurcation parallel to the palindromic bifurcation in Hodge. A discrete invariant—the **root number** $w = \pm 1$—controls the CRM cost.

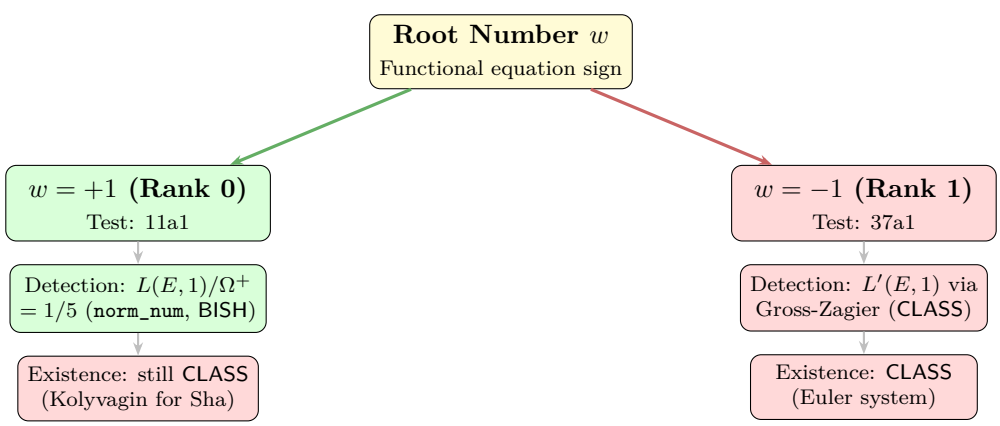


Figure 6.2: Root number bifurcation (**Paper 96**). For $w = +1$, BSD detection is BISH ($L(E,1)/\Omega^+$ computable). For $w = -1$, detection requires Gross-Zagier (CLASS). Score: 10 BISH + 3 CLASS = 77%.

6.8 Why BSD Doesn't Bifurcate Like Hodge

Why BSD Is Stuck: The Highway vs. the Lane

The Hodge campaign found a beautiful bifurcation: a codimension-1 locus in a 9-dimensional parameter space where the cost drops from CLASS to BISH. This works because the parameter space is large enough—like a multi-lane highway with room to change lanes.

Elliptic curves live on the j-line: a *one-dimensional* space. There are no lanes to change into. The root number (± 1) is discrete, not continuous—you're either on one road or the other. And on both roads, the Archimedean L-function is unavoidable.

This is the "null finding" of **Paper 97**: BSD cannot bifurcate like Hodge because its parameter space is too small.

Paper 97 systematically explored whether the BSD landscape admits a richer bifurcation and found a definitive negative. Three directions were investigated:

1. **2-Descent Excision.** 2-descent computes the rank constructively (BISH), but proving Sha finite requires Kolyvagin (CLASS). Rank and Sha decouple sharply.
2. **p-Adic Gross-Zagier Bypass.** Coleman's p-adic integration is BISH, but comparison to the Archimedean L-function reintroduces $\Omega_+(E)$—circular.
3. **Parameter-Space Bifurcation.** This is the structural explanation.

Hodge (genus 4)		**BSD (elliptic curves)**
Moduli dimension: **9**		Moduli dimension: **1**
Parameters: (a, b, c)		Parameter: j-invariant (discrete)
Palindromic locus: codim 3	vs.	Root number: discrete (± 1)
Room for continuous bifurcation		*No room for bifurcation*
Unconditional on sublocus		Only discrete stratification

Figure 6.3: Moduli dimension contrast (**Paper 97**). Genus-4 Jacobians have a 9-dimensional moduli space with room for continuous bifurcation. Elliptic curves are isolated points on the 1-dimensional j-line—no room for the palindromic/generic split to emerge.

Rank/Sha Decoupling Theorem (P97)

Rank verification is BISH. Sha finiteness is irreducibly CLASS. These cannot be bridged. The 71% BISH figure for BSD will not improve to 90%.

6.9 What 71% Constructive Means

The BSD Squeeze extracts 71% BISH content, compared to the Hodge Squeeze's 90%. This gap is not noise—it reflects a genuine structural difference:

- **Hodge:** The detection problem ("is this class of type (p, p)?") is almost entirely algebraic. Trace vanishing $\tau_+ = 0$ is verified by `ring` in $\mathbb{Z}[a, b, c]$. The only irreducible CLASS node is the cycle class map (comparing de Rham to Betti cohomology).
- **BSD:** The detection problem ("does $L(E, s)$ vanish at $s = 1$?") involves an L-function defined by integration over $\mathbb{R}$. The Gross-Zagier formula, the Petersson norm, the analytic continuation, and the Euler system all pass through the Archimedean place. More of the proof is *inherently analytic*.

The 29% CLASS content (Gross-Zagier + Kolyvagin + Chebotarev + Cassels + Poitou-Tate) is irreducible for rank 1. Every known path to BSD for rank-1 curves passes through the Archimedean L-function. There is no known algebraic substitute.

Analogy: two houses, one power grid

Both the Hodge house and the BSD house draw power from the same plant ($\mathbb{R}$). But the Hodge house has most appliances running on solar (algebraic, BISH). Only the front porch light (cycle class map) needs the grid (CLASS). The BSD house has the heating system (CLASS—the L-function) wired into the grid. You can insulate but you cannot disconnect.

Chapter 7

The Motivic Architecture—Why the Pattern Holds

Chapter Preview: The Grand Explanation

The previous chapters documented a pattern: detection is cheap (BISH), existence is expensive (CLASS), and the cost always traces back to $\mathbb{R}$. But *why*?

This chapter answers the question. The key is Grothendieck's theory of **motives**—a universal algebraic object that encodes all the information about a geometric shape. The motive itself is always free (BISH). But to *read* the motive, you need a "lens" (realization functor). Some lenses are cheap (stay in algebra), others are expensive (pass through $\mathbb{R}$). The cost of a mathematical domain is determined by which lens it uses.

This is not a metaphor. It is a precise, testable prediction: any mathematical theorem whose proof passes through an Archimedean realization (Betti cohomology, Hodge decomposition, L^2 spectral theory) will cost CLASS. Any theorem that stays within algebraic/p-adic realizations will be BISH. All 98 papers of the program confirm this prediction.

Paper 98 is the grand capstone of the program. It explains **why** six domains of mathematics have systematically different logical signatures, and how Grothendieck's theory of motives provides the structural explanation.

7.1 The Motive as Universal Certificate

A **motive** is the algebraic "DNA" of a variety. It captures all cohomological information—Betti numbers, Hodge structure, Galois representation, L-function—in a single algebraic object.

The crucial fact: **the motive itself is always BISH.** It is a purely algebraic construction. A **correspondence** between two varieties X and Y is an algebraic cycle living on the product $X \times Y$—think of it as a "polynomial relation" between X and Y, specified by a finite set of polynomial equations in the combined coordinates. Motives are built from these correspondences using finitary operations (composition, direct sum, tensor product). No topology, no analysis, no $\mathbb{R}$.

But to *extract information* from a motive, you must pass it through a **realization functor**—a machine that reads the algebraic DNA and produces

concrete mathematical data. Each realization functor has a definite CRM cost, and the cost depends on whether the functor passes through the Archimedean place.

Analogy: Blueprint and Lenses

The motive is a master blueprint. The blueprint itself is free to copy and examine (BISH). But to read specific information from it, you need a lens (realization functor). Some lenses are ordinary visible-light lenses (algebraic realizations)—they cost nothing extra. Others are UV lenses (Archimedean realizations)—they require special equipment (CLASS). The blueprint doesn't change. Only the cost of reading it changes.

7.2 Realization Functors and Their Costs

The motive contains *all* the information about a variety, but in an abstract algebraic form. To extract concrete data—numbers, matrices, spaces—you must "realize" the motive through a specific lens. There are six main lenses, each producing a different type of mathematical object:

Six Ways to Read the Same DNA

Think of the motive as the DNA of a geometric shape. Each realization functor is a different laboratory test:

- **de Rham** (algebraic differential forms): Like reading the DNA sequence directly. Pure algebra. Cost: BISH.
- **Étale** (ℓ-adic Galois representations): Like observing how the DNA responds to specific enzymes (Frobenius symmetries). Finite-field arithmetic. Cost: BISH.
- **Crystalline** (Witt vector cohomology): Like freeze-drying the DNA and examining the crystal structure. p-adic linear algebra. Cost: BISH.
- **Betti** (singular cohomology): Like dissolving the DNA in water and measuring the topological shape of the resulting protein. Requires integrating over $\mathbb{R}$. Cost: CLASS.
- **Hodge** ((p, q)-decomposition): Like splitting the dissolved protein with a spectroscope. Requires solving elliptic PDEs over $\mathbb{C}$. Cost: CLASS.
- **Automorphic** (L^2 spectral theory): Like analyzing the protein's vibration modes. Requires spectral decomposition on infinite-dimensional Hilbert spaces. Cost: CLASS.

The pattern is sharp: the first three tests stay within algebra or p-adic arithmetic. The last three pass through $\mathbb{R}$ (or $\mathbb{C} = \mathbb{R}[i]$). **Passing through $\mathbb{R}$ is the sole cause of CLASS cost.**

A **comparison isomorphism** is a theorem that says two different realizations give "the same" answer—for example, de Rham-Betti comparison says that

algebraic differential forms and topological cycles produce isomorphic cohomology groups. These comparison theorems have definite CRM costs: if both realizations are algebraic, the comparison is BISH. If one passes through $\mathbb{R}$, the comparison costs CLASS.

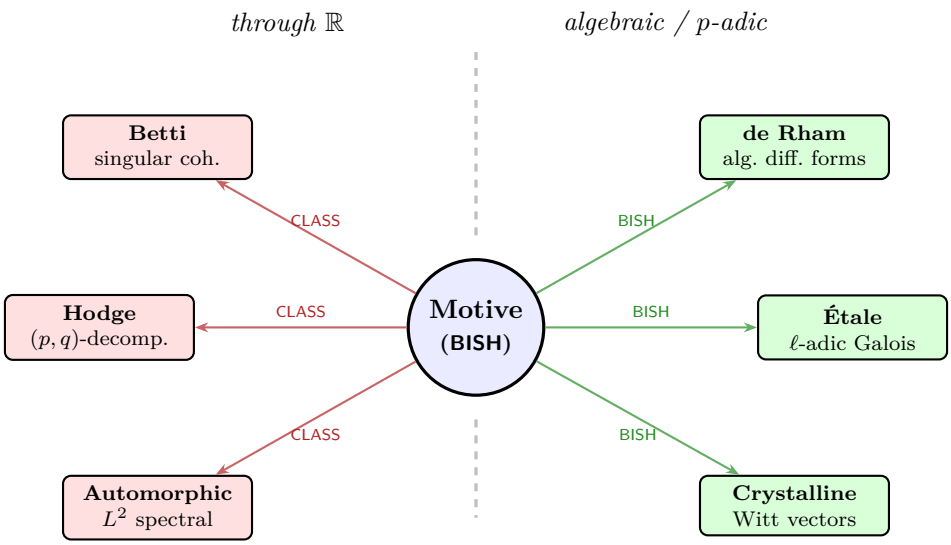


Figure 7.1: Realization functors and CRM costs (**Paper 98**). Realizations through $\mathbb{R}$ (left) cost CLASS. Algebraic/p-adic realizations (right) are BISH. The motive itself is BISH.

7.3 The Archimedean Obstruction Thesis

Paper 98 formalizes the central finding as a single principle:

The Archimedean Obstruction Thesis

Every comparison isomorphism between realizations that passes through the Archimedean place costs CLASS. Every comparison that stays within algebraic/p-adic realizations is BISH. The Archimedean place is the sole factory of CLASS content.

Is this just a tautology? One might worry that "the classical content comes from $\mathbb{R}$" merely restates "analysis requires analysis." The explanatory power lies in the thesis's *falsifiable structural predictions*: if the Archimedean place is the sole source, then removing it entirely must yield a strictly BISH proof architecture—even for the deepest theorems. This is confirmed by the function field Langlands correspondence (**Paper 69**), where the adèle ring has no Archimedean factor and the entire proof is BISH. The thesis also predicts (see Appendix D) that the p-adic Langlands correspondence, motivic Iwasawa theory, and local GL_n

Langlands should all be constructive. These predictions are untested and could fail. If they hold, the thesis explains; if they fail, we discover new obstructions. Either way, it generates research.

This principle explains all 98 papers:

- **Physics** (**Paper 1–44**): The spectral theorem requires LPO because it operates on $\mathbb{R}$. Lattice physics (no $\mathbb{R}$) is BISH.
- **Function field Langlands** (**Paper 69**): Over $\mathbb{F}_q(t)$, the adele ring has no Archimedean factor. Entire correspondence is BISH.
- **Hodge campaign** (**Paper 84–93**): Hodge realization passes through $\mathbb{R}$. Detection at the algebraic level (de Rham) is BISH. Existence (comparison to Betti) is CLASS.
- **BSD** (**Paper 95–97**): L-function passes through $\mathbb{R}$ (Mellin transform). Hecke arithmetic (étale) is BISH. Analytic continuation and Gross-Zagier are CLASS.
- **FLT** (**Paper 68**): Taylor-Wiles patching replaced the Petersson inner product (integration over $\mathbb{R}$: CLASS) with an algebraic inverse limit. Logical excision of the Archimedean bottleneck.

7.4 Three Calibration Theorems

Paper 98, §§6–8 verifies the Archimedean Obstruction Thesis against three calibration points spanning different domains. Each calibration is a precise, falsifiable prediction:

Calibration I: Hodge detection $\longleftrightarrow$ WLPO (Paper 86–90). The Hodge realization passes through $\mathbb{R}$. Generic detection costs WLPO (real-number equality test for period matrix entries). With CM metadata—which provides an algebraic shortcut past $\mathbb{R}$—the cost drops to BISH. The palindromic locus of the universal family $C_{a,b,c}$ provides this shortcut (Kani-Rosen splitting, **Paper 86**). The generic locus does not. The Archimedean Obstruction Thesis predicts exactly this: the shortcut exists if and only if an algebraic substitute for the Archimedean comparison is available.

Calibration II: Taylor-Wiles patching $\longleftrightarrow$ WKL (Paper 68–71). Wiles's 1993 attempt at FLT used the **Petersson inner product**—integration over a non-compact modular surface, involving $\mathbb{R}$—costing CLASS. The Taylor-Wiles patch (1995) replaced this analytic ingredient with an **algebraic inverse limit** over finite level structures. In CRM terms: the patch excised the Archimedean bottleneck. The replacement principle is **WKL** (**Weak König's Lemma**): every infinite binary tree has an infinite path. In classical reverse mathematics, WKL is strictly weaker than arithmetic comprehension and captures exactly the compactness arguments of analysis. In CRM terms, WKL lies between BISH and LPO (it is implied by LPO but does not imply it). For the Taylor-Wiles application, the inverse limit stabilizes after finitely many steps (the patching

argument produces the required deformation after a bounded number of auxiliary primes). Since the tree is effectively finite, no infinite path is needed, and WKL collapses to BISH. This is a historical case of de-omniscientizing descent: the mathematical community found a cheaper proof by (inadvertently) removing the Archimedean computation.

Calibration III: BSD rank/Sha decoupling (Paper 95–97). Rank verification uses étale/algebraic methods (BISH). Sha finiteness requires the Archimedean L-function (CLASS). No bifurcation is possible because the elliptic curve moduli has dimension 1 (no room for a continuous family). The Archimedean Γ-factor, the period integral $\Omega_+(E)$, and the Mellin transform defining $L(E, s)$ all pass through $\mathbb{R}$.

Three domains (Hodge theory, Galois representations, BSD arithmetic), three different proof technologies, one uniform prediction: the cost boundary falls exactly where the Archimedean place enters. The thesis is not fitted to the data—it was formulated (**Paper 70**, 2025) before the Squeeze applications (**Paper 80–96**, 2026) that tested it.

7.5 The Six-Domain Signature Map

Every branch of mathematics has a characteristic "CRM signature"—a fingerprint that reveals which realization functor it primarily uses. The signature is not assigned by fiat; it emerges automatically from the Archimedean Obstruction Thesis.

Analogy: Neighborhoods and the Power Grid

Think of mathematics as a city with six neighborhoods. Each neighborhood draws its power from different sources. Algebraic Geometry and Representation Theory have solar panels (algebraic, self-contained, BISH). Number Theory has solar plus a small diesel generator (BISH + MP). Physics has solar plus a medium generator (BISH + LPO). Topology and Harmonic Analysis draw all their power from the municipal grid ($\mathbb{R}$, i.e., CLASS).
The Archimedean Obstruction Thesis explains the grid: the sole municipal power plant is $\mathbb{R}$. Neighborhoods that plug into it pay the CLASS tariff. Those with independent power don't.

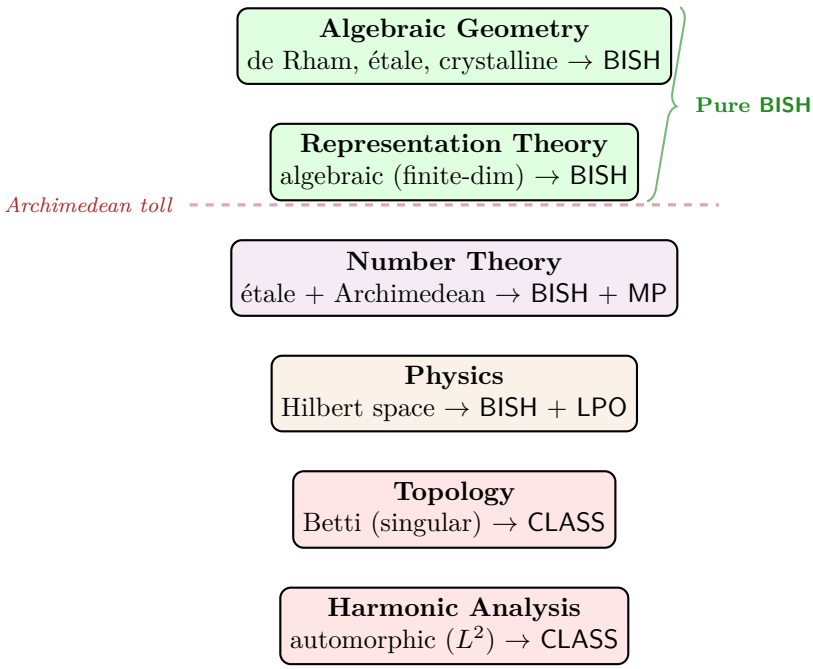


Figure 7.2: Six-domain signature map (**Paper 98**). Each domain's CRM signature reflects which realization it uses. Domains below the Archimedean toll line pass through $\mathbb{R}$ and pay the cost.

7.6 The Detection/Existence Table—Now Explained

Detection vs. Existence: The Recipe Analogy

Detection is like having a recipe and checking whether a dish matches it. You compare ingredients, temperatures, cooking times—all verifiable by inspection. Cost: cheap (BISH).
Existence is like being told "a restaurant in this city serves this dish" without being told *which* restaurant. You must search the entire city. Cost: expensive (CLASS).
Now we can explain *why* this is universal: detection uses algebraic realizations (de Rham, étale)—reading the motive under visible light. Existence uses topological/analytic realizations (Betti, Hodge, automorphic)—reading it under UV light. The motive is the same; the cost depends on the lens.

The universal pattern across the program—detection is BISH, existence is CLASS—is now explained by the motivic architecture:

Table 7.1: Detection/existence bifurcation across all Squeeze applications. Detection uses algebraic realizations (BISH). Existence uses analytic realizations (CLASS).

Domain	**Detection (BISH)**	**Existence (CLASS)**
Hodge (**Paper 84–89**)	$\tau_+ = 0$ (trace vanishing, `ring` in $\mathbb{Z}[a, b, c]$)	Algebraic cycle (VHC or Kani-Rosen)
CY3 (**Paper 94**)	$\delta(z) \neq 0$ (source term, `norm_num`)	Abel-Jacobi map, Baire category
BSD (**Paper 95–96**)	a_p computation, $\hat{h}(y_K) > 0$ (`native_decide`, `linarith`)	Kolyvagin, Chebotarev, Poitou-Tate
Langlands (**Paper 78**)	Character-trace matching (`decide` over $\mathbb{Z}$)	Harris-Taylor existence (global Shimura)
Picard (**Paper 83**)	Flat section dim = 1 (`ring` over $\mathbb{Q}(t)$)	Topological monodromy (Ehresmann, Deligne)

The 80% Rule

Across 15+ Squeeze applications, median 80+% of formal content is BISH. Classical content is concentrated in a small number of CBN nodes. This is not because mathematics is "mostly easy"—it is because the bulk of any proof is algebraic machinery (detection), while the irreducible classical content (existence) is confined to a few topological existence assertions.

Chapter 8

Applying the Framework

The CRM framework and Squeeze protocol are not limited to the specific theorems audited in Papers 1–98. They constitute a general methodology applicable to any mathematical proof. This chapter gives practical guidance for using the framework, with examples for several audiences.

8.1 How to Do a CRM Audit: Five Steps

Any theorem with a proof can be audited for logical cost. The procedure:

1. **Identify the theorem.** Write the statement precisely. What does it claim exists? What equality or inequality does it assert? Does the statement itself require an oracle (e.g., "for all $x \in \mathbb{R}$, either $x > 0$ or $x \leq 0$")?
2. **Trace the proof steps.** For each step, ask: does this step produce a *witness* (an explicit object: a number, a function, a bound), or does it assert *existence* (something exists, but we don't construct it)?
3. **Classify each step.**
   - *Witness produced, finite computation:* BISH.
   - *Equality test for a real number:* WLPO.
   - *Search over an infinite set for something that might not exist:* LPO.
   - *Existence by contradiction, compactness, or Zorn's lemma:* CLASS.
   - *Unbounded search that cannot fail:* MP.
4. **Find the CBN.** The Classical Boundary Node is where the proof transitions from BISH to CLASS. This is typically a single step: an appeal to analytic continuation, a limit existence theorem, a compactness argument, or a density result.
5. **Measure the conservation gap.** Compare the logical cost of the *statement* to the logical cost of the *proof*. If the statement is BISH but the proof is CLASS, the gap is nonzero—and a simpler proof may exist.

8.2 Cross-Domain Applications

For Product Managers: Technical Debt as Conservation Gap

The conservation gap has a direct analogue in software engineering: **technical debt**. A codebase where the implementation (proof) is more complex than the specification (statement) has a conservation gap. The CRM audit procedure maps directly:

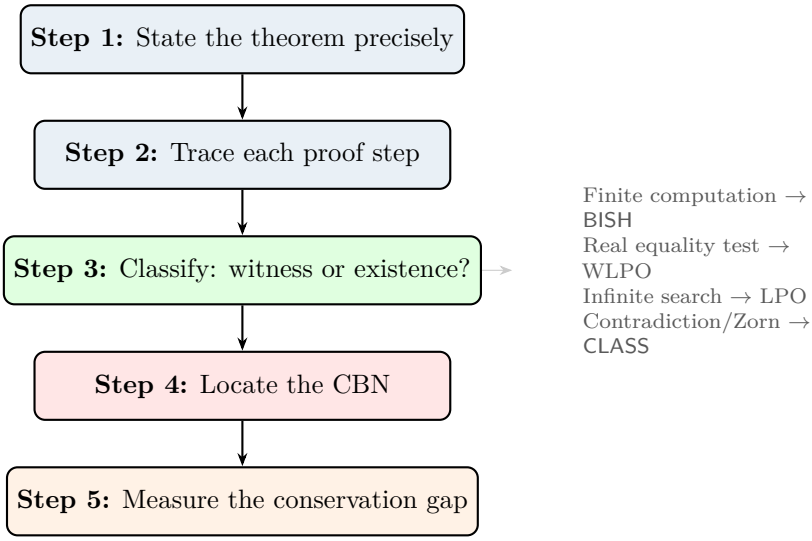


Figure 8.1: The five-step CRM audit procedure. Steps 1–2 are mathematical analysis. Steps 3–5 are CRM classification.

- *Core logic* (business rules, algorithms) = BISH. These are deterministic, testable, and efficient. **Leave them alone.**
- *Infrastructure scaffolding* (caching layers, indeterminate-timeout operations, legacy adapters) = possibly higher-cost. Audit for simplification.
- *Conservation gap* = the scaffolding that exists because of historical accident, not structural necessity. Reducing this gap is the highest-leverage refactoring.

The CRMLint Squeeze protocol has a software analogue: **dependency analysis**. Trace each module's dependencies. Flag the ones that introduce the most complexity. Ask: is this dependency *structural* (the feature requires it) or *scaffolding* (a simpler implementation exists)?

For Physicists: The Measurement Problem Dissolved

Paper 44 showed that the "measurement problem" in quantum mechanics—the apparent collapse of the wave function—is not a single problem but three logically distinct interpretations with different CRM signatures:

- **Copenhagen:** BISH + LPO. The collapse postulate is an oracle (LPO: "does the detector click?") applied to the quantum state.
- **Many-Worlds:** BISH + LPO. No collapse, but decoherence requires the spectral theorem (LPO) to split Hilbert space into branches.
- **Bohmian:** BISH + MP. The guiding equation $m\dot{x} = \hbar \operatorname{Im}(\nabla\psi/\psi)$ is deterministic, but computing the particle's trajectory requires iterating the velocity field to convergence. Since the velocity depends on the global wave function (which may have arbitrarily fine structure), you know the iteration terminates (the trajectory exists by Picard-Lindelöf) but you don't know when to within

any fixed bound—this is exactly Markov's Principle.

All three are logically distinct but empirically equivalent (they agree on all measurable predictions at BISH + LPO). CRM dissolves the metaphysical debate: the three interpretations are three ways of *distributing* the same logical cost across the formalism.

For Graduate Students: Reading Proofs with CRM Eyes

Every proof you read can be audited informally:

- Every "there exists" without an explicit construction = CLASS.
- Every appeal to compactness, Zorn's lemma, or the axiom of choice = CLASS.
- Every limit whose convergence rate is unknown = at least MP.
- Every spectral decomposition over $\mathbb{R}$ = LPO (spectral theorem).
- Everything else = BISH.

The CRM hierarchy gives you a new annotation layer for proofs. When you encounter a difficult proof, ask: is the difficulty *structural* (the theorem genuinely needs this logical power) or *scaffolding* (the author chose a classical shortcut)?

For AI/ML Practitioners: Scaffolding vs. Structure in Model Architectures

The CRM finding that proof complexity $\neq$ logical cost has a direct analogue in neural network design. Complex architectures (many layers, attention heads, skip connections) may be *scaffolding*: the underlying function learned is simpler than the architecture used to learn it. Knowledge distillation, pruning, and neural architecture search are informal versions of the Squeeze protocol: compress the model, test whether the compressed version achieves the same performance, and identify which components are structural vs. scaffolding.

A deeper connection: CRM distinguishes *logical cost* (what principles a result needs) from *proof complexity* (how many steps the proof takes). Similarly, in ML, *sample complexity* (how much data a function needs) differs from *model complexity* (how many parameters the architecture has). A 10-billion-parameter model that implements a function learnable by a 1-million-parameter model has a "conservation gap"—and that gap is the target for distillation. CRM-trained AI could learn to predict which mathematical proofs have analogous gaps, guiding automated simplification.

8.3 CRMLint at Scale: The Mathlib Vision

The ultimate application of the Squeeze protocol is to **audit all of Mathlib**—the 150,000+ declaration library of formally verified mathematics in Lean 4.

Running CRMLint on Mathlib would produce a **proof technique atlas**: for every declaration, the CRM classification and conservation gap. This atlas would reveal:

- Which proof techniques are scaffolding (used everywhere, never structural).
- Where conservation gaps predict simpler proofs exist.
- Which domains of mathematics are "logically cheap" (mostly BISH) and which are "logically expensive" (depend on CLASS principles).

The 98-paper CRM program is the **calibration dataset** for this larger project. Its 14 Lean bundles provide ground truth: we know the correct CRM classification for each theorem, verified by human audit. Any automated tool must reproduce these classifications to be trusted.

The CRMLint Vision

CRMLint on Mathlib = `npm audit` for mathematical proofs. Conservation gaps = technical debt. AI trained on CRM-classified proofs can learn to *predict* where simpler proofs exist and *generate* search targets for proof simplification.

Chapter 9

Conclusion

9.1 What the Programme Proved

The 98-paper CRM program established, with machine-checked proofs:

1. **The physics ceiling is BISH + LPO** (**Paper 29, 35**). Every empirical physical theory calibrated (quantum mechanics, general relativity, QCD, statistical mechanics, scattering theory) fits within this ceiling. LPO enters through the spectral theorem for unbounded self-adjoint operators over $\mathbb{R}$. Fan Theorem and Dependent Choice are dispensable (**Paper 30, 31**).
2. **Five great conjectures all exhibit de-omniscientizing descent** (**Paper 45–49**). Full cost: LPO or CLASS. With algebraic metadata: BISH or BISH + MP. The metadata is always a positive-definite structure from $\mathbb{R}$.
3. **DPT axioms are canonical** (**Paper 72–74**). Each is biconditional with decidability in its domain: Standard Conjecture D $\Leftrightarrow$ morphism decidability; Algebraic Spectrum $\Leftrightarrow$ eigenvalue decidability; Archimedean Polarization $\Leftrightarrow$ cycle-search decidability. No alternative axiom substitutes for any of them.
4. **The Squeeze protocol automates CRM discovery** (**Paper 76–96**). Scaffold $\to$ X-Ray $\to$ Excise $\to$ Squeeze. Across 15+ applications, >80% of formal content is BISH. CLASS content is confined to a small number of CBN nodes: analytic continuation, Euler systems, topological existence assertions.
5. **The motivic architecture explains why** (**Paper 98**). Motives are BISH. Each realization functor has a definite CRM cost. Realizations that pass through $\mathbb{R}$ cost CLASS. Everything else is BISH. The Archimedean Obstruction Thesis is a single principle that explains all 98 papers.

9.2 What Remains Open

The conservation conjecture. Is the CLASS scaffolding in Fargues-Scholze proofs eliminable? DPT and condensed mathematics partition arithmetic into complementary sectors (discrete/BISH vs. continuous/CLASS). The interface between them—five specific questions in **Paper 75**, §6.5—is open.

CRMLint at scale. What does the full atlas of Mathlib look like? How many declarations have conservation gaps? Can AI exploit these gaps to find simpler proofs?

Standard Conjecture D for abelian fourfolds. Axiom 1 of DPT. Proved for abelian varieties of dimension ≤ 3 (Lieberman, Tate). Open for dimension 4+.

Intermediate axiom sets. Are there natural logical systems strictly between

BISH and LPO? WKL (Weak König's Lemma) appears in the Taylor-Wiles calibration (**Paper 98**). Is there a natural family of such intermediate systems?

9.3 The Deeper Claim

CRM is not a restriction of mathematics. It is a sharper lens.

The program did not weaken any theorem. It did not replace any proof. It measured something that was previously invisible: the logical cost of mathematical operations across the domains it surveyed—mathematical physics, arithmetic geometry, representation theory, and selected areas of algebraic geometry. Within these domains, it found consistent structure:

- The $h = f$ **identity**—self-intersection equals conductor on CM abelian fourfolds (**Paper 56–58**)—discovered by tracing where equality testing enters.
- The **algebraic-vs-transcendental boundary** (**Paper 69**)—discovered by comparing number fields ($\mathbb{R}$) to function fields (no $\mathbb{R}$).
- The **detection/existence bifurcation** (**Paper 84–96**)—discovered by separating algebraic computation from topological existence.
- The **palindromic/root-number parallel** (**Paper 86, 96**)—discovered by tracing where discrete invariants control CRM cost.

None of these were discovered by asking classical questions. They were discovered by asking **"what does this cost?"**—a question classical mathematics never asks.

98 papers. One question. One answer.

The logical cost of mathematics is the logical cost of $\mathbb{R}$.

Appendix A

Complete Paper List

No.	Title	Status	DOI suffix
1	[Withdrawn]	—	—
2	The Bidual Gap and WLPO	Active	18501689
3	[Withdrawn]	—	—
4	Axiom Calibration for Quantum Spectra	Active	17118223
5	Schwarzschild Curvature Verification	Active	18489703
6	Heisenberg Uncertainty (v2)	Active	18519836
7	Physical Bidual Gap, Trace-Class Operators	Active	18527559
8	1D Ising Model and LPO	Active	18516813
9	Ising Formulation-Invariance	Active	18517570
10	Logical Geography of Mathematical Physics	Active	18636180
11	Entanglement, CHSH, Tsirelson Bound	Active	18527676
12	Constructive History of Mathematical Physics	Active	18636250
13	Event Horizon as Logical Boundary	Active	18529007
14	Quantum Decoherence	Active	18569068
15	Noether Theorem	Active	18572494
16	Born Rule (Technical Note)	Active	18575377
17	Bekenstein-Hawking Formula	Active	18597306
18	Yukawa RG Constructive Stratification	Active	18626839
19	WKB Tunneling and LLPO	Active	18602596
20	Observable-Dependent Logical Cost, 1D Ising	Active	18603079
21	Bell Nonlocality and LLPO	Active	18603251
22	Markov's Principle and Radioactive Decay	Active	18603503
23	Fan Theorem and Optimization	Active	18604312
24	Kochen-Specker Contextuality and LLPO	Active	18604317
25	Choice Axis (CC, DC) and Ergodic Theorems	Active	18615453
26	Bidual Gap Arithmetic Route, WLPO	Active	18615457
27	Bell Angle Optimisation via IVT, LLPO	Active	18615459
28	Newton vs. Lagrange vs. Hamilton	Active	18616620
29	Fekete's Subadditive Lemma and LPO	Active	18643617
30	Physical Dispensability of the Fan Theorem	Active	18638394
31	Physical Dispensability of Dependent Choice	Active	18645578
32	QED One-Loop: The Landau Pole	Active	18642598
33	QCD One-Loop and Confinement	Active	18642610
34	Scattering Amplitudes Are Constructively Computable	Active	18642612
35	Logical Constitution of Empirical Physics: Metatheorem	Active	18642616
36	Spectral Gap Undecidability: Cubitt = LPO	Active	18642620

No.	Title	Status	DOI suffix
37	Undecidability Landscape Is LPO	Active	18642802
38	Wang Tiling and Origin of Physical Undecidability	Active	18642804
39	Beyond LPO: Thermodynamic Stratification	Active	18642806
40	Logical Constitution of Physical Reality (Monograph)	Active	18654773
41	AdS/CFT Diagnostic	Active	18654780
42	Cosmological Constant Problem	Active	18654789
43	What the Ceiling Means: Constructive Schools	Active	18665418
44	Measurement Problem Dissolved	Active	18671162
45	Weight-Monodromy Conjecture and LPO	Active	18676170
46	Tate Conjecture and LPO	Active	18682285
47	Fontaine-Mazur Conjecture and LPO	Active	18682788
48	BSD Conjecture and LPO	Active	18683400
49	Hodge Conjecture and Constructive Omniscience	Active	18683802
50	Three Axioms for the Motive	Active	18705837
51	Constructive Archimedean Rescue in BSD	Active	18732168
52	Decidability Transfer: Conj D, Abelian 3-folds	Active	18732559
53	CM Decidability Oracle	Active	18713089
54	Bloch-Kato Calibration	Active	18732964
55	K3 Surfaces, Kuga-Satake, and DPT	Active	18733731
56	Exotic Weil Self-Intersection on CM Abelian 4-folds	Active	18734021
57	Exotic Weil Self-Intersection Across Nine Heegner Fields	Active	18735172
58	Class Number Correction for Exotic Weil Classes	Active	18734718
59	De Rham Decidability and DPT Completeness	Active	18735931
60	[Retired—merged into Paper 59]	—	—
61	Lang's Conjecture as the MP→BISH Gate	Active	18736959
62	[Retired—merged into Paper 63]	—	—
63	The Intermediate Jacobian Obstruction	Active	18736965
64	Uniform p-Adic Decidability for Elliptic Curves	Active	18737090
65	Self-Intersection Patterns Beyond Cyclic Cubics	Active	18743151
66	Form-Class Resolution for Non-Cyclic Totally Real Cubics	Active	18745722
67	The Motive Is a Decidability Certificate (Monograph)	Active	18776113
68	Fermat's Last Theorem Is BISH	Active	18838541
69	Function Field Langlands Is BISH	Active	18749757
70	The Archimedean Principle (Capstone)	Active	18750992
71	Archimedean Principle in Cryptography	Active	18752015
72	The DPT Characterization Theorem	Active	18765393
73	Standard Conjecture D Is Necessary	Active	18765700
74	Algebraic Spectrum Is Necessary	Active	18773827
75	Conservation Test: Genestier-Lafforgue LLC	Active	18773831
76	CRMLint: Automated CRM Logical Cost Analysis	Active	18779362
77	Explicit Hodge Decompositions for E^4	Active	18779210

No.	Title	Status	DOI suffix
78	Explicit Local Langlands, Wildly Ramified	Active	18789008
79	[Unassigned]	—	—
80	Algebraic Gauss-Manin via Griffiths Pole Reduction	Active	18791985
81	Fixed Part Theorem via Tensor Gauss-Manin	Active	18801814
82	Picard-Fuchs Monodromy via Kovacic's Algorithm	Active	18801822
83	Generic Picard Number of $E_t \times E_t$	Active	18801826
84	Exotic Weil Classes as Flat Sections	Active	18802819
85	Universal Trace Vanishing for Exotic Weil Classes	Active	18806608
86	Hodge Conjecture via Kani-Rosen Splitting	Active	18809895
87	Omniscience Cost of the Hodge Conjecture	Active	18809903
88	Fermat Domination and Variational Hodge Conj.	Active	18809905
89	Absolute Hodge, Universal Hyperelliptic Weil Locus	Active	18809909
90	The Squeeze Is a Microscope (Synthesis)	Active	18809911
91	Logical Cost of Unconditional Hodge (Markman Audit)	Active	18809917
92	Cohomological Flatness, Genera 5–8 (Zariski Grid)	Active	18821004
93	Structural Vanishing Theorem	Active	18816697
94	Griffiths Group of the Mirror Quintic	Active	18821015
95	BSD Squeeze: Gross-Zagier-Kolyvagin	Active	18821019
96	Root Number Bifurcation: Rank 0 BSD	Active	18820969
97	BSD Landscape Survey (Null Finding)	Internal	—
98	The Motivic CRM Architecture (Capstone)	Active	—

All DOIs carry the prefix `https://doi.org/10.5281/zenodo.`
Series DOI: `10.5281/zenodo.17054050`

Appendix B

Glossary

Archimedean place
: The completion of $\mathbb{Q}$ at infinity, i.e., the real numbers $\mathbb{R}$. Distinguished from finite (p-adic) places by having infinite unit index $u(\mathbb{R}) = \infty$.

BISH
: Bishop's constructive mathematics. The base level of the hierarchy. Every existence claim has an explicit witness. Every algorithm terminates with a known bound.

Biconditional
: An "if and only if" result: $A \Leftrightarrow B$. When a DPT axiom is biconditional with decidability, it is both sufficient *and* necessary.

Chebotarev density
: A theorem on the distribution of primes: for any conjugacy class in a Galois group, there exist infinitely many primes whose Frobenius lands in that class. The proof requires analytic L-functions on $\mathbb{R}$: CLASS.

CM
: Complex multiplication. An abelian variety has CM when its endomorphism ring contains a number field of degree $= 2 \cdot \dim(A)$. CM provides algebraic metadata that drops Hodge detection from WLPO to BISH.

Comparison isomorphism
: A theorem establishing a canonical bijection between two different cohomology theories applied to the same variety. The CRM cost depends on whether the comparison passes through $\mathbb{R}$.

CBN
: Classical Boundary Node. The specific point in a proof where a classical (non-constructive) principle is invoked. Identified by CRMLint during the X-Ray step.

CLASS
: Full classical logic: excluded middle, axiom of choice, Zorn's lemma. Existence without witnesses.

Conservation gap
: The difference between the logical cost of a theorem's statement and the logical cost of its proof. A nonzero gap suggests a simpler proof exists.

CRM
: Constructive Reverse Mathematics. The framework for measuring the logical cost of mathematical theorems.

CRMLint
: A Lean 4 metaprogram (940 lines, 0 `sorry`) that automates CRM analysis by tracing classical dependencies in proof terms.

DC Dependent Choice. The principle that dependent sequences of choices can be made. Physically dispensable (**Paper 31**).

De-omniscientizing descent
The process by which a theorem's logical cost drops from LPO/CLASS to BISH when appropriate algebraic metadata is provided.

DPT Decidable Polarized Tannakian. Three axioms characterizing constructive decidability for motivic cycle-search. Each is biconditional with decidability in its domain.

Euler system
A coherent family of Galois cohomology classes indexed by auxiliary primes. Used by Kolyvagin to prove BSD for rank ≤ 1. Irreducibly CLASS.

Kani-Rosen splitting
A decomposition of the Jacobian of a curve into a product of lower-dimensional abelian varieties, induced by a group of automorphisms. Used in the Hodge campaign to force algebraicity unconditionally.

FT Fan Theorem. A continuity principle for functions on Cantor space. Physically dispensable (**Paper 30**).

Heegner point
A special algebraic point on an elliptic curve constructed via CM theory and the modular parametrization. The computation is BISH.

LLPO Lesser Limited Principle of Omniscience. Can determine which of two binary sequences (at most one with a 1) contains the 1.

LPO Limited Principle of Omniscience. Can scan an infinite binary sequence and determine whether it contains a 1. Existential search over infinity.

Modular symbol
The ratio $L(E, 1)/\Omega^+$, a rational number computed from Hecke eigenvalues. Provides BISH BSD detection for rank-0 curves.

Motive The algebraic "DNA" of a variety. Captures all cohomological information in a single algebraic object. Always BISH.

MP Markov's Principle. If an unbounded search cannot fail forever, it terminates—but you don't know when.

Mumford-Tate group
The smallest algebraic subgroup of $\mathrm{GL}(H^k(X, \mathbb{Q}))$ whose $\mathbb{R}$-points contain the Hodge structure. Its invariants characterize Hodge classes.

Northcott property
A height function has Northcott if $\{P : h(P) \leq B\}$ is finite for every bound B. Converts unbounded search (LPO) to bounded search (MP).

Oracle A hypothetical device that answers a question no algorithm can

answer. LPO, WLPO, and CLASS each correspond to a specific type of oracle.

Palindromic
A polynomial whose coefficient sequence reads the same forwards and backwards ($a_k = a_{n-k}$). In the Hodge campaign, palindromic symmetry of the defining polynomial produces the reciprocal involution and Kani-Rosen splitting.

Realization functor
A machine that converts a motive into concrete mathematical data (cohomology, period matrix, Galois representation). Each realization has a definite CRM cost.

Rosati involution
A positive involution on the endomorphism algebra of a polarized abelian variety, analogous to the adjoint operator on a Hilbert space. It provides the positive-definite pairing exploited by the Archimedean place.

Spectral theorem
The decomposition of a self-adjoint operator on a Hilbert space into a family of projections onto eigenspaces. Requires LPO because it searches for eigenvalues on $\mathbb{R}$.

Squeeze protocol
Four-step procedure (Scaffold, X-Ray, Excise, Squeeze) for extracting constructive content from classical proofs.

$u(\mathbb{R}) = \infty$ The unit index of the real numbers at the Archimedean place is infinite. Forces positive-definite forms in every dimension.

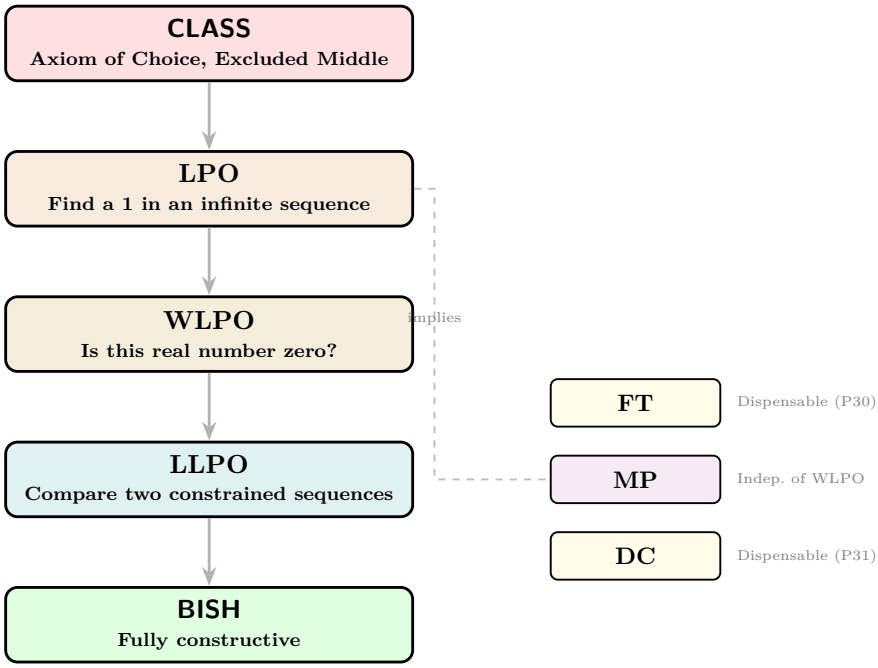
VHC Variational Hodge Conjecture (Grothendieck, 1966). Hodge classes that extend in families remain algebraic. Used on the generic complement in the Hodge campaign.

WLPO Weak LPO. Can decide whether a real number equals zero. An equality test for the continuum, strictly weaker than LPO.

WKL Weak König's Lemma. Every infinite binary tree has an infinite path. Captures compactness arguments. Lies between BISH and LPO in strength.

Appendix C

The Full Hierarchy



Programme Findings:

Physics ceiling	= BISH + LPO	(P29, 35)
Arithmetic + metadata	= BISH + MP	(P48, 51, 61)
Function field Langlands	= BISH	(P69)
FLT	= BISH	(P68)
Hodge test (uniform)	= WLPO	(P87)
Hodge test (with CM)	= BISH	(P87)
BSD detection (rank 0)	= BISH	(P96)
BSD detection (rank 1)	= CLASS	(P95)
Motive itself	= BISH	(P98)
Archimedean realizations	= CLASS	(P98)

Figure C.1: The complete CRM hierarchy with program findings. Each floor strictly implies those below. FT and DC are physically dispensable.

Appendix D

Research Frontiers

What Comes Next?

A geological survey does not end with the map. Once you know where the hard rock is and where the soft ground lies, you know where to build tunnels—and where tunnels are impossible. The CRM program has produced the map. This appendix identifies the tunnels: specific mathematical targets where the Archimedean Obstruction Thesis makes falsifiable predictions about what cheaper proofs should look like.

Each target below is a **prediction**: either the theorem can be proved at the predicted CRM level (confirming the thesis), or it cannot (revealing a new obstruction beyond the Archimedean place). Either outcome advances mathematics.

D.1 Constructive p-adic Langlands

The Neighbourhood with Its Own Generator

The p-adic Langlands correspondence (Breuil, Colmez, Emerton) is the number-theoretic neighbourhood that never connects to the municipal grid. It matches p-adic Galois representations with p-adic Banach space representations of $\mathrm{GL}_2(\mathbb{Q}_p)$—*everything stays p-adic.* No real integration, no spectral theory on $\mathbb{R}$, no Archimedean place.

It is as if an entire district of the city built its own solar farm and disconnected from the power plant entirely.

Prediction. The p-adic Langlands correspondence should be $\mathsf{BISH} + \mathsf{WKL}_0$ (WKL for inverse limits in Banach space completions). No CLASS content should arise. A CRMLint Squeeze of Colmez's proof would be a natural future test of the Archimedean Thesis.

Why it matters. If confirmed, this would be the strongest evidence yet that the Archimedean place is the *sole* source of classical content: even the deepest representation-theoretic correspondence is constructive when $\mathbb{R}$ is absent.

D.2 Condensed Mathematics as a New Lens

A Better Microscope

Scholze's condensed mathematics replaces messy topological spaces with clean algebraic objects (condensed sets = sheaves on profinite sets). In our lens-and-motive language, this is a *brand new realization functor*—a seventh lens for reading the motive's DNA.
The exciting question: does this new lens avoid the Archimedean place? If condensed cohomology can substitute for Betti cohomology in the comparison isomorphisms, it would be like replacing the UV light (expensive, CLASS) with a new wavelength that reads the same information at the cost of ordinary visible light (BISH).

Obstacle (identified in Paper 98). Full condensed mathematics does *not* avoid CLASS. The category of condensed abelian groups has enough projectives because extremally disconnected spaces are projective in compact Hausdorff spaces (Gleason's theorem). Gleason's theorem requires the Boolean Prime Ideal Theorem (BPI). BPI is strictly weaker than the full Axiom of Choice, but still non-constructive—it is equivalent to the ultrafilter lemma, which asserts the existence of ultrafilters without constructing them. For the condensed mathematics application, BPI suffices to produce enough projectives, but it remains CLASS. Condensed mathematics trades *analytic* CLASS axioms (real integration) for *set-theoretic* CLASS axioms (infinite choice). The CRM cost is not reduced.

The fix: light condensed sets. Clausen–Scholze's *light* condensed sets—sheaves on totally disconnected spaces of countable cofinality—avoid Gleason's theorem entirely. The residual cost is sequential completions, which require WKL_0 (compactness for inverse limits) or DC (dependent choice for sequential constructions). Since WKL_0 and DC are independent over BISH, determining which governs light condensed cohomology is itself a CRM question.

Revised prediction. Light condensed de Rham $\to$ light condensed Betti has cost $\mathsf{BISH} + \mathsf{WKL}_0/\mathsf{DC}$, strictly below the CLASS cost of the classical comparison. The Conservation Conjecture (**Paper 67**) becomes testable: is this residual cost optimal?

D.3 Motivic Iwasawa Theory

The p-adic Shortcut to BSD

Recall from Chapter 6 that BSD's classical bottleneck is the Archimedean L-function $L(E, s)$, defined by a Mellin transform over $\mathbb{R}$. But there is a p-adic version of the L-function that avoids $\mathbb{R}$ entirely. Iwasawa theory

studies this p-adic analogue.
Imagine the BSD question ("does this curve have rational points?") as asking "how tall is this building?" The classical approach measures it from a hilltop across town (looking through $\mathbb{R}$). Iwasawa theory measures it from the basement (p-adic, staying inside the building). If the basement measurement is enough, the hilltop trip is unnecessary.

Prediction. $\mathsf{BISH} + \mathsf{WKL}_0$ for the algebraic side of the Iwasawa Main Conjecture. The CLASS content enters only through the *interpolation step*—constructing the p-adic L-function by interpolating complex L-values. A purely p-adic formulation of BSD that avoids complex L-values entirely could achieve $\mathsf{BISH} + \mathsf{WKL}_0$.

Evidence. Paper 96's rank-0 test case (11a1, $w = +1$) is suggestive: the modular symbol ratio $L(E, 1)/\Omega^+$ is already algebraic (BISH by `norm_num`).

D.4 Trace Formula Excision

Replacing the Factory's Assembly Line

The Arthur–Selberg trace formula is the *universal* CLASS bottleneck. It is the assembly line at the $\mathbb{R}$ power plant: every automorphic result passes through it. The bottleneck is the spectral decomposition of $L^2(G(\mathbb{Q})\backslash G(\mathbb{A}))$ at the Archimedean place $G(\mathbb{R})$.
Over function fields, this problem vanishes: the geometric trace formula (Frenkel–Langlands–Ngô) replaces spectral analysis with the Hitchin fibration. It is as if the factory replaced its diesel-powered assembly line with a solar-powered one.
Over number fields, no such replacement exists. The Archimedean places are structurally required. Finding an algebraic bypass would be the mathematical equivalent of room-temperature superconductivity: extremely valuable, widely sought, and currently beyond known techniques.

Assessment: irreducible CLASS (Paper 98). Over number fields, the "base curve" is $\mathrm{Spec}(\mathcal{O}_K) \cup \{\infty\}$, and the Archimedean places $\{\infty\}$ are structurally required by Arakelov geometry to compactify the arithmetic curve. There is no purely algebraic "arithmetic Hitchin fibration" that ignores infinity. The global trace formula over $\mathbb{Q}$ is an irreducible CLASS node.

The local avenue. The Fargues–Scholze geometrization of the *local* Langlands correspondence uses the Fargues–Fontaine curve (purely p-adic, no Archimedean place). This achieves $\mathsf{BISH} + \mathsf{WKL}_0$ for local results, but does not excise the Archimedean place from *global* automorphic arguments.

D.5 Higher-Dimensional Hodge Bifurcations

From One Dial to Many

The palindromic bifurcation in Chapter 5 worked because the genus-4 family had three tuning dials (a, b, c)—enough room for a symmetry locus $\{a = c\}$ of dimension 2. Elliptic curves (Chapter 6) had only one dial (j-invariant), so no bifurcation was possible.

What about mathematical objects with *more* dials? Higher-dimensional varieties have larger moduli spaces, which means more room for symmetry subloci. The prediction: wherever there are enough dials, constructive Hodge loci should exist.

Three concrete targets:

- **Abelian varieties of dimension g:** Moduli dimension $g(g+1)/2$. For $g \geq 3$, positive-dimensional symmetry loci should exist, supporting constructive Hodge class detection.
- **Calabi-Yau threefolds:** The mirror quintic (**Paper 94**) has 1 modulus (ψ). But the full quintic family has $h^{2,1} = 101$ moduli. Palindromic-type loci in 101-dimensional moduli are plausible targets.
- **K3 surfaces:** Moduli dimension 20. The Kuga–Satake construction (**Paper 55**) provides algebraic metadata analogous to CM. The Kuga–Satake locus should support a Hodge bifurcation.

D.6 Explicit Langlands Beyond GL_2

Scaling Up the Translation Manual

Paper 78 found that the local Langlands correspondence for $\mathrm{GL}_2(\mathbb{Q}_2)$ is BISH—the character-trace matching is pure algebra. The global correspondence, by contrast, requires the trace formula (CLASS).

Think of it as a translation dictionary between two languages. The local dictionary (one city's dialect) can be written by hand (BISH). The global dictionary (all dialects everywhere) requires a massive statistical survey (CLASS, via the trace formula).

Can the hand-written approach scale to GL_n? The local data is still algebraic (Bernstein–Zelevinsky classification, Bushnell–Kutzko types). The Archimedean Thesis predicts: BISH for local GL_n.

Prediction. The local Langlands correspondence for GL_n over p-adic fields is BISH. The CLASS content enters only through the global correspondence (trace formula). A CRMLint Squeeze of the Harris–Taylor proof for GL_n would quantify the CLASS content precisely.

D.7 Opportunity Summary

Target	Current	Predicted	Method
p-adic Langlands (Colmez)	Unaudited	$\mathsf{BISH} + \mathsf{WKL}_0$	CRMLint Squeeze
Condensed comparison	Unaudited	$\leq \mathsf{CLASS}$	Conservation test
p-adic BSD ($\mathbb{Z}_p$-ext.)	CLASS	$\mathsf{BISH} + \mathsf{WKL}_0$	Avoid complex L-values
Trace formula excision	CLASS	WKL_0	Hitchin bypass
CY3 Hodge ($h^{2,1} = 101$)	CLASS	Bifurcation	Palindromic loci
K3 Kuga–Satake	CLASS	Bifurcation	Algebraic metadata
GL_n local Langlands	CLASS	BISH (local)	Bushnell–Kutzko

Table D.1: Research targets identified by the Archimedean Obstruction Thesis (**Paper 98**). "Current" is the CRM cost of existing proofs; "predicted" is the framework's prediction for the optimal achievable cost. Each is a falsifiable prediction.

From Catalogue to Predictions

The motivic CRM architecture transforms the program from a catalogue of individual results into a **predictive theory**. Each entry in Table D.1 is falsifiable: either the target theorem can be proved at the predicted level (confirming the Archimedean Thesis), or it cannot (revealing a new obstruction beyond the Archimedean place). Either outcome advances mathematics.

Appendix E

Controversies Resolved

The motivic framework from **Paper 98** resolves several issues that arose during the program.

E.1 `Classical.choice` in $\mathbb{R}$-Theorems

The confusion. Every Lean theorem mentioning $\mathbb{R}$ shows `Classical.choice` in `#print axioms`, even proofs using only `ring` or `norm_num`. Does this mean they are not constructive?

The resolution. No. `Classical.choice` is a *Lean infrastructure artifact*: Mathlib's construction of $\mathbb{R}$ (Cauchy completion, quotient field, decidable equality on supports) injects this axiom into the type signature. A theorem proved by `ring` operates in the de Rham realization (algebraic polynomial identity): it is BISH regardless of what the axiom checker reports.

Analogy: The Factory Stamp

Every product shipped from a factory carries the factory's quality stamp, even products that never went through the quality testing line. The stamp (`Classical.choice`) is on the package because the *packaging material* ($\mathbb{R}$ as defined in Mathlib) comes from the factory. The product inside (the polynomial identity proved by `ring`) was assembled entirely by hand.

The principle. CRM classification is determined by *proof content* (which realization functors and comparison isomorphisms are used), not by the Lean axiom checker output.

E.2 DPT vs. Fargues–Scholze: Complementary, Not Competing

The confusion. The DPT axiom scheme (**Paper 50–53, 72–74**) and the Fargues–Scholze geometrization appear to address similar problems. Are they competing frameworks?

The resolution. They are complementary and *logically forced* to partition along the Archimedean boundary:

- **DPT** operates on the Archimedean side: it provides algebraic oracles (Conjecture D, algebraic spectrum, Archimedean polarization) that bypass the three

Archimedean comparison isomorphisms.

- **Fargues–Scholze** operates on the non-Archimedean side: it works entirely within p-adic Hodge theory, diamonds, and perfectoid spaces—avoiding $\mathbb{R}$ entirely.

However, Fargues–Scholze does not achieve BISH for free. It trades the analytic cost of $\mathbb{R}$ for the topological cost of non-Archimedean completeness (completing $\overline{\mathbb{Q}_p}$ to $\mathbb{C}_p$, functional analysis on p-adic Banach spaces). This requires WKL_0 for compactness and completions. The Archimedean CLASS is eliminated, but a WKL_0 residue remains.

E.3 FLT vs. BSD: Why Their CRM Profiles Diverge

The puzzle. Both FLT and BSD concern elliptic curves over $\mathbb{Q}$, yet their CRM profiles are qualitatively different. FLT has an enormous BISH core with a single contested node (§2.5); BSD is structurally CLASS with classical content distributed across multiple components (**Paper 95–96**).

The explanation.

- **FLT** asserts *non-existence*: the proof proceeds by contradiction (Ribet level-lowering to $N = 2$, no such form exists). The entire argument from Frey curve to contradiction routes through étale and crystalline realizations (BISH), with potential CLASS content concentrated at a single node—the dihedral base case via Hecke theta series (contested: see §2.5).
- **BSD** asserts an *equality*: $\mathrm{ord}_{s=1} L(E, s) = \mathrm{rk}\, E(\mathbb{Q})$. The L-function uses the Mellin transform at the Archimedean place. CLASS content is *distributed* across the analytic side (the L-function *is* the Archimedean realization) and the Galois-cohomological side (Kolyvagin's Euler system requires Chebotarev density).

The structural difference: for FLT, the potential CLASS content is *concentrated in a single contested node* that **Paper 68** argues can be bypassed; for BSD, the CLASS content is *distributed and structural*. This explains why **Paper 96** found no bifurcation for BSD: there is no single node to excise.

E.4 Hodge Bifurcation vs. BSD Rigidity

The puzzle. The Hodge conjecture admits a constructive/classical bifurcation (palindromic locus, **Paper 86–89**). BSD does not (**Paper 96**). Why?

The explanation. Bifurcation requires two ingredients:

1. A continuous family of mathematical objects (moduli space of dimension ≥ 2).
2. A positive-dimensional sublocus where extra symmetry provides an algebraic bypass.

For Hodge: the universal hyperelliptic family has 3 parameters (a, b, c), and the palindromic locus $\{a = c\}$ has dimension 2. For BSD: the moduli of elliptic

curves is the j-line (dimension 1), and proper algebraic subvarieties are isolated points (CM curves).

The Highway vs. the Path

As we said in Chapter 6: the Hodge family lives on a multi-lane highway (9-dimensional moduli for genus-4 Jacobians), so there is room for a "constructive lane" (the palindromic locus). Elliptic curves live on a single-track path (j-line, dimension 1). There is no room for a constructive lane—you are either on the path or off it.

Appendix F

Logos: Philosophical Foundations

Ἐν ἀρχῇ ἦν ὁ Λόγος,
καὶ ὁ Λόγος ἦν πρὸς τὸν Θεόν,
καὶ Θεὸς ἦν ὁ Λόγος.

In the beginning was the Word, and the Word was with God, and the Word was God.

— John 1:1

The author's original motivation for this program was a search for Λόγος—rational structure at the foundations of physics and mathematics. That commitment is recorded in the epigraph and in Paper 40. This appendix sets the program's findings in the context of current debates in the philosophy of mathematics and physics, and explains how CRM addresses open questions that the field's leading thinkers have identified.

F.1 The Open Questions in Foundations

The foundations of mathematics and physics are, as of 2026, in a state of productive tension. Several distinct research programs offer competing or complementary views of what mathematics *is*, what physics *needs* from it, and whether the logical framework matters. The key positions:

Wigner's unreasonable effectiveness. Eugene Wigner's 1960 observation—that mathematics is "unreasonably effective" in the natural sciences—remains unexplained. Why should abstract structures invented by pure mathematicians turn out to describe the physical world? Wigner offered no mechanism. The question has been sharpened but not resolved.

Penrose's mathematical Platonism. Roger Penrose (2004) argues for a three-world picture: a mathematical world (Platonic), a physical world, and a mental world, with mysterious connections between them. The physical world appears to obey mathematical laws, but Penrose emphasizes that we lack an explanation for *which* mathematics is physically relevant. His tilting toward non-computability (via Gödel–Turing arguments about consciousness) places him in tension with constructive approaches, yet his insistence that mathematics has objective structure aligns with what CRM discovers.

Tegmark's mathematical universe. Max Tegmark (2008, 2014) radicalizes Wigner: the physical world does not merely *use* mathematics—it *is* a mathematical structure. The Mathematical Universe Hypothesis (MUH) implies that all consistent mathematical structures are equally real. CRM provides a concrete objection: not all logical frameworks are equally useful for physics. The physical world selects BISH + LPO with perfect precision, not the full classical hierarchy. If the universe is a mathematical structure, it is a very specific one, and the specificity is measurable.

Voevodsky's univalent foundations. Vladimir Voevodsky (Fields Medal, 2002) spent his final years developing Homotopy Type Theory (HoTT) as an alternative foundation for mathematics, motivated partly by his own experience of errors in published proofs. His program shares two convictions with CRM: (1) formal verification matters, and (2) the choice of logical foundation is not neutral. HoTT replaces set theory with type theory and treats equality as a homotopical concept. CRM is orthogonal: it does not change the foundation but *measures* the logical cost within the existing one. The two programs are complementary—HoTT asks "what foundation should we use?"; CRM asks "what does the current foundation actually cost?"

Feferman's predicativism. Solomon Feferman (1998, 2005) argued that all "scientifically applicable" mathematics is predicatively reducible—roughly, that physics never needs the full power of set theory. This is close in spirit to what CRM finds, but with a crucial difference. Feferman's criterion is predicativity (avoidance of impredicative definitions). CRM's criterion is constructivity (avoidance of omniscience principles). The two notions overlap but do not coincide: impredicative definitions can be constructive (as in Lean's `Prop` universe), and predicative mathematics can be non-constructive (classical arithmetic is predicative but uses LEM). CRM provides the finer instrument.

Connes's noncommutative geometry. Alain Connes (Fields Medal, 1982) has argued that noncommutative geometry reveals the "fine structure" of spacetime at the Planck scale. His spectral approach to geometry replaces point-set topology with operator algebras. The spectral theorem—the engine of Connes's entire program—costs exactly LPO in CRM (**Paper 2**). This is not a criticism of Connes but a calibration: his program lives at the physics ceiling (BISH + LPO), and the non-constructive content enters precisely where he crosses from algebra to analysis (the spectral decomposition of unbounded operators on $\mathbb{R}$).

Scholze's condensed mathematics. Peter Scholze (Fields Medal, 2018) and Dustin Clausen developed condensed mathematics to repair the categorical deficiencies of classical topology. Their framework replaces topological spaces with condensed sets (sheaves on profinite sets). As discussed in §3.6 and Appendix D, condensed mathematics and CRM govern complementary territories: Scholze's framework controls the p-adic/geometric sector; CRM controls the algebraic/Archimedean sector. The conservation conjecture—whether CLASS scaffolding in condensed proofs is eliminable—is the open interface.

Maddy's naturalism. Penelope Maddy (1997, 2011) argues that mathematical methodology should be judged by *mathematical* standards, not philosophical ones. The working mathematician does not need a foundational justification to use the axiom of choice. CRM agrees with the methodological point but adds an empirical one: even if mathematicians do not *need* to justify their axioms philosophically, they can *measure* the logical cost of each axiom empirically, and the measurement reveals structure invisible to both philosophical argument and mathematical practice.

F.2 What CRM Contributes

The program addresses these debates by providing something none of them had: **empirical data on logical cost at scale**.

Against logical arbitrariness. The strongest finding is negative: the logical structure of mathematics is *not* arbitrary. If theorems required random assortments of logical principles, CRM would have found a scattered, incoherent map. Instead, it found that all non-constructive content in both physics and arithmetic geometry traces to a single source ($\mathbb{R}$), concentrates at a single mechanism (comparison isomorphisms through the Archimedean place), and is classified by a single hierarchy (Bishop's). This is evidence that the logical structure of mathematics has objective content—supporting Penrose's Platonism, refining Tegmark's MUH, and extending Feferman's predicativism to a finer resolution.

A mechanism for Wigner. CRM offers a partial explanation for unreasonable effectiveness: mathematics is effective in physics because *physics uses very little of it*. The physics ceiling is $\mathsf{BISH} + \mathsf{LPO}$—constructive mathematics plus one omniscience principle. The vast majority of classical mathematics (CLASS) is scaffolding that physics never touches. What Wigner observed as "unreasonable effectiveness" is partly an artifact of not having measured logical cost: mathematics *seems* unreasonably powerful because we attribute to physics the full cost of CLASS, when physics only uses the first two floors.

Calibration for formal verification. Voevodsky's vision of machine-checked mathematics is now reality (Lean, Coq, Isabelle, Agda). But formal verification so far has focused on *correctness* (is the proof valid?) rather than *cost* (what axioms does the proof need?). CRM adds the cost dimension. The 96,200 lines of Lean code in this series are not just verified proofs—they are verified *calibrations*, each tagged with its exact position in Bishop's hierarchy. This provides a template for what large-scale formal verification could become: not just a certificate of truth, but a map of logical dependencies.

Constructivity as empirical hypothesis. Feferman conjectured that all scientifically applicable mathematics is predicatively reducible. CRM tests and refines a parallel conjecture: all *empirically testable* physics is constructive up to LPO. Across 42 papers and every major domain of physics—QED, QCD, general relativity, statistical mechanics, quantum information, the cosmological constant

problem—the ceiling holds. This is no longer a philosophical position; it is an empirical finding, verified line by line in Lean.

The Archimedean thesis as organizing principle. None of the programs above—Penrose, Tegmark, Voevodsky, Feferman, Connes, Scholze, Maddy—identified $\mathbb{R}$ as the single source of logical cost. CRM did. The Archimedean Obstruction Thesis (§1.5) states: non-constructive content enters if and only if the proof path passes through the Archimedean place. This gives the debates a focal point. Instead of arguing in the abstract about whether classical logic is "necessary," one can now ask: does this specific theorem cross the Archimedean boundary? If yes, classical content is expected. If no, look for a constructive proof.

F.3 The Author's Motivation

The Greek word Λόγος—word, reason, proportion, account—is what motivated this program. The author went looking for rational structure at the foundations of physics and mathematics, believing it was there to be found. The 98 papers and 96,200 lines of Lean code are what he found. The reader is free to interpret the finding as evidence for mathematical Platonism, as a contribution to the Wigner problem, as a refinement of Feferman's predicativism, or as confirmation of John 1:1. The proofs do not depend on the interpretation.

This monograph covers Papers 1–98 of the Constructive Reverse Mathematics Series by Paul Chun-Kit Lee (NYU).

All claims are backed by machine-checked proofs in Lean 4 (96,200+ lines, zero `sorry`). All papers are published with DOIs on Zenodo.

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